

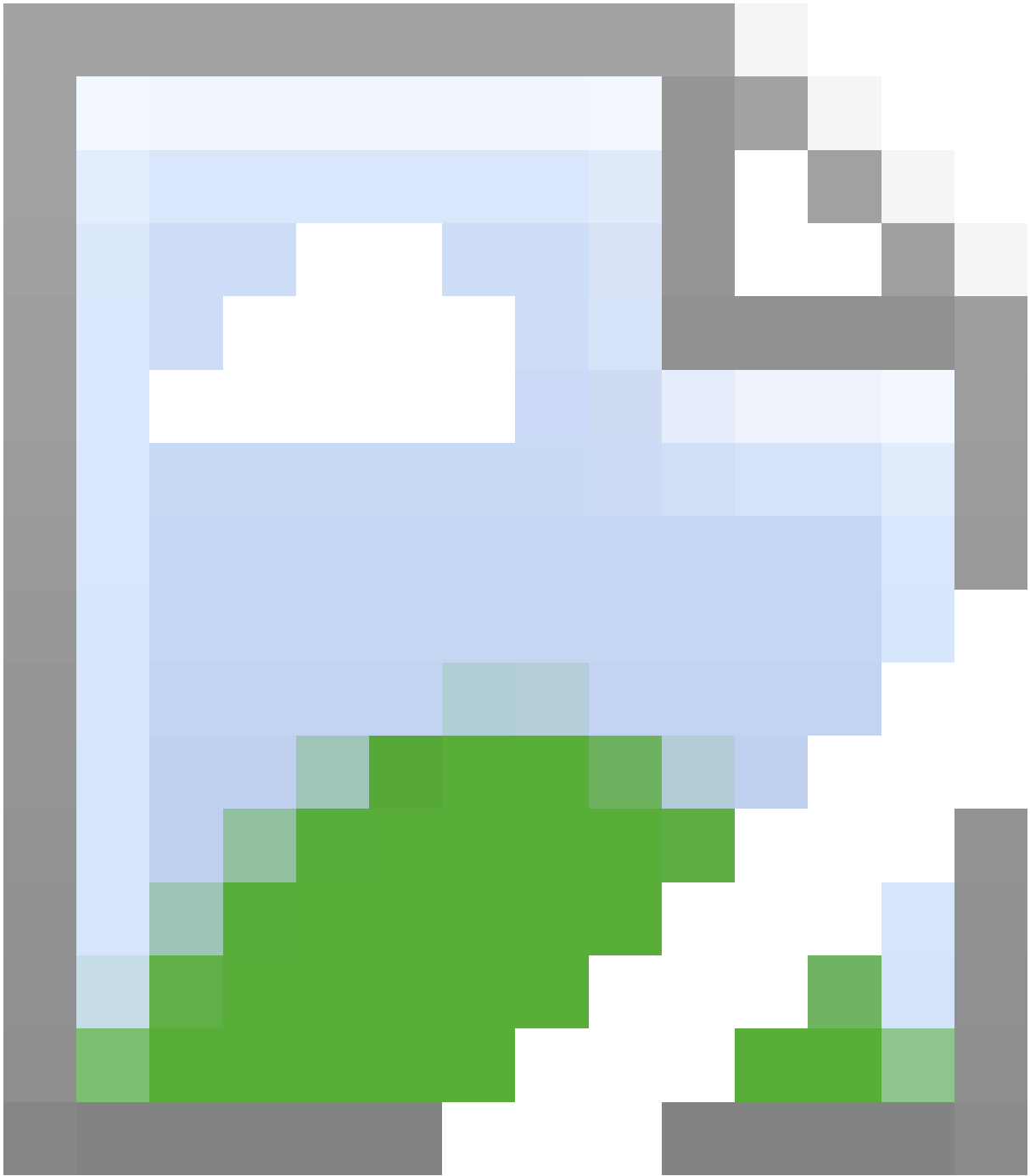
# Inferential and Predictive Statistics for Business

## Module 1



**ILLINOIS**

Gies College of Business





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# Preface

Thank you for choosing a Gies eBook.

This Gies eBook is based on an extended video lecture transcript made from Module 1 of Professor Fataneh Taghaboni-Dutta's [\*Inferential and Predictive Statistics for Business\*](#) on Coursera. The Gies eBook provides a reading experience that covers all of the information in the MOOC videos in a fully accessible format. The Gies eBook can be used with any standards-based e-reading software supporting the ePub 3.0 format.

Each Gies eBook is broken down by lessons that are navigable using our e-reader's table of contents feature. Within each lesson the following sequence of content will always occur:

- Lesson title
- A link to the web-based videos for each lesson (You must be online to view.)

Within the lesson, every time there is a slide change or a switch to the next informative video scene, you will be presented with:

- Thumbnail image of the current slide or video scene
- Any text present on the slide in the video is recreated below the thumbnail in a searchable, screen reader-ready format.
- Extended text description of the important visuals such as graphs and charts presented in the slides.
- Any tabular data from the video is recreated and properly labeled for screen reader navigation and reading.
- All math equations are presented in MathML that provides both content and presentation if on screen.
- Transcript that captures all of the original speech in the video labeled by the person speaking.

All Gies eBooks are designed with accessibility and usability as a priority. This design is intended to serve all readers in a flexible manner regardless of their choice of digital reading tools.

If you have any questions or suggestions for improvement for this Gies eBook, please contact [Giesbooks@illinois.edu](mailto:Giesbooks@illinois.edu)

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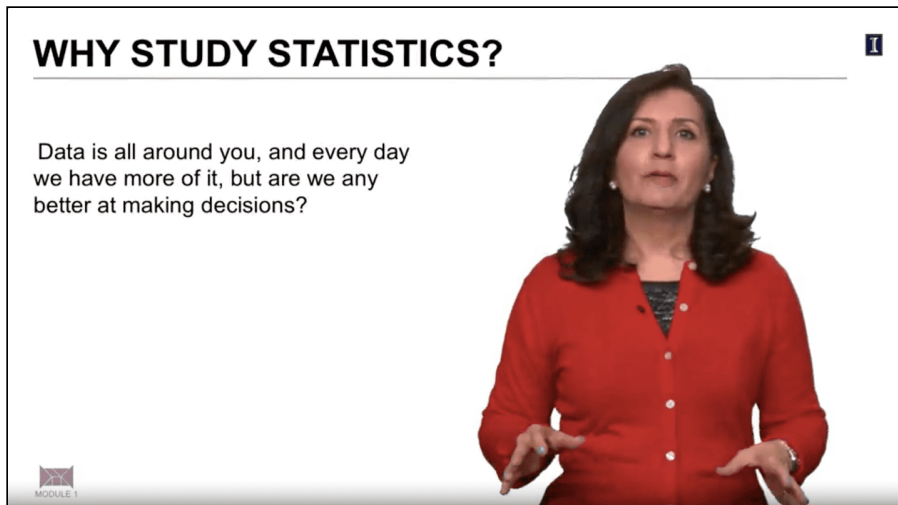
# **Module 1 Inferential and Predictive Statistics for Business**

## About the Course

### Welcome to Inferential and Predictive Statistics for Business!

[Media Player for Video](#) 

### Why Study Statistics? (1 of 3) - Slide 1



Data is all around you, and every day we have more of it, but are we any better at making decisions?

### Transcript

There's data all around you. All day long, you hear about GDP inflation rate, unemployment rate, or in a more fun setting, you may be hearing about some athlete's performance numbers or chance of winning in lottery sometimes in the same sentence as being hit by lightning. In every walk of life, we hear about numbers. These numbers are given to inform us, so that we can make better decision. But are we better decision-makers? Do we all know how to use these numbers?

## Why Study Statistics? (2 of 3) - Slide 2

### WHY STUDY STATISTICS?

Data analysis is important to every field.

- Agro business
- Health care
- Financial institutions
- Hospitality industries
- Retail
- Manufacturing
- Education
- Etc., etc.



MODULE 1

Data analysis is important to every field.

- Agro business
- Health care
- Financial institutions
- Hospitality industries
- Retail
- Manufacturing
- Education
- Etc., etc.

### Transcript

Statistics can be used to explain anything from medicine to agriculture to weather to economics to sports to social behavior. You name it. And statistics can help us identify factors that are influencing outcomes. And learning how these factors influence these outcomes can help us shape the outcomes, so they become more favorable to us.

## Why Study Statistics? (3 of 3) - Slide 3

### WHY STUDY STATISTICS?

Statistics helps us make sense of the data and is one of the most important skills sought by employers in their new hires.

Listed as the number one "hot" skill by LinkedIn\*

"...the sexy job in the next 10 years will be statisticians" (Hal Varian, Chief Economist at Google)\*\*



MODULE 1

\*Murthy, 2014  
\*\*Lohr, 2009

Statistics helps us make sense of the data and is one of the most important skills sought by employers in their new hires.

- Listed as the number one "hot" skill by LinkedIn\*
- "... the sexy job in the next 10 years will be statisticians" (Hal Varian, Chief Economist at Google)\*\*

Sources: \*Murthy, 2014 and \*\*Lohr, 2009

### Transcript

Because statistics can help us make sense of the data, it is one of the most important skills sought by employers in their new hires. You see statistics listed as the number one hot skill by LinkedIn, or being called the sexy job of the next decades. All organizations are looking for these data scientists.

## So What is Statistics? (1 of 2) - Slide 4

### SO WHAT IS STATISTICS?

**DONE THE RIGHT WAY,**  
Statistics involves using mathematical methods on data – but this also means knowing how to get data that is actually useful and finally interpreting the results.



MODULE 1

**Done the Right Way,** Statistics involves using mathematical methods on data—but this also means knowing how to get data that is actually useful and finally interpreting the results.

### Transcript

So what is statistics? This is not a trivial question. Statistics is one field that can easily be misused, either intentionally by unethical or unintentionally by the unaware. When it is done the right way, the data is collected in the right way and analysis done in the right way. And finally the results are presented in the right way, it can be a great tool in understanding the world around us, and benefit by making better decision.

## So What is Statistics? (2 of 2) - Slide 5

### SO WHAT IS STATISTICS?

**WHAT CAN GO WRONG?**  
Any or all of these:  
Data that we have is not good  
Method used is inappropriate  
The interpretation is incorrect



MODULE 1

## What Can Go Wrong?

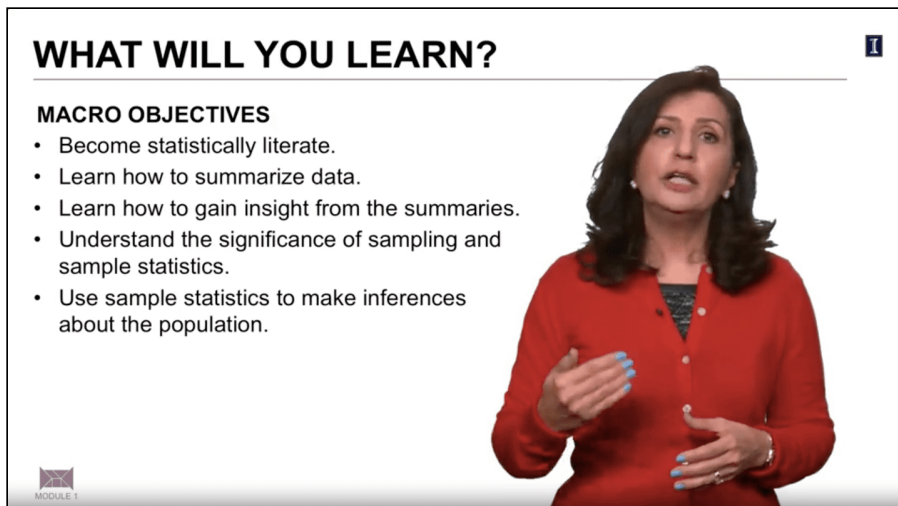
Any or all of these:

- Data that we have is not good
- Method used is inappropriate
- The interpretation is incorrect

## Transcript

And when it's not done the right way, because we start off with bad data, or use the wrong methodology, or misinterpret the result, we can actually do harm.

## What Will You Learn? - Slide 6



**WHAT WILL YOU LEARN?**

**MACRO OBJECTIVES**

- Become statistically literate.
- Learn how to summarize data.
- Learn how to gain insight from the summaries.
- Understand the significance of sampling and sample statistics.
- Use sample statistics to make inferences about the population.

MODULE 1

### Macro Objectives

- Become statistically literate.
- Learn how to summarize data.
- Learn how to gain insight from the summaries.
- Understand the significance of sampling and sample statistics.
- Use sample statistics to make inferences about the population.



## Transcript

So now let's focus on this course. First, I will be completely honest with you. You will not be a data scientist or a statistician after completing this course. That takes many courses and much more training. However, you're taking this course, I assume, because you want to understand the topic better. We need a lot of people with this skill set. Most managers can hire outstanding data scientists, but they themselves don't understand some of the basics then how would they direct these data scientists to explore the right questions? Or how would they ask the right questions when the results are presented. And ultimately, how would they use the results to make better decisions. This is a course for that manager, who wants to get the basics of statistical analysis and procedures at the macro level, the learning objectives of this course is for you to become statistically literate, learn how to summarize data. Learn how to gain insights from the summaries, understanding the significance of sampling and sample statistics, and using sample statistics to make inferences about the population.

## How Will You Learn - Slide 7

A video lecture slide titled "HOW WILL YOU LEARN?". On the right side, a woman with dark hair, wearing a red cardigan, is speaking and gesturing with her hands. On the left side, there is text and a bulleted list. The text says "You will engage in a variety of activities in order to better understand the subject matter; these include:". The list includes "Video lectures", "Excel illustrations", "Graded quizzes", and "Peer reviewed assignments". In the bottom left corner, there is a small logo with the text "MODULE 1". In the top right corner, there is a small blue square icon with the number "1".

**HOW WILL YOU LEARN?**

You will engage in a variety of activities in order to better understand the subject matter; these include:

- Video lectures
- Excel illustrations
- Graded quizzes
- Peer reviewed assignments

MODULE 1

You will engage in a variety of activities in order to better understand the subject matter; these include:

- Video lectures
- Excel illustrations
- Graded quizzes
- Peer reviewed assignments

## Transcript

So how will you be able to get these skills? Each module starts with an overall introduction of topic we are learning about. The video lectures will provide topic specific information. These video lectures will teach the theoretical concepts, which we will then use to apply to various problems and settings. The course objective is to use large data sets where we can apply every concepts we are learning. For this reason, the Excel illustration videos will show you how to apply the methodology, and then how to interpret the results. Here our goal is not to just produce the results. Our goal is to know what the results actually mean. So then we can make better decisions. To make sure that you have mastered the material, you will be given quizzes. Since the statistical concepts covered in this course all build on each other, it is imperative that you get a good understanding of each subject before you move on. The self assessment will allow you to know if you are ready to move on, or if you need to review the concepts covered in more detail. If you feel as do you have not mastered the material, it may be a good time for you to use one of the free textbooks that you haven't given links to as an additional resource. You will also be given questions or scenarios, which are less quantitative in nature and or ask to explain or critique. These are part of the peer-reviewed assignments. You will be asked to post your response and you are asked to read other students' responses and post comments on their quality. This activity will sharpen your skills in reacting to reports that are based on data analysis. And is where you are integrating many of the skills needed by a decision maker. I hope you stay with me and work through the material. You have not pick an easy topic. If it were easy, there wouldn't be a shortage for the skill and employment market. But I can promise you that we will be focus on applying the tools we've learned. Even that I'm going through some dry concepts as sometimes classified by others, not me. I will only go through those only by as much as necessary for you to understand the results and not beyond that. If you stay with me and remain intelligent, you too will see the beauty and the power of this fascinating science.

## References - Slide 8

### REFERENCES

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#### Academic Citations

Lohr, S. (2009, August 5). For today's graduate, just one word: statistics. *The New York Times*. Retrieved from [http://www.nytimes.com/2009/08/06/technology/06stats.html?\\_r=4&](http://www.nytimes.com/2009/08/06/technology/06stats.html?_r=4&)

Murthy, S. (2014, December 17). The 25 hottest skills that got people hired in 2014 [Blog post]. Retrieved from <https://blog.linkedin.com/2014/12/17/the-25-hottest-skills-that-got-people-hired-in-2014>



### Academic Citations

Lohr, S. (2009, August 5). For today's graduate, just one word: statistics. *The New York Times*. Retrieved from [http://www.nytimes.com/2009/08/06/technology/06stats.html?\\_r=4&](http://www.nytimes.com/2009/08/06/technology/06stats.html?_r=4&)

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## **Transcript**

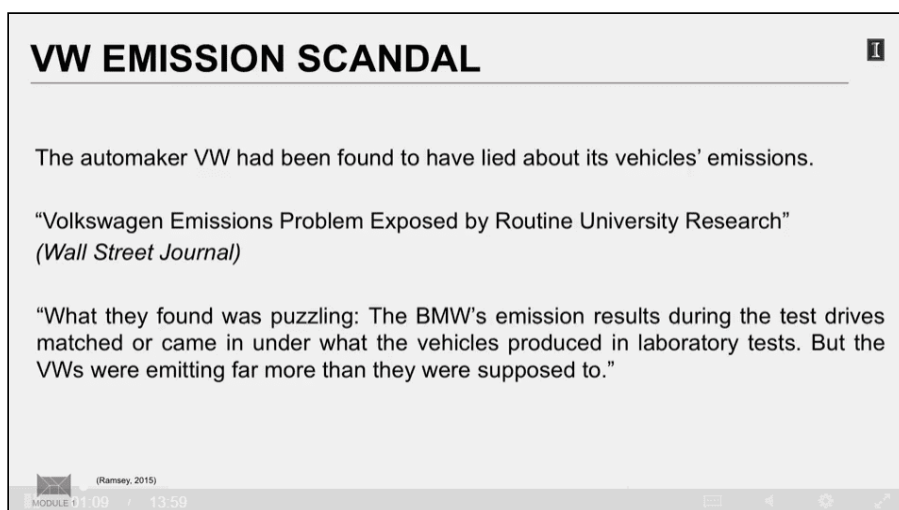
No instruction provided on this slide.

# Lesson 1-1 Formulating the Hypothesis

## Lesson 1-1.1 Formulating the Hypothesis

[Media Player for Video](#) 

### VW Emission Scandal - Slide 9



The screenshot shows a presentation slide with the title "VW EMISSION SCANDAL" in bold. Below the title, there is a paragraph of text. At the bottom of the slide, there is a small logo and the text "(Ramsey, 2015)". The slide is part of a presentation, as indicated by the "MODULE 01 09" and "13:59" text at the bottom.

**VW EMISSION SCANDAL**

The automaker VW had been found to have lied about its vehicles' emissions.

"Volkswagen Emissions Problem Exposed by Routine University Research"  
(*Wall Street Journal*)

"What they found was puzzling: The BMW's emission results during the test drives matched or came in under what the vehicles produced in laboratory tests. But the VWs were emitting far more than they were supposed to."

(Ramsey, 2015)

MODULE 01 09 / 13:59

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(Ramsey, 2015)

## Transcript

You might have heard about Volkswagen intentionally creating tests that misrepresented the emission from its diesel cars. However, you might have not known how they were caught. A researcher at West Virginia University had a grant to study a few diesel cars. This included diesel cars from VW, among others like BMW. However, the sample from Volkswagen would not give emissions that would fall within the confidence interval, and this caught the attention of the researchers. And when they repeated their tests, and were not able to replicate VW's result is when it became apparent that VW must have misrepresented their results. Going against a giant car company is never easy, but the statistics used was not something they could ignore. That's when they went to the United States Environmental Protection Agency, and the rest, as they say, is history. This is not the only time that something like this has happened. Just recently, another car company, Mitsubishi Motors, admitted to falsifying fuel economy tests.

## Other Examples - Slide 10

### OTHER EXAMPLES

- Lance Armstrong
- Falsified data was used in an Alzheimer's disease study involving major pharmaceutical firms.
- You, or someone you know, had a medical test that gave the wrong results.



(Agence France Presse, 2014)

MODULE 1

- Lance Armstrong
- Falsified data was used in an Alzheimer's disease study involving major pharmaceutical firms.
- You, or someone you know, had a medical test that gave the wrong results.

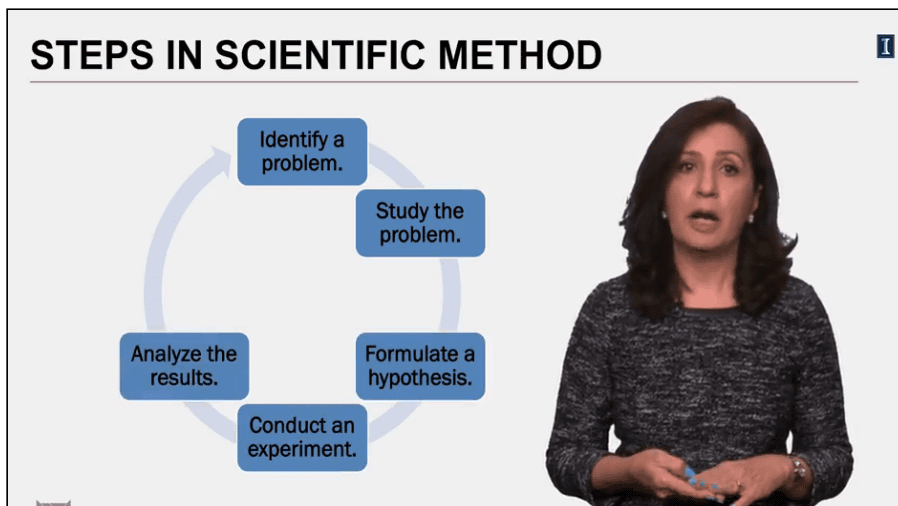
(Agence France Presse, 2014)

## Transcript

Finding that a claim or a finding is not true is not limited to auto industries. Lance Armstrong was a celebrated cyclist who had won several consecutive Tour de France, until finally, there was a blood test that could detect banned substances in his blood samples. Which finally resulted in getting him banned from competing in all sports, and getting stripped of all of his achievements. Of course, he is not the only sports figure to have been found guilty of doping. More disturbing is when we see falsification about drugs, their efficacy or side effects, like where falsified data was used in Alzheimer's disease study involving major pharmaceutical firms. Or maybe someone you know, or even yourself, have done some kind of a medical test, and the results of the test eventually turned out to be wrong.

So, what do all these examples have in common? Well, being that we are learning about statistical models, it must be something about statistics. That's your first clue. Second clue, well, that's the title of this module. All of these examples illustrate the use of hypothesis testing, which builds on what you learned in the first course, and this is an extension of the confidence interval, and thus falls under inferential statistics. Hypothesis testing is an important step in every scientific study.

## Steps in Scientific Method (1 of 2) - Slide 11

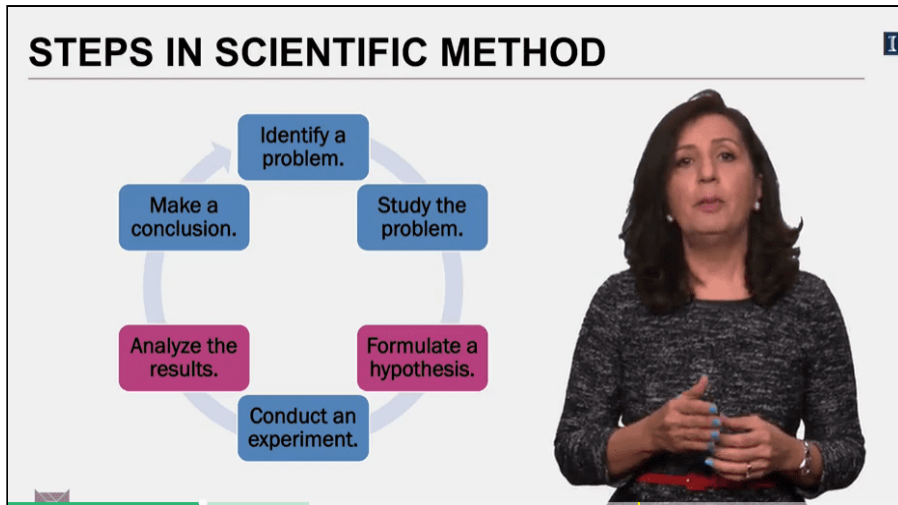


This slide shows a diagram of the scientific method. Beginning at the top and moving clockwise, the steps include: Identify a problem, Study the problem, Formulate a hypothesis, Conduct an experiment, and Analyze the results.

## Transcript

In every scientific study, you start by identify a problem, study the problem, literature research or some preliminary testing, etcetera. Formulate a hypothesis, conduct an experiment, analyze the results, and reach a conclusion.

## Steps in Scientific Method (2 of 2) - Slide 12



This slide shows the diagram of the scientific method from [Slide 11 Steps in Scientific Method \(1 of 2\)](#). There is an additional step at the end: Make a conclusion.

Step 3: Formulate a hypothesis and Step 5: Analyze the results have been highlighted.

### Transcript

Our focus in this module will be on how to formulate a hypothesis, analyzing the results and making your conclusion. The hypothesis testing process begins with formulating the hypothesis, so let's begin with that.

## Formulate a Hypothesis (1 of 2) - Slide 13

**FORMULATE A HYPOTHESIS**

There are two types of hypotheses:

1. **Null hypothesis ( $H_0$ )**
  - What you expect to happen
2. **Alternate hypothesis ( $H_a$  or  $H_1$ )**
  - Everything else

The slide features a presenter on the right side.

There are two types of hypotheses:

1. **Null hypothesis ( $H_0$ )** - What you expect to happen
2. **Alternate hypothesis ( $H_A$  or  $H_1$ )** - Everything else

## Transcript

There are two types of hypotheses, null hypothesis and alternate hypothesis. Together, they will cover all possibilities. Null hypothesis, denoted by  $H$  subscript 0, is what you expect to happen. For example, the researcher in West Virginia University was expecting to get the results that VW had reported. The alternate hypothesis, denoted by  $H$  subscript A or  $H$  subscript 1, will be everything else that can happen. In VW's case, the emission was higher than was reported by the car's manufacturer. So, once you formulate the null hypothesis, the alternate is just its complement.

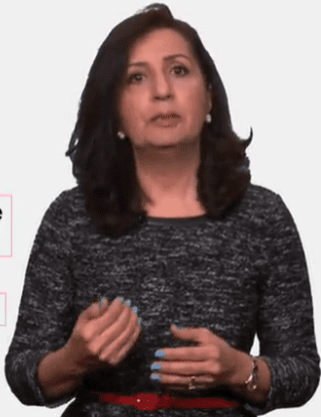
## Formulate a Hypothesis (2 of 2) - Slide 14

### FORMULATE A HYPOTHESIS

Null hypothesis ( $H_0$ )  
Alternate hypothesis ( $H_A$  or  $H_1$ )

$H_0$  and  $H_A$  must be mutually exclusive (both cannot be true) and collectively cover all possibilities.

$H_0$  is a belief that we try to reject, using sample evidence.



Null Hypothesis ( $H_0$ )

Alternate hypothesis ( $H_A$  or  $H_1$ )

- $H_0$  and  $H_A$  must be mutually exclusive (both cannot be true) and collectively cover all possibilities.
- $H_0$  is a belief that we try to reject, using sample evidence.

## Transcript

Null and alternate hypotheses must be mutually exclusive, both cannot be true and collectively cover all possibility. Null hypothesis is a belief that we try to reject using sample evidence.



## Example 1 - Population Mean (1 of 3) - Slide 15

### EXAMPLE 1 – POPULATION MEAN



The average class of 2015 graduates with an average of \$35,000 in debt.

"University of Good Deal" believes their students have a lower debt amount.

SPARSHOTT (2015)



The average class of 2015 graduates with an average of \$35,000 in debt.

"University of Good Deal" believes their students have a lower debt amount.

(Sparshott, 2015)

### Transcript

One issue that's getting attention in the United States is the cost of higher education. One recent article in "Wall Street Journal" claimed that the average debt for the class of 2015 to be \$35,000. Many colleges are trying to reduce this burden by offering financial aid to their students, and they may want to highlight to prospective students that their students graduate with less debt than the national average. But first, this belief that my college does better must be tested. One way is to conduct a hypothesis test to check whether the graduates of university, like Good Deal Universities, are better off or not. So, we have two claims here. Claim one is that the graduates of this university, Good Deal Universities, are not any different than the rest of all graduates. So, the average debt for these graduates is also \$35,000. Basically, there is no reason to believe that we have a different debt coming out of this particular university versus any other. So, this is the current belief, and thus, it's our null hypothesis.

## Example 1 - Population Mean (2 of 3) - Slide 16

### EXAMPLE 1 – POPULATION MEAN

**$H_0$ :** Average debt for Good Deal graduates is \$35,000

**$H_a$ :** Average debt for Good Deal graduates is less than \$35,000

The average class of 2015 graduates with an average of \$35,000 in debt.

“University of Good Deal” believes their students have a lower debt amount.

(Sparshott, 2015)



$H_0$ : Average debt for Good Deal graduates is \$35,000

$H_A$ : Average debt for Good Deal graduates is less than \$35,000

The average class of 2015 graduates with an average of \$35,000 in debt.

“University of Good Deal” believes their students have a lower debt amount.

(Sparshott, 2015)

### Transcript

Claim two is what the Good Deal University is making. This claim needs to be tested against the null hypothesis, and thus, it is the alternate hypothesis. To debunk the null hypothesis, Good Deal University needs to provide data which debunks the current belief in favor of what they believe is the case for their graduates. Now, let's see how we show these two competing claims in notation.

## Example 1 - Population Mean (3 of 3) - Slide 17

**EXAMPLE 1 – POPULATION MEAN**

$H_0$ : Average debt for Good Deal graduates is \$35,000  
 $H_a$ : Average debt for Good Deal graduates is less than \$35,000

$H_0: \mu \geq \$35,000$   
 $H_a: \mu < \$35,000$

(Sparshott, 2015)

MODULE 1



$H_0$ : Average debt for Good Deal graduates is \$35,000

$H_A$ : Average debt for Good Deal graduates is less than \$35,000

$H_0 : \mu \geq \$35,000$

$H_A : \mu < \$35,000$

(Sparshott, 2015)

### Transcript

To show you how this is done, it may be easier to start with the alternate hypothesis, which is the average debt is less than \$35,000. Now, recall that the null hypothesis and the alternate hypothesis must be mutually exclusive, and collectively cover all possibilities. This means that the null hypothesis must be stated as the average debt is \$35,000 or more. Imagine this scenario, a chocolate manufacturing plant is producing bags of chocolate that are labeled at 300 grams.

## Example 2 - Population Mean - Slide 18

### EXAMPLE 2 – POPULATION MEAN

A chocolate manufacturing plant is producing bags of chocolate that are labeled at 300 grams. Quality control takes samples of 50 bags to check for accuracy.

$H_0: \mu = 300$   
 $H_a: \mu \neq 300$





A chocolate manufacturing plant is producing bags of chocolate that are labeled at 300 grams. Quality control takes samples of 50 bags to check for accuracy.

$$H_0 : \mu = 300$$

$$H_A : \mu \neq 300$$

### Transcript

The quality control takes samples of 50 bags to check for accuracy. This is a typical quality control example, happening across industries. Engineers provide specifications, and the job of the quality engineer is to make sure that the specs are met. So, every time quality control checks are done, we are conducting hypothesis testing. Here, our current belief will be that the bags are filled correctly, and the alternate would be that they are not.

## One-tail and Two-tail Tests - Slide 19

### ONE-TAIL AND TWO-TAIL TESTS

**One-tail test:** the null hypothesis allows any value of a parameter larger (or smaller) than a specified value.

$H_0: \mu \geq \$35,000$

**Two-tail test:** the null hypothesis states a specific value for the population parameter.



**One-tail test:** the null hypothesis allows any value of a parameter larger (or smaller) than a specified value.

$$H_0 : \mu \geq \$35,000$$

**Two tail test:** the null hypothesis states a specific value for the population parameter.

### Transcript

We have a one-tail test when the null hypothesis allows any value of a parameter larger or smaller than a specified value, like we did in case of student debt, when we said average was more than \$35,000. In this case, the null hypothesis allows for 35,000 or any value greater than 35,000. Two-tail tests, on the other hand, has a null hypothesis that states a specific value for the population parameter, as was the example for where the bag's weight is set at 300 grams.

## Equal Sign (1 of 2) - Slide 20

### EQUAL SIGN

We believe the freshman entering our university have an average ACT score of more than 24.

$$\mu > 24$$



We believe the freshman entering our university have an average ACT score of more than 24.

$$\mu > 24$$

### Transcript

Let's do this example. We believe that the freshman entering our university has an average ACT score of more than 24, that's our current belief. Has it remained the same, or has this changed? If you look here, the current belief is that  $\mu$ , the average – population average is greater than 24. However, since the equality is missing, then we will make this our alternate hypothesis. And the null will be –  $H_0$  is that  $\mu$  is less than or equal to 24.

## Equal Sign (2 of 2) - Slide 21

### EQUAL SIGN

We believe the freshman entering our university have an average ACT score of more than 24.

$$H_0: \mu \leq 24$$
$$H_a: \mu > 24$$



We believe the freshman entering our university have an average ACT score of more than 24.

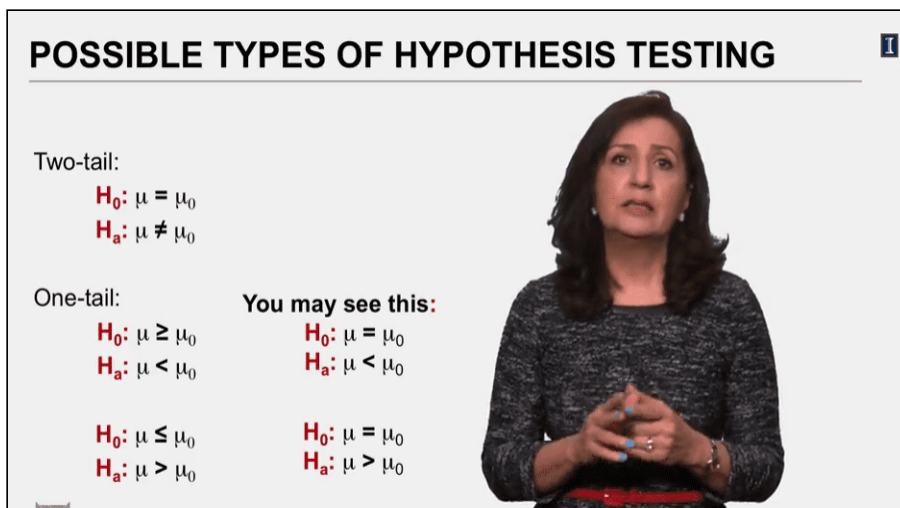
$$H_0 : \mu \leq 24$$

$$H_A : \mu > 24$$

## Transcript

This is because we need to have the equals sign to appear in the null hypothesis. This is a necessary part of formulating the null and alternate hypotheses. When you're testing the mean, then you have three possible sets of hypotheses. One is in the form of a two-tail test, which is the mean is equal to some hypothesized mean, or it's not. The other one is one of the two forms of one-tail test.

## Possible Types of Hypothesis Testing - Slide 22



**POSSIBLE TYPES OF HYPOTHESIS TESTING**

Two-tail:

- $H_0: \mu = \mu_0$
- $H_a: \mu \neq \mu_0$

One-tail:

- $H_0: \mu \geq \mu_0$
- $H_a: \mu < \mu_0$

**You may see this:**

- $H_0: \mu \leq \mu_0$
- $H_a: \mu > \mu_0$

**You may see this:**

- $H_0: \mu = \mu_0$
- $H_a: \mu < \mu_0$

- $H_0: \mu = \mu_0$
- $H_a: \mu > \mu_0$

Two-tail:

$$H_0 : \mu = \mu_0$$

$$H_A : \mu \neq \mu_0$$

One-tail:

$$H_0 : \mu \geq \mu_0$$

$$H_A : \mu < \mu_0$$

$$H_0 : \mu \leq \mu_0$$

$$H_A : \mu > \mu_0$$

**You may see this:**

$$H_0 : \mu = \mu_0$$

$$H_A : \mu < \mu_0$$

$$H_0 : \mu = \mu_0$$

$$H_A : \mu > \mu_0$$

## Transcript

Because the equals sign is always in the null hypothesis, most statistical software programs will only ask for the formulation of alternate hypothesis only with three choices, being not equal, less than, and greater than. Based on what you pick, the null will be inferred. Sometimes, people get lazy and write the hypothesis like this. But, again, this is technically incorrect, but it is done out of laziness. Because based on the alternate, and the fact that the null and the alternate cover all possibilities, everyone knows that a one-tail test will be conducted.

## Let's Practice - Slide 23

### LET'S PRACTICE

A company's website promotes to its potential job applicants that its employees are so happy that on average they stay with the company for 25 years. This claim has been made for many years. The company has hired a new HR director. The new HR director wants to make sure that the claim is still true.

What is  $H_0$  and  $H_a$ ?



A company's website promotes to its potential job applicants that its employees are so happy that on average they stay with the company for 25 years. This claim has been made for many years. The company has hired a new HR director. The new HR director wants to make sure that the claim is still true.

What is  $H_0$  and  $H_A$ ?



## Transcript

So now let's practice. Consider the following. A company's website promotes to its potential job applicants that their employers are so happy that on average they stay with the company for 25 years. This claim has been made for many years. The company has hired a new HR director. The new HR director wants to make sure that the claim is still true. What is the null and alternate hypothesis here?

## Example 3 - Population Proportion (1 of 2) - Slide 24

**EXAMPLE 3 – POPULATION PROPORTION**

A manager is considering moving her retail store to a new location where it is believed that stores similar to hers have higher customer traffic. The monthly increase in rent would mean that her customer traffic must increase by at least 20%. Should she move to this location?



A manager is considering moving her retail store to a new location where it is believed that stores similar to hers have higher customer traffic. The monthly increase in rent would mean that her customer traffic must increase by at least 20%. Should she move to this location?

## Transcript

This is a test about the population mean. The claim is that the average length of tenure is 25 years, and the alternate is that the average has changed. Thus, this is a hypothesis test about the population mean, and it's a two-tail test. Now, let's look at another example. A manager's considering moving her retail store to a new location where it is believed that similar stores to her have higher customer traffic. The monthly increase in rent would mean that at least her customer traffic must increase by 20% should she move to this location. Clearly, if the customer traffic is not increased, then the probability of more sales will not be there to offset the additional cost. In this example, the question is about the population proportion.

## Example 3 - Population Proportion (2 of 2) - Slide 25

**EXAMPLE 3 – POPULATION PROPORTION**

$p$ : the percent increase in customer traffic

$H_0: p \geq 0.20$   
 $H_a: p < 0.20$



$p$ : the percent increase in customer traffic

$$H_0 : p \geq 0.20$$

$$H_A : p < 0.20$$

### Transcript

Stating the null and the alternate hypotheses for population proportion follows the same principles as we followed for the population mean. If  $p$  represents the percent increase in customer traffic, then this store owner is going to test the belief that the proportion of customer traffic will go up at least by 20%, which means the alternate is that the proportion is below 20%. Remember that the owner will decide against this move only if the customer traffic increases by less than 20%. This is another example of one-tail test. You may find it easier to state the alternate hypothesis first, and then based on that, write the null. So, here's another example for population proportion. A nutrition supplement claims that 80% of its customers report improved sense of physical wellbeing. A newspaper investigator is wondering if that is so or not.

## Example 4 - Population Proportion (1 of 3) - Slide 26

### EXAMPLE 4 – POPULATION PROPORTION

A nutrition supplement claims that 80% of its customers report an improved sense of physical well-being. A newspaper investigative reporter is interested in testing this claim.

$p$ : the percent who reported an improved sense of physical well-being

$H_0: p = 0.80$   
 $H_a: p \neq 0.80$



A nutrition supplement claims that 80% of its customers report an improved sense of physical well-being. A newspaper investigative reporter is interested in testing this claim.

$p$ : the percent who reported an improved sense of physical well-being

$$H_0 : p = 0.80$$

$$H_A : p \neq 0.80$$

### Transcript

So, based on the claim, we can say that  $p$  represents the proportion of the population using the supplement who reported an improved sense of physical wellbeing. Based on this, one could write the null hypothesis as  $p$  is equal to .8, and the alternate would be  $p$  is not 80%, and this will be a two-tail test. Now, let's reword the example we just saw. This time, a nutrition supplement claims that at least 80% of its customers report improved sense of physical wellbeing.

## Example 4 - Population Proportion (2 of 3) - Slide 27

### EXAMPLE 4 – POPULATION PROPORTION

A nutrition supplement claims that **at least** 80% of its customers report an improved sense of physical well-being. A newspaper investigative reporter is interested in testing this claim.

$p$ : the percent who reported an improved sense of physical well-being

$H_0: p \geq 0.80$   
 $H_a: p < 0.80$



A nutrition supplement claims that **at least** 80% of its customers report an improved sense of physical well-being. A newspaper investigative reporter is interested in testing this claim.

$p$ : the percent who reported an improved sense of physical well-being

$$H_0 : p \geq 0.80$$

$$H_A : p < 0.80$$

### Transcript

The newspaper reporter now is going to investigate this claim. In this case, then, the null hypothesis will be  $p$  is equal or more than .8, and the alternate would be  $p$  is less than 80%, which will make this a one-tail test. Now, let's really look at this problem again. This time, the nutrition supplement has claimed that 80% as the percentage of proportion that claim an increased sense of wellbeing.

## Example 4 - Population Proportion (3 of 3) - Slide 28

### EXAMPLE 4 – POPULATION PROPORTION

A nutrition supplement claims that 80% of its customers report an improved sense of physical well-being. A newspaper investigative reporter thinks **this is too high** and would like to study it.

$p$ : the percent who reported an improved sense of physical well-being

$H_0: p \geq 0.80$   
 $H_a: p < 0.80$



A nutrition supplement claims that 80% of its customers report an improved sense of physical well-being. A newspaper investigative reporter thinks **this is too high** and would like to study it.

$p$ : the percent who reported an improved sense of physical well-being

$$H_0 : p \geq 0.80$$

$$H_A : p < 0.80$$

### Transcript

The reporter thinks this is too high, and would like to study this. Now, the reporter is putting out an alternate hypothesis, a theory that its proportion is less than 80%. Then the null has to be formed as will be  $p$  is equal or more than .8%. Through these three ways of thinking about the same example, I hope you see that thinking about the research question should guide you in developing the null and alternate hypothesis.

## Hypothesis Testing - Slide 29

### HYPOTHESIS TESTING

**Step 1:**

State the null hypothesis and the alternative hypothesis.



**Step 1:** State the null hypothesis and the alternative hypothesis

### Transcript

All scientific studies start with asking a question about a variable of interest. We start the testing by formulating the question in a form of a hypothesis that gives us the null hypothesis and the alternate hypothesis. The null hypothesis is the statement that is believed to be true. The data collection and analysis is based on the null hypothesis. If the data collected contradicts the null hypothesis, then we will reject the null hypothesis. We will learn how to do this in the other lessons in this module.

### Academic Citations:

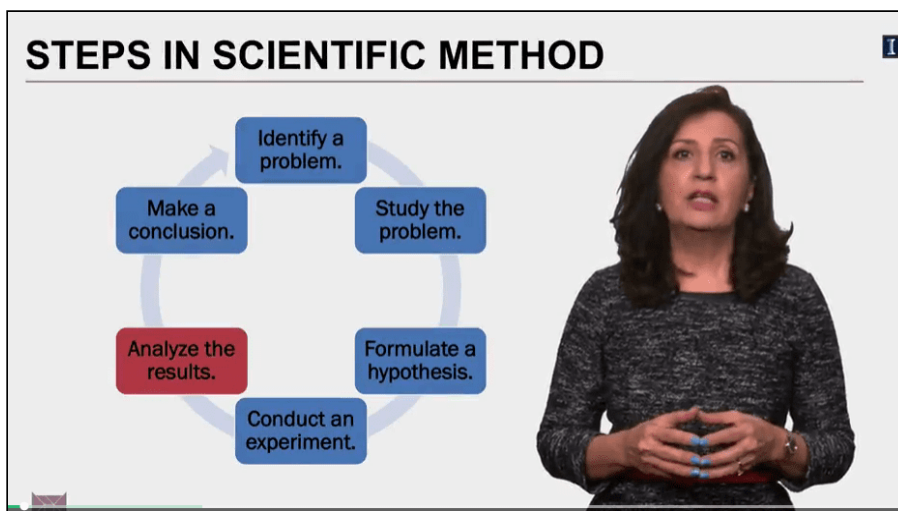
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## Lesson 1-2 Analyzing the Result

### Lesson 1-2.1 Analyzing the Result

[Media Player for Video](#) 

### Steps in Scientific Method - Slide 30



This slide shows the diagram of the scientific method from [Slide 12 Steps in Scientific Method \(2 of 2\)](#). Step 5: Analyze the results is highlighted.

### Transcript

Once we have formulated the hypotheses then we can gather data and after that we can start analyzing. Conducting an experiment goes back to the principles we discussed in the earlier course about how to produce data and conduct sample studies.

## Significance Level (1 of 2) - Slide 31

### SIGNIFICANCE LEVEL

Specify the level of significance  $\alpha$ .

Confidence level =  $1 - \alpha$



Specify the level of significance  $\alpha$

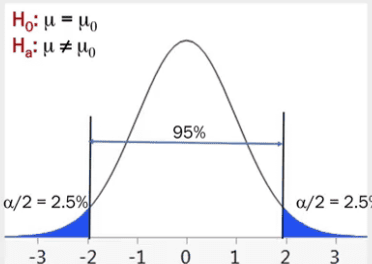
Confidence level =  $1 - \alpha$

### Transcript

Here we're assuming that we have followed the proper steps in attaining our data, so we are focusing on how to proceed with the data analysis. The first thing is to set the significance level referred to as  $\alpha$ . This will be used for testing the hypothesis. I hope you remember that we heard about  $\alpha$  when we learned about confidence interval in the previous course. Confidence level is one minus level of significance.

## Significance Level (2 of 2) - Slide 32

### SIGNIFICANCE LEVEL



$H_0: \mu = \mu_0$   
 $H_a: \mu \neq \mu_0$

A chocolate manufacturing plant is producing bags of chocolate that are labeled at 300 grams. Quality control takes samples of 50 bags to check for accuracy.

$H_0: \mu = 300$   
 $H_a: \mu \neq 300$



A chocolate manufacturing plant is producing bags of chocolate that are labeled at 300 grams. Quality control takes samples of 50 bags to check for accuracy.

$$H_0 : \mu = 300$$

$$H_A : \mu \neq 300$$

Bell curve with tails shaded on right and left, labeled with the following: alpha divided by 2 equals 2.5 percent. Area under curve between two shaded areas labeled 95 percent.

$$H_0 : \mu = \mu_0$$

$$H_A : \mu \neq \mu_0$$

## Transcript

So, for instance, when  $\alpha$  is .05 then confidence level is 95%, take  $\alpha$  of .05 and a two-tail test where we are testing the equal to not equal.  $\alpha$  is the probability of taking a sample from the blue shaded area (Areas at either tail of the curve). This is a sample that doesn't represent the population. It could lead us to making the wrong conclusion. So, before we can talk about the analysis, let's look at where our analysis could lead us to. I'm going to go back to one of the examples we had in the previous lesson where a chocolate manufacturer wants to see if the bags weight 300 grams as they are supposed to. When we do our analysis, we can make two mistakes, Type I and Type II errors. Type I error also known as  $\alpha$  is when we reject the null hypothesis incorrectly. And Type II error is when we don't reject or retain the null hypothesis incorrectly.

## Type I and Type II Errors - Slide 33

| TYPE I AND TYPE II ERRORS                                         |                  |                          |                           |
|-------------------------------------------------------------------|------------------|--------------------------|---------------------------|
| Type I error: Reject $H_0$ incorrectly (also known as $\alpha$ )  |                  |                          |                           |
| Type II error: Retain $H_0$ incorrectly. (also known as $\beta$ ) |                  |                          |                           |
| $H_0: \mu = 300$<br>$H_a: \mu \neq 300$                           |                  |                          |                           |
| Decision                                                          |                  | Reality                  |                           |
|                                                                   |                  | $H_0: \mu = 300$ is True | $H_0: \mu = 300$ is False |
| Reject                                                            | $H_0: \mu = 300$ | Type I Error             | ✓                         |
| Retain                                                            | $H_0: \mu = 300$ | ✓                        | Type II Error             |

Type I error: Reject  $H_0$  incorrectly (also known as  $\alpha$ )

Type 2 Error: Retain  $H_0$  incorrectly (also known as  $\beta$ )

$$H_0 : \mu = 300$$

$$H_A : \mu \neq 300$$

Type I and II errors

|                                      | Reality - $H_0 : \mu = 300$ is<br>True | Reality - $H_0 : \mu = 300$ is<br>False |
|--------------------------------------|----------------------------------------|-----------------------------------------|
| Decision - Reject: $H_0 : \mu = 300$ | Type I error                           | ✓                                       |
| Decision - Retain: $H_0 : \mu = 300$ | ✓                                      | Type II error                           |

## Transcript

Let's look at the example where we were checking the weights of chocolate bags. We believe that the bags weight 300 grams. Thus, we are taking samples to check on this. The possible conclusion we can have are summarized in this table. There are two possibilities about our production process. One is that in reality the bags do weight 300 grams and the other possibility is that they don't. If the production process is working correctly, but our sample shows otherwise, then we will end up rejecting the null hypothesis. This is a mistake and is known as a Type I error. We can also have a production system that is not working properly. Thus, the bags are not really 300 grams. Now if you take a sample and that leads us to believe that the system is working properly, then we have made Type II error, notation here being  $\beta$ . The other two possibilities represent correctly reject a false null hypothesis or don't reject the hypothesis which is true.

## Manufacturing Setting - Slide 34

### MANUFACTURING SETTING

Type I error: **Manufacturer's risk**  
Shut down production unnecessarily

Type II error: **Consumer's risk**  
Fail to stop production to make adjustments



Type I error: ***Manufacturer's risk*** - Shut down production unnecessarily

Type II error: ***Consumer's risk*** - Fail to stop production to make adjustments

## Transcript

In different settings the probability that we can make a mistake is called different things. For example, in manufacturing and quality control type one error is referred to as manufacturers risk; this is when a production line is shut down unnecessarily. Type II error is known as the consumer's risk, this is when they fail to stop a production line that really needs adjustments and defective products are passed on to the consumers.

## Medical Setting - Slide 35

### MEDICAL SETTING

Type I error: **False positive**  
The test result indicates a disease is present for a healthy patient.

Type II error: **False negative**  
The test fails to detect the disease that is actually present.



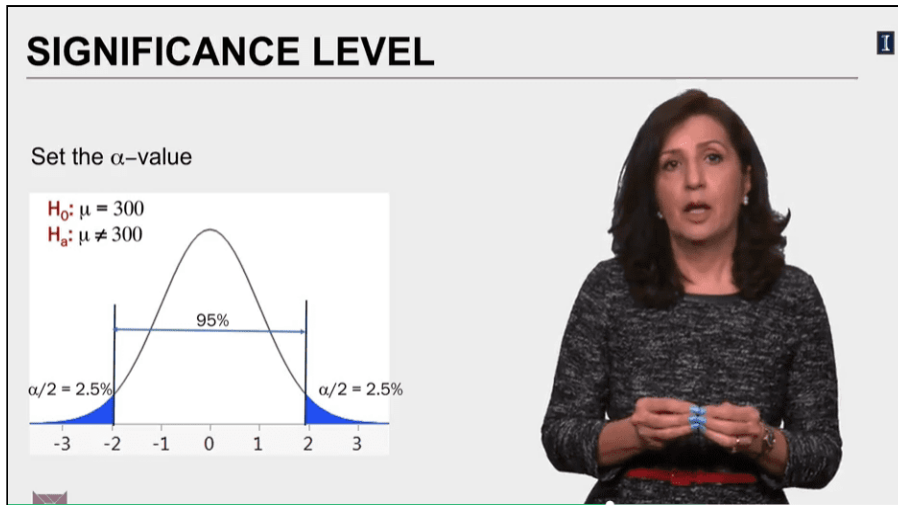
Type I error: **False positive** - The test result indicates a disease is present for a healthy patient.

Type II error: **False negative** - The test fails to detect the disease that is actually present.

### Transcript

We also refer to these errors differently in medical setting. In medical setting Type I error is called a false positive, which is based on lab results falsely calling a patient sick but in fact they are healthy. And Type II error is called false negative, when the lab results will lead a physician to think a patient is disease free, when in fact they have a disease. You can think of Lance Armstrong who was being declared drug free when in fact he was using performance enhancing drugs. The lab results were giving false negative results. So now that you know we can make mistakes in our analysis, we can go back to  $\alpha$ . Before any analysis is to begin we need to set this value. A statistical test relies on the sampling distribution of the statistics that estimate the parameter specified in the null, and alternative hypothesis.  $\alpha$  represents the chance of getting a sample that differs from the null hypothesis by as much as the sample we have, if null hypothesis and reality is true.

## Significance Level - Slide 36



Set the  $\alpha$  value

Bell curve with tails on right and left shaded and labeled 2.5 percent.

$$H_0 : \mu = 300$$

$$H_A : \mu \neq 300$$

### Transcript

In case of a chocolate manufacturer, the production line will be stopped to be fixed if the mean of the sample we have taken will fall in either direction into one of the blue shaded area. We will want to stop the production line and make adjustments. If the bags are too heavy or too light, we can by chance take a sample that is that far from the mean and lead us to stop the production when in reality the production line is just fine. When we set  $\alpha$  at .05 then the chance is evenly split between the two tails, that is what is called a two-tail test.

## Typical Values - Slide 37

### TYPICAL VALUES

- Low alpha gives small chance of rejecting a true  $H_0$
- Typically,  $\alpha = 0.05$ 
  - Strong evidence is required to reject  $H_0$
  - Usually choose  $\alpha$  between 0.01 and 0.10
  - $\alpha = 0.01$  requires very strong evidence to reject  $H_0$



Low alpha gives small chance of rejecting a true  $H_0$

Typically,  $\alpha = 0.05$

- Strong evidence is required to reject  $H_0$
- Usually choose  $\alpha$  between 0.01 and 0.10
- $\alpha = 0.01$  requires very strong evidence to reject  $H_0$

### Transcript

$\alpha$  is usually set to a low value to minimize the probability of rejecting the true null hypothesis. In business,  $\alpha$  is typically set to .05. And in this setting, it takes strong evidence to reject the null hypothesis. Setting of .01 and .10 are also sometimes used for  $\alpha$ .  $\alpha$  of .01 would be considered a low value, which requires strong evidence to reject the null hypothesis and .10 would be considered a high value. There is a relationship between  $\alpha$  and  $\beta$ , Type I and Type II errors. For a fixed sample size the smaller we make  $\alpha$ , the larger  $\beta$  becomes.

## Trade off between $\alpha$ and $\beta$ - Slide 38

### TRADE OFF BETWEEN $\alpha$ AND $\beta$

For a given sample size, reducing  $\alpha$  will increase  $\beta$ :

$$\alpha \downarrow \Rightarrow \beta \uparrow$$

If we can increase the sample size, we can reduce both  $\alpha$  and  $\beta$  simultaneously:

$$n \uparrow \Rightarrow \alpha \downarrow \text{ and } \beta \downarrow$$



For a given sample size, reducing  $\alpha$  will increase  $\beta$ :

$$\alpha \downarrow \rightarrow \beta \uparrow$$

If we can increase the sample size, we can reduce both  $\alpha$  and  $\beta$  simultaneously:

$$n \uparrow \rightarrow \alpha \downarrow \text{ and } \beta \downarrow$$

### Transcript

That is the lower the probability of rejecting a true null hypothesis, the higher the probability of failing to reject a false null hypothesis. Note that while we set and therefore know the value of  $\alpha$ , we don't set the value of  $\beta$ .  $\beta$  is calculated based on several parameters. Since you know by choosing  $\alpha$  you're setting the probability of Type I error, you may want to set this value very low. But before deciding what the  $\alpha$  value should be, you should know that there is a relationship between Type I error,  $\alpha$  and Type II error,  $\beta$ . For a fixed sample size the smaller we make  $\alpha$ , the larger  $\beta$  becomes. That is the lower the probability of rejecting a null hypothesis manufacturers risk, the higher the probability of rejecting a false null hypothesis consumers risk. As I mentioned before we don't set the value of  $\beta$ , but we can calculate for it. Here again, as a general rule the test will be it's better for larger sample sizes and as the sample size grows the probability of making either mistakes,  $\alpha$  or  $\beta$  would reduce. But here again, is the question of cost.



## Steps in Testing the Mean - Slide 39

### STEPS IN TESTING THE MEAN

1. State  $H_0$  and  $H_a$ .
2. Specify the significance level  $\alpha$ .
3. Calculate the p-value for the sample data.
4. Reject or not reject  $H_0$ .



1. State  $H_0$  and  $H_A$
2. Specify the significance level  $\alpha$
3. Calculate the p-value for the sample data
4. Reject or not reject  $H_0$

### Transcript

After stating the null hypothesis and specifying the level of significance we can move on to the next step which is calculating the p-value for the sample data. Once we have the p-value the only thing remaining is to make a decision about our null hypothesis, that is to reject or not reject. We will explore these two steps in the next lesson.

## Lesson 1-3 Two-Tail Test for Mean

### Lesson 1-3.1 Two-Tail Test for Mean

[Media Player for Video](#) 

#### Steps in Testing the Mean - Slide 40

**STEPS IN TESTING THE MEAN** 



1. State  $H_0$  and  $H_a$ .
2. Specify the significance level  $\alpha$ .
3. Calculate the p-value for the sample data.
4. Reject or not reject  $H_0$ .

1. State  $H_0$  and  $H_A$
2. Specify the significance level  $\alpha$
3. Calculate the p-value for the sample data
4. Reject or not reject  $H_0$

#### Transcript

In this lesson, we will start by focusing on step 3 and learn how to calculate what is known as the p-value for the sample data that we have collected.

## p-value - Slide 41

### p-VALUE

p-Value: The largest chance of a Type I error if  $H_0$  is rejected based on the observed test statistic.

**Type I error:** Rejecting  $H_0$  when it should not be rejected.



p-Value: The largest chance of a Type I error if  $H_0$  is rejected based on the observed test statistic

**Type I error:** Rejecting  $H_0$  when it should not be rejected.

### Transcript

P-value is the largest chance of having Type I error occur, that is null hypothesis is rejected based on observed test of statistics. Remember, Type I error is the probability of rejecting a null hypothesis when it should not have been rejected because in reality it's actually true. We know this could happen, because there is a chance that we have a sample that is not really a good representative sample which leads us to make a wrong conclusion. Let me explain this concept to an example. Let's go back to this example we did in an earlier lesson and expand on it. This is where we want to ensure that the bag of chocolates is getting filled correctly to 300 grams. If too much, then we will be losing money. Since we are charging the customer as if the bag only has 300 grams of chocolate. Too little, and our company can be accused of defrauding the customers. So, this is a two-tail test.

## Example 1 - Population Mean - Slide 42

**EXAMPLE 1 – POPULATION MEAN**

A chocolate manufacturing plant is producing bags of chocolate that are labeled at 300 grams. Quality control takes samples of 50 bags to check for accuracy. Analysis is done at 5% level of significance.

$H_0: \mu = 300$   
 $H_a: \mu \neq 300$

$\bar{x} = 294.6$   
 $s = 11$   
 $n = 50$



A chocolate manufacturing plant is producing bags of chocolate that are labeled at 300 grams. Quality control takes samples of 50 bags to check for accuracy. Analysis is done at 5% level of significance.

$$H_0 : \mu = 300$$

$$H_A : \mu \neq 300$$

$$\bar{x} = 294.6$$

$$s = 11$$

$$n = 50$$

### Transcript

To do this testing, we are taking samples of 50 bags and based on what we get, we will decide if the production line is working properly or if it needs adjustments.

## Steps in Testing the Mean - Slide 43

### STEPS IN TESTING THE MEAN

1. State  $H_0$  and  $H_a$ .  
 $H_0: \mu = 300$   
 $H_a: \mu \neq 300$
2. Specify the significance level  $\alpha$ .  
 $\alpha = 0.05$
3. Calculate the p-value for the sample data.  
 $\bar{x} = 294.6$   
 $s = 11$
4. Reject or not reject  $H_0$ .

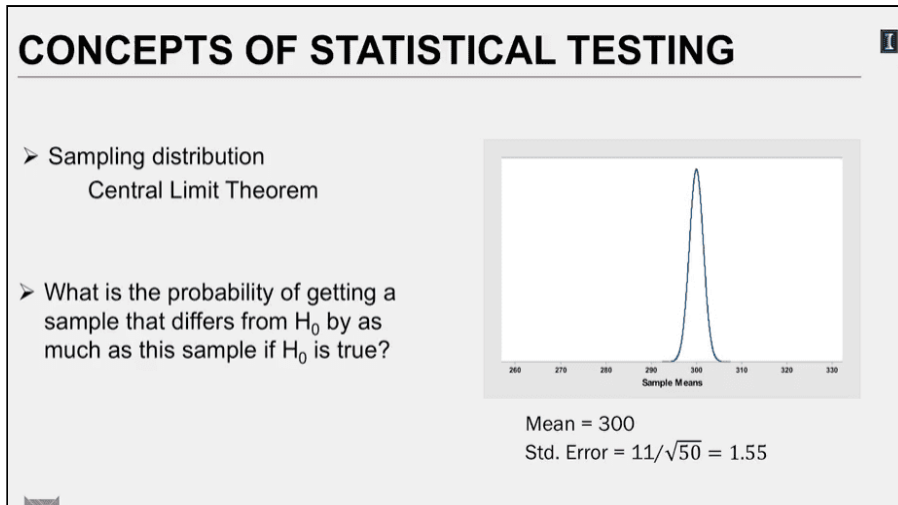


1. State  $H_0$  and  $H_A$ :  $H_0: \mu = 300$  and  $H_A: \mu \neq 300$
2. Specify the significance level  $\alpha$ :  $\alpha = 0.05$
3. Calculate the p-value for the sample data:  $\bar{x} = 294.6$  and  $s = 11$
4. Reject or not reject  $H_0$

### Transcript

The latest sample has given us the following results. Sample mean is about 294.6 and standard deviation of the sample is about 11. We will follow the steps for performing hypothesis testing of stating the hypothesis, specify the significance level and move to calculating the p-value which will be based on sample mean and standard deviation. But before we can actually calculate the p-value, I would like to review the principles behind what is being done.

## Concepts of Statistical Testing (1 of 3) - Slide 44



Sampling distribution - Central Limit Theorem

What is the probability of getting a sample that differs from  $H_0$  by as much as this sample if  $H_0$  is true?

This slide contains a line graph with a very narrow normal curve with tall peak around 300 on the x-axis.

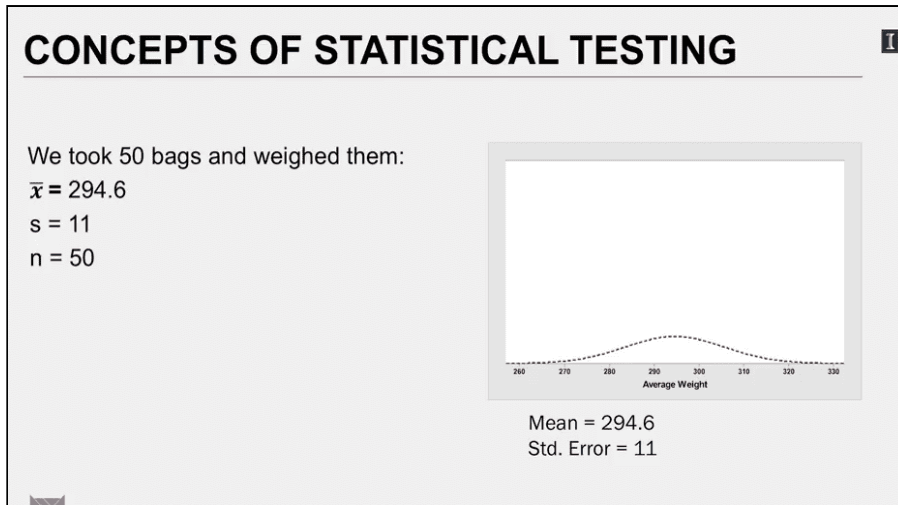
Mean = 300

$$\text{Std. Error} = \frac{11}{\sqrt{50}} = 1.55$$

### Transcript

Statistical tasks rely on sampling distribution of the statistics that estimates the parameters specified in the null and alternate hypotheses. This is based on what we learned in the first course. Central Limit Theorem, which showed that if we repeatedly take samples from a population, in this case, the population of bags of chocolates we are producing, then the means of these samples will cluster around the center and the variability is determined by the standard error. This is a distribution of the sample means. If we repeatedly took samples from our production line and plotted the sample means. This curve peaks at 300 and has a standard error of 1.55, so, now we have taken one sample. What we are contemplating is to know what is the chance of getting a sample that differs from our null hypothesis by as much as this one sample that we have if the null is actually true.

## Concepts of Statistical Testing (2 of 3) - Slide 45



We took 50 bags and weighed them:

$$\bar{x} = 294.6$$

$$s = 11$$

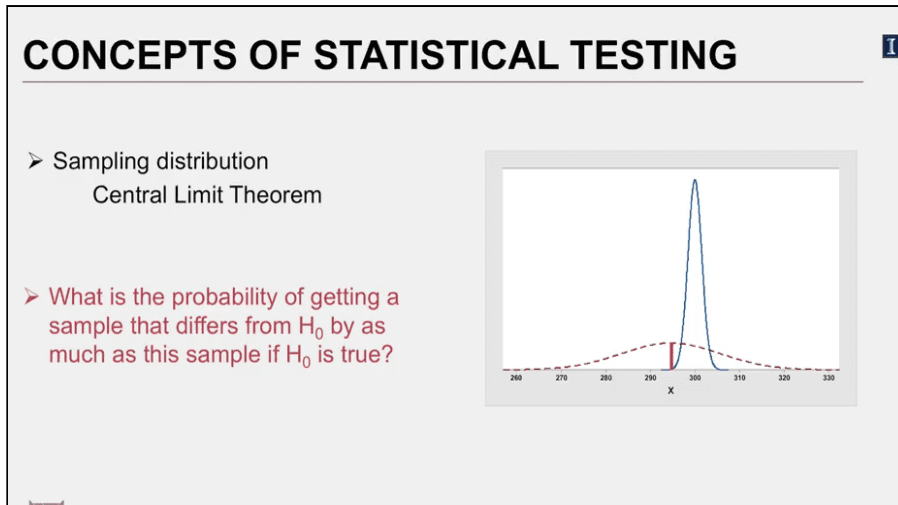
$$n = 50$$

This slide contains a line graph with a widely spread normal curve with a short peak between 290 and 300 on the x-axis. Mean equals 294.6 and Std. Error equals 11.

### Transcript

Let's first look at our sample and the distribution of weight for the 50 bags that we weighed. Our sample of 50 bags gave a mean of 294.8 and a standard deviation of 11. This is how the weight of these 50 bags were distributed. The distribution of this sample shows a lot more variability. Looking at the two distributions at the same, you can see that our samples mean falls way out to left of the hypothesized mean of 300.

## Concepts of Statistical Testing (3 of 3) - Slide 46



Sampling distribution - Central Limit Theorem

**What is the probability of getting a sample that differs from  $H_0$  by as much as this sample if  $H_0$  is true?**

This slide contains an image of the line graph presented on [Slide 44 - Concepts of Statistical Testing \(1 of 3\)](#) superimposed over the line graph presented on [Slide 45 - Concepts of Statistical Testing \(2 of 3\)](#)

### Transcript

The answer to the question of what is the probability of getting a sample that differs from the hypothesized mean by as much as this one sample. If the null hypothesis is indeed true, this answer is what we refer to as the p-value. So, p-value is probability of finding a sample like the one we have.

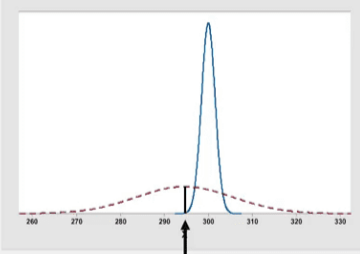


## Calculate p-value (1 of 2) - Slide 47

### CALCULATE p-VALUE

$$\text{Test Statistics : } t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$
$$t = \frac{294.6 - 300}{11 / \sqrt{50}} = -3.47$$

| D                                  | E | F |
|------------------------------------|---|---|
| =T.DIST(                           |   |   |
| T.DIST(x, deg_freedom, cumulative) |   |   |



Sample mean is -3.47 standard error away from hypothesized mean of 300

Test statistics:  $t = \frac{\bar{x} - \mu_0}{s \div \sqrt{n}}$

$$t = \frac{294.6 - 300}{11 \div \sqrt{50}} = -3.47$$

Sample mean is -3.47 standard error away from hypothesized mean of 300.

=T.DIST(x, deg\_freedom, cumulative).

### Transcript

Okay, so let's now see how to find the p-value of our sample. And to do this, first we have to calculate the t-value, known as the test of statistics which tells us how many standard errors we are away from the mean. When we are testing the mean, we will use the t-distribution, the other bell-shaped curve with similar properties as the normal distribution except it's spread is partly controlled by its degrees of freedom which is sample size minus one. So, in this example, we put 294.6 for the sample mean and subtracted from the hypothesized mean of 300. Then the difference is divided by the standard error which is a sample standard deviation of 11 divided by the square root of the sample size of 50. The test of statistics or the t-value for our sample, is negative 3.47. So, the sample we have falls 3.47 standard deviations away from the center of the distribution. Do you remember what we call observations which are more than 3 standard deviations away from the mean?

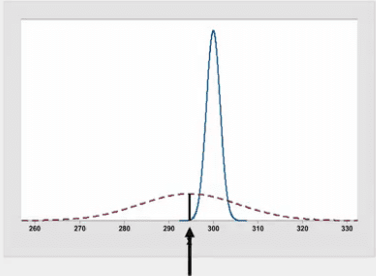
## Calculate p-value (2 of 2) - Slide 48

### CALCULATE p-VALUE

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$
$$t = \frac{294.6 - 300}{11 / \sqrt{50}} = -3.47$$

=T.DIST(-3.47,49,1)

|             | E       | F |
|-------------|---------|---|
| p - value = | 0.00055 |   |



Sample mean is -3.47 standard error away from hypothesized mean of 300

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

$$t = \frac{294.6 - 300}{11 / \sqrt{50}} = -3.47$$

This slide contains the same double line graph presented on [Slide 46 - Concepts of Statistical Testing \(3 of 3\)](#)

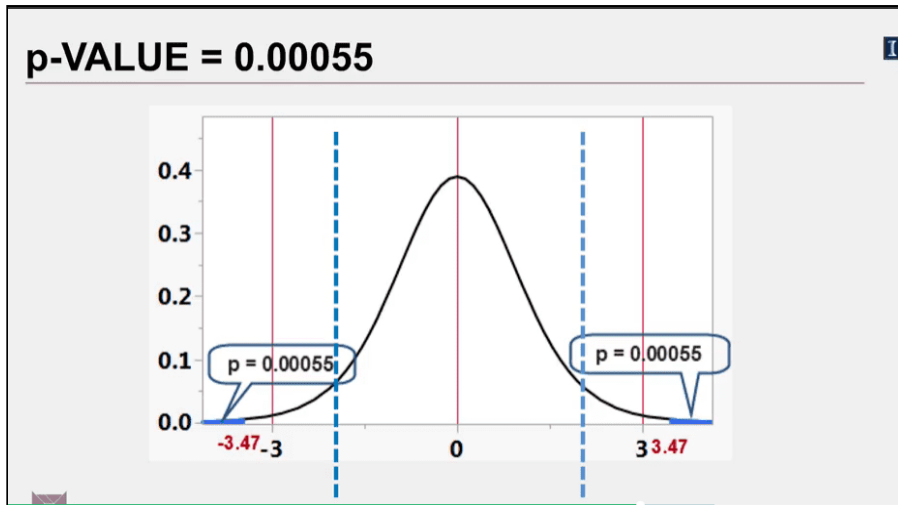
Sample mean is -3.47 standard error away from hypothesized mean of 300

p-value = 0.00055

### Transcript

Outliers. These have very low probability of occurring. So, right away you know that you have a sample that is way in the left tail. The actual probability of finding a sample, this is this far to the left is found by using the Excel function shown here. This function will return the probability to the left of a given test or t-value. It has three arguments. The first one is the t-value you have calculated. Second argument is the degrees of freedom which is sample size minus 1 and the last argument, 1 means that we want cumulative probability. You will also use 1 for the last argument in this class. Of course, there are Excel illustration videos which I take you through every step, so, to be sure to watch those. Now, back to here, placing the appropriate values for these arguments returns the p-value of 0.00055. This is the probability of drawing a sample like ours which is pretty low, 5 out of 100,000. Let me explain more what this means in pictures.

## p-value = 0.00055 (1 of 3) - Slide 49



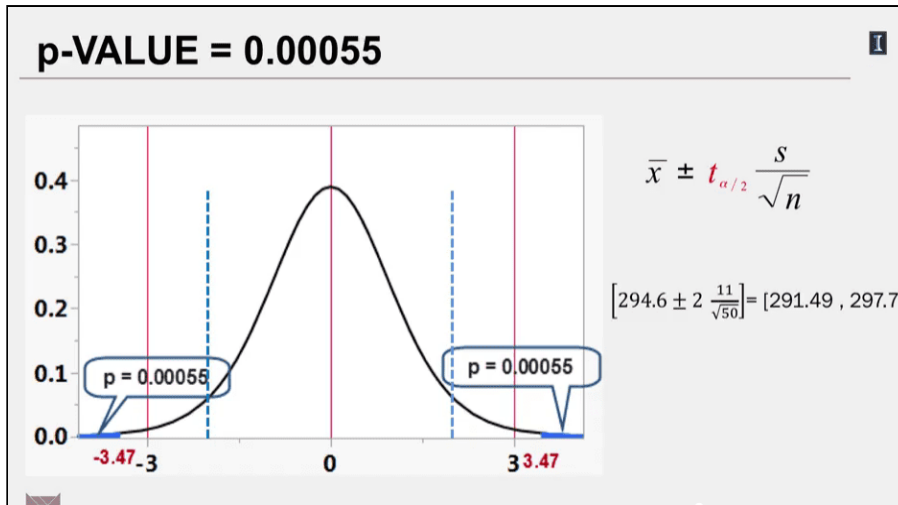
This slide contains a line graph with a normal curve with peak at 0 on the x-axis and between 0.3 and 0.4 on the y-axis.

$p = 0.00055$  is labeled on both tails of the graph near -3.47 and 3.47

### Transcript

Clearly our sample falls way out in the tail on the left side and the probability of finding a sample like this or even more extreme is extremely small, again, 55 out of 100,000. For us, our alpha sets the limit up to which it will accept as reasonable differences we will notice from sample to sample. We used  $\alpha 0.05$  and that is roughly about two standard errors. So, now that we have the p-value, we are ready to move on to making conclusions from our sample results. So, the significance level, sets the level of deviation from mean we accept as natural variation and acceptable.

## p-value = 0.00055 (2 of 3) - Slide 50



This slide contains the same line graph presented on [Slide 49 - p-value = 0.00055 \(1 of 3\)](#).

$$\bar{x} \pm t_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$$

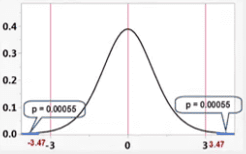
$$\left[ 294.6 \pm 2 \frac{11}{\sqrt{50}} \right] = [291.49, 297.7]$$

### Transcript

Anything outside of those bounds, means that we believe that it's too much difference between what we see from our sample information and what we thought we should see. We could use this to even find the confidence interval for the population mean. An  $\alpha$  of 5% confidence level is 95% and that is roughly two standard errors. We can use this to get a quick estimate for the confidence interval which would be between 294, 0.49 and 297.7. So, the bags mean weight appears to be below 300 grams. Given what we see, of course, there are two possibilities. One, known hypothesis is true. We are very unlucky to find this sample or two, sample data provides evidence against the known hypothesis and it should be rejected. The most plausible explanation is a second conclusion.

## p-value = 0.00055 (3 of 3) - Slide 51

**p-VALUE = 0.00055**



1.  $H_0$  is true; we are very unlucky to find this sample.
2. Sample data provides evidence against  $H_0$  and it should be rejected.



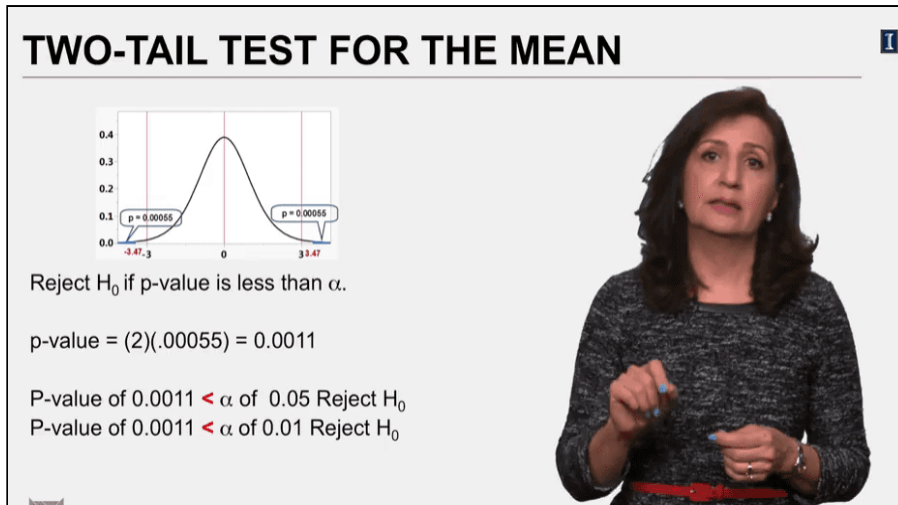
This slide contains the same line graph presented on [Slide 49 - p-value = 0.00055 \(1 of 3\)](#)

1.  $H_0$  is true; we are very unlucky to find this sample
2. Sample data provides evidence against  $H_0$  and it should be rejected

### Transcript

However, if you feel uncomfortable with this conclusion, then you can take another sample to see what happens. This is exactly what happened in the Volkswagen case. When the researcher at first found sample information that was way out, first they decided that it must be their bad luck, so they repeated their experiment. And time and time again, they would get results that would point them to rejected Volkswagen's claims. So, ultimately, they decided to report their results and as you know, it was Volkswagen who, that had lied. So, the correct conclusion was indeed to reject the known hypothesis. For a two-tail test, you can reject a known hypothesis if sample mean is too far to the right or to the left. In our example, we would reject a known hypothesis if the mean weight was too low or too high. Because in both cases, the production process would require adjustments. The rule is to reject the known hypothesis if p-value is less than  $\alpha$ .

## Two-tail Test for the Mean - Slide 52



This slide contains the same line graph presented on [Slide 49 - p-value = 0.00055 \(1 of 3\)](#).

- Reject  $H_0$  if p-value is less than  $\alpha$
- P-value =  $(2) \times (0.00055) = 0.0011$
- P-value of 0.0011  $<$   $\alpha$  of 0.05 Reject  $H_0$
- P-value of 0.0011  $<$   $\alpha$  of 0.01 Reject  $H_0$

### Transcript

In the two-tail test, both the  $\alpha$  and p-values split between the two-tails. So, p-value we need to compare is 2 times 0.00055 or 0.0011 which is less than 0.05. And thus, reject the null hypothesis. You would actually reject even at 0.01 level of significance. This means that the line manager must stop the production and make adjustments. Otherwise, on average, they will continue producing bags that will weigh less than 300 grams.

## Lesson 1-3.2 Two-Tail Test for Mean in Excel

[Media Player for Video](#) 

## Example 1 - Population Mean - Slide 53

**EXAMPLE 1 – POPULATION MEAN**

A chocolate manufacturing plant is producing bags of chocolate that are labeled at 300 grams. Quality control takes samples of 50 bags to check for accuracy. Analysis is done at 5% level of significance.

$H_0: \mu = 300$   
 $H_a: \mu \neq 300$

$\bar{x} = 294.6$   
 $s = 11$   
 $n = 50$

A chocolate manufacturing plant is producing bags of chocolate that are labeled at 300 grams. Quality control takes samples of 50 bags to check for accuracy. Analysis is done at 5% level of significance.

$$H_0: \mu = 300$$

$$H_A: \mu \neq 300$$

$$\bar{x} = 294.6$$

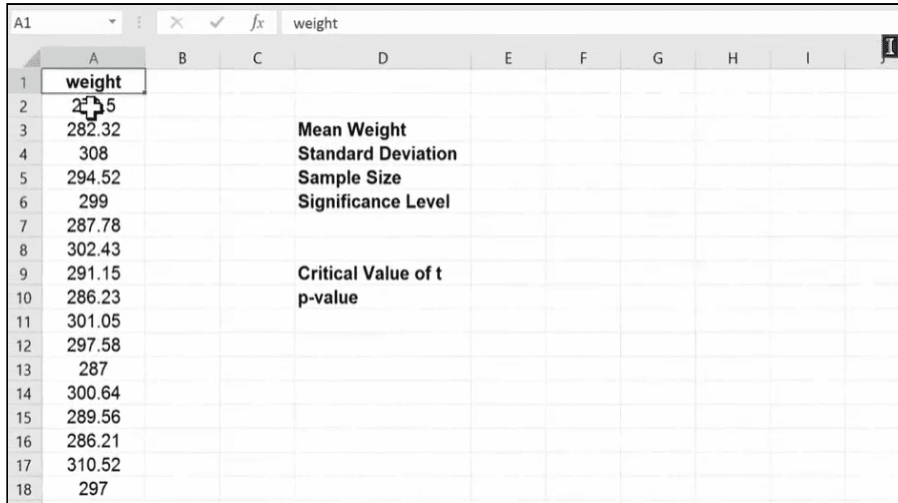
$$s = 11$$

$$n = 50$$

### Transcript

So, now let's go back to the example that we have in lesson 3 of the module, looking at the chocolate manufacturing that wants to make sure that the chocolate bags are being filled to the right amount. So, this is an example that we wanted the chocolate bags to weigh 300 grams and we are taking 50 bags to check whether or not the production line is doing it correctly or not and we are testing at a 5% level of significance. In this case, we want the average to be 300 and no difference. So, basically, we don't want the bags to be overfilled or under filled and I showed how that data gave me the sample mean as well as the standard deviation.

## Excel Worksheet (1 of 11) - Slide 54



|    | A      | B | C | D                   | E | F | G | H | I |
|----|--------|---|---|---------------------|---|---|---|---|---|
| 1  | weight |   |   |                     |   |   |   |   |   |
| 2  | 282.32 |   |   |                     |   |   |   |   |   |
| 3  | 282.32 |   |   | Mean Weight         |   |   |   |   |   |
| 4  | 308    |   |   | Standard Deviation  |   |   |   |   |   |
| 5  | 294.52 |   |   | Sample Size         |   |   |   |   |   |
| 6  | 299    |   |   | Significance Level  |   |   |   |   |   |
| 7  | 287.78 |   |   |                     |   |   |   |   |   |
| 8  | 302.43 |   |   |                     |   |   |   |   |   |
| 9  | 291.15 |   |   | Critical Value of t |   |   |   |   |   |
| 10 | 286.23 |   |   | p-value             |   |   |   |   |   |
| 11 | 301.05 |   |   |                     |   |   |   |   |   |
| 12 | 297.58 |   |   |                     |   |   |   |   |   |
| 13 | 287    |   |   |                     |   |   |   |   |   |
| 14 | 300.64 |   |   |                     |   |   |   |   |   |
| 15 | 289.56 |   |   |                     |   |   |   |   |   |
| 16 | 286.21 |   |   |                     |   |   |   |   |   |
| 17 | 310.52 |   |   |                     |   |   |   |   |   |
| 18 | 297    |   |   |                     |   |   |   |   |   |

This slide contains a screen shot of an Excel worksheet with Column A weight populated with numbers to row 50.

Column D contains the following labels: Mean Weight, Standard Deviation, Sample Size, Significance Level, Critical Value of t, and p-value.

[Download the Chocolate Weights excel file \(Refer to Data - First worksheet\)](#) 

### Transcript

What I'm going to show you here is the sample that we have and how these numbers were derived. So, I'm going to click on the tab that says data. So, here's my weights for every bag that has been measured. So, this is the first bag and we will have 50 of these in this sample. So, I'm going to calculate the mean weight.



## Excel Worksheet (2 of 11) - Slide 55

|    |        |   |   |                     |                                  |   |   |
|----|--------|---|---|---------------------|----------------------------------|---|---|
| A2 |        |   |   |                     |                                  |   |   |
|    | A      | B | C | D                   | E                                | F | G |
| 1  | weight |   |   |                     |                                  |   |   |
| 2  | 278.5  |   |   |                     |                                  |   |   |
| 3  | 282.32 |   |   | Mean Weight         | =AVERAGE(A2                      |   |   |
| 4  | 308    |   |   | Standard Deviation  | AVERAGE(number1, [number2], ...) |   |   |
| 5  | 294.52 |   |   | Sample Size         |                                  |   |   |
| 6  | 299    |   |   | Significance Level  |                                  |   |   |
| 7  | 287.78 |   |   |                     |                                  |   |   |
| 8  | 302.43 |   |   |                     |                                  |   |   |
| 9  | 291.15 |   |   | Critical Value of t |                                  |   |   |
| 10 | 286.23 |   |   | p-value             |                                  |   |   |
| 11 | 301.05 |   |   |                     |                                  |   |   |
| 12 | 297.58 |   |   |                     |                                  |   |   |
| 13 | 287    |   |   |                     |                                  |   |   |

This slide contains the same Excel worksheet presented on [Slide 54 - Excel Worksheet \(1 of 11\)](#). In Column E, next to the label Mean Weight, the formula =AVERAGE is being used.

### Transcript

So, the mean weight is simply the average. Click on the first value, control, shift down, close the parenthesis, return and scroll back up and I will see the value of 294.609.

## Excel Worksheet (3 of 11) - Slide 56

|    |        |   |   |                     |                                  |   |   |
|----|--------|---|---|---------------------|----------------------------------|---|---|
| A2 |        |   |   |                     |                                  |   |   |
|    | A      | B | C | D                   | E                                | F | G |
| 1  | weight |   |   |                     |                                  |   |   |
| 2  | 278.5  |   |   |                     |                                  |   |   |
| 3  | 282.32 |   |   | Mean Weight         | 294.609                          |   |   |
| 4  | 308    |   |   | Standard Deviation  | =STDEV.S(A2                      |   |   |
| 5  | 294.52 |   |   | Sample Size         | STDEV.S(number1, [number2], ...) |   |   |
| 6  | 299    |   |   | Significance Level  |                                  |   |   |
| 7  | 287.78 |   |   |                     |                                  |   |   |
| 8  | 302.43 |   |   |                     |                                  |   |   |
| 9  | 291.15 |   |   | Critical Value of t |                                  |   |   |
| 10 | 286.23 |   |   | p-value             |                                  |   |   |
| 11 | 301.05 |   |   |                     |                                  |   |   |
| 12 | 297.58 |   |   |                     |                                  |   |   |
| 13 | 287    |   |   |                     |                                  |   |   |

This slide contains the same Excel worksheet presented on [Slide 54 - Excel Worksheet \(1 of 11\)](#). In Column E, next to the label Standard Deviation, the formula =STDEV.S is being used.

## Transcript

Standard deviation is just standard deviation, which is stdev.s for sample. Pick the first value, I have to scroll up and make sure that I am taking all of the values.

## Excel Worksheet (4 of 11) - Slide 57

|    |        |   |   |                     |          |   |   |
|----|--------|---|---|---------------------|----------|---|---|
| E4 |        |   |   |                     |          |   |   |
|    |        |   |   |                     |          |   |   |
|    | A      | B | C | D                   | E        | F | G |
| 1  | weight |   |   |                     |          |   |   |
| 2  | 278.5  |   |   |                     |          |   |   |
| 3  | 282.32 |   |   | Mean Weight         | 294.609  |   |   |
| 4  | 308    |   |   | Standard Deviation  | 10.98013 |   |   |
| 5  | 294.52 |   |   | Sample Size         |          |   |   |
| 6  | 299    |   |   | Significance Level  |          |   |   |
| 7  | 287.78 |   |   |                     |          |   |   |
| 8  | 302.43 |   |   |                     |          |   |   |
| 9  | 291.15 |   |   | Critical Value of t |          |   |   |
| 10 | 286.23 |   |   | p-value             |          |   |   |
| 11 | 301.05 |   |   |                     |          |   |   |
| 12 | 297.58 |   |   |                     |          |   |   |
| 13 | 287    |   |   |                     |          |   |   |

This slide contains the same Excel worksheet presented on [Slide 54 - Excel Worksheet \(1 of 11\)](#). Mean Weight now reads 294.609 and Standard Deviation reads 10.98013.

## Transcript

So, from here to down to 50 and then return and that is 10.98. So, let me just slip back to the problem and you'll see these values.

## Excel Worksheet (5 of 11) - Slide 58

A chocolate manufacturing plant is producing bags of chocolate that are labeled at 300 grams. Quality control takes samples of 50 bags to check for accuracy. Analysis is done at 5% level of significance.

$$H_0: \mu = 300$$

$$H_a: \mu \neq 300$$

$$\bar{x} = 294.6$$

$$s = 11$$

$$n = 50$$

A chocolate manufacturing plant is producing bags of chocolate that are labeled at 300 grams. Quality control takes samples of 50 bags to check for accuracy. Analysis is done at 5% level of significance.

$$H_0: \mu = 300$$

$$H_A: \mu \neq 300$$

$$\bar{x} = 294.6$$

$$s = 11$$

$$n = 50$$

### Transcript

In the problem, I have rounded the standard deviation from 10.98 to 11 and the mean was 294.6 and this is what we have got from our analysis here. I can always take the sample size by saying what is the count and again, this is useful if you don't know already what the sample size is.

## Excel Worksheet (6 of 11) - Slide 59

|        |                     |                              |  |
|--------|---------------------|------------------------------|--|
| weight |                     |                              |  |
| 278.5  |                     |                              |  |
| 282.32 | Mean Weight         | 294.609                      |  |
| 308    | Standard Deviation  | 10.98013                     |  |
| 294.52 | Sample Size         | =count(A2                    |  |
| 299    | Significance Level  | COUNT(value1, [value2], ...) |  |
| 287.78 |                     |                              |  |
| 302.43 |                     |                              |  |
| 291.15 | Critical Value of t |                              |  |
| 286.23 | p-value             |                              |  |
| 301.05 |                     |                              |  |
| 297.58 |                     |                              |  |
| 287    |                     |                              |  |
| 300.64 |                     |                              |  |
| 289.56 |                     |                              |  |
| 286.84 |                     |                              |  |

This slide contains the same Excel worksheet presented on [Slide 54 - Excel Worksheet \(1 of 11\)](#). In Column E, next to the label Sample Size, the formula =count is being used, the result is 50. Significance Level is 0.05.

### Transcript

No instruction provided on this slide

## Excel Worksheet (7 of 11) - Slide 60

|                     |                          |  |
|---------------------|--------------------------|--|
| Mean Weight         | 294.609                  |  |
| Standard Deviation  | 10.98013                 |  |
| Sample Size         | 50                       |  |
| Significance Level  | 0.05                     |  |
| Critical Value of t | = (E3-300)/(E4/sqrt(E5)) |  |
| p-value             |                          |  |

Handwritten notes and formula:

$$t = \frac{\bar{X} - \mu}{s / \sqrt{n}}$$

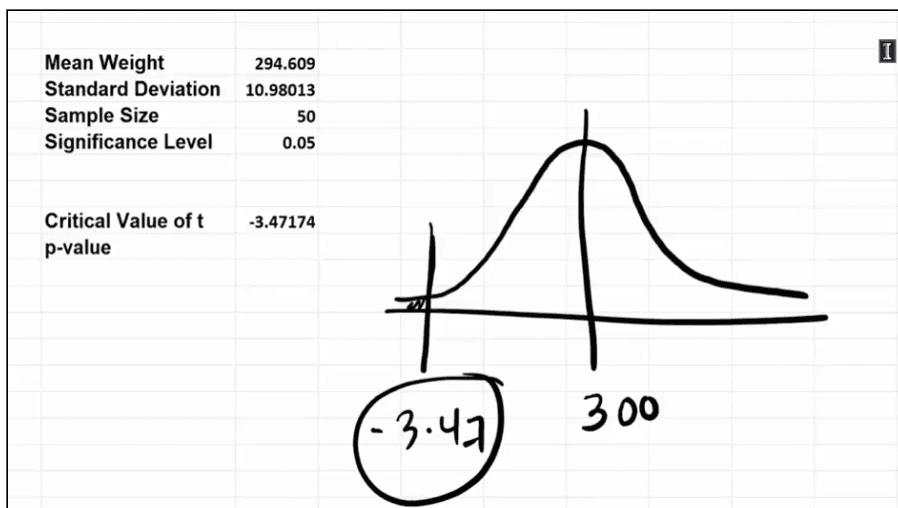
Annotations:  $\bar{X} = 294.6$ ,  $\mu = 300$ ,  $s = 11$ ,  $n = 50$

Critical value of t.  $t = \frac{\bar{x} - \mu}{s \div \sqrt{n}} = \frac{(294.6 - 300)}{(11 \div \sqrt{50})}$

## Transcript

So, it's a good thing to do and counts will come back and say that you have 50 values. Level of significance is what we said we want to have which was 5%. So, I'm going to enter .05 here. So, now I'm going to calculate the critical value of t. Critical value of t had the formula that was  $t$  is equal to  $\bar{x}$  minus  $\mu$  divided by the standard error which would have been my standard deviation divided by the square root of  $n$  and  $\bar{x}$  here is 294.6, which is the value I get there.  $\mu$  is my hypothesized mean of 300. S, I just calculated to be 11 and  $n$  is 50. So, this is what I'm going to put in the equation right here. So, I simply am going to now say this is equal to, and I'm going to repeat what we have here,  $\bar{x}$  which is 294.6 and it says right here minus  $\mu$ , and that was my hypothesized mean. I didn't put it anywhere here so I'm just going to enter it, divided by, and the denominator is the standard deviation divided by the square root of  $n$  and that would be 50 right here. And this is the equation I'm using and the values for those notations are found in the addresses I've given here.

## Excel Worksheet (8 of 11) - Slide 61



This slide contains the same Excel worksheet presented on [Slide 54 - Excel Worksheet \(1 of 11\)](#). Critical Value of t now reads -3.47174.

To the right of the slide is a line graph with peak at 300, left tail marked negative 3.47.

## Transcript

So, press enter and it turns out to be -3.47. So, what does that mean? Well, let me show you graphically what that means. That means if I draw my normal distribution right there, the sample that I have found, if this is 300 and this is the hypothesized mean, the sample I have found has given me a mean that falls -3.47 to the left of this. So, what I want to know is that what is the probability of finding a sample like we did, like here, that is this far away from our hypothesized mean. And for that, I'm just going to use the t.dist and I'm going to put this value in there. We are this many away.

## Excel Worksheet (9 of 11) - Slide 62

|                     |                   |
|---------------------|-------------------|
| Mean Weight         | 294.609           |
| Standard Deviation  | 10.98013          |
| Sample Size         | 50                |
| Significance Level  | 0.05              |
| Critical Value of t | -3.47174          |
| p-value             | =t.dist(E9,E5-1,1 |

T.DIST(x, deg\_freedom, cumulative)  
 (...) TRUE - cumulative distribution  
 (...) FALSE - probability density function

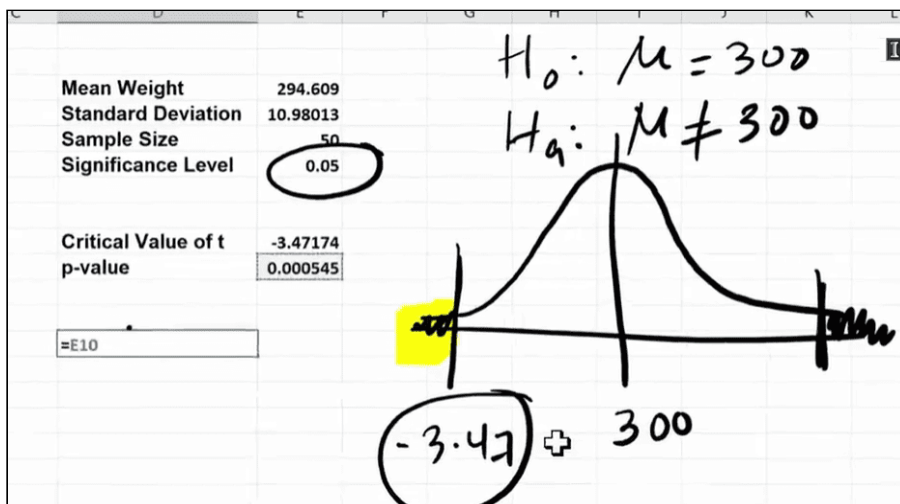
This slide contains the same Excel worksheet presented on [Slide 54 - Excel Worksheet \(1 of 11\)](#). In Column E, next to the label p-value, the formula =t.dist is being used.

p-value: equals T.DIST(t-value, deg\_freedom, cumulative). equals T.DIST(-3.47, 49, 1) or 0.000545.

### Transcript

Degrees of freedom is always your sample size minus 1 so I can either right here write 49, or I can just click on this and say minus 1 and the last argument is always 1. So, I'm going to just say 1 and it turns out to be .0005.

## Excel Worksheet (10 of 11) - Slide 63



This slide contains the same Excel worksheet and line graph presented on [Slide 61 - Excel Worksheet \(8 of 11\)](#).

$$H_0: \mu = 300$$

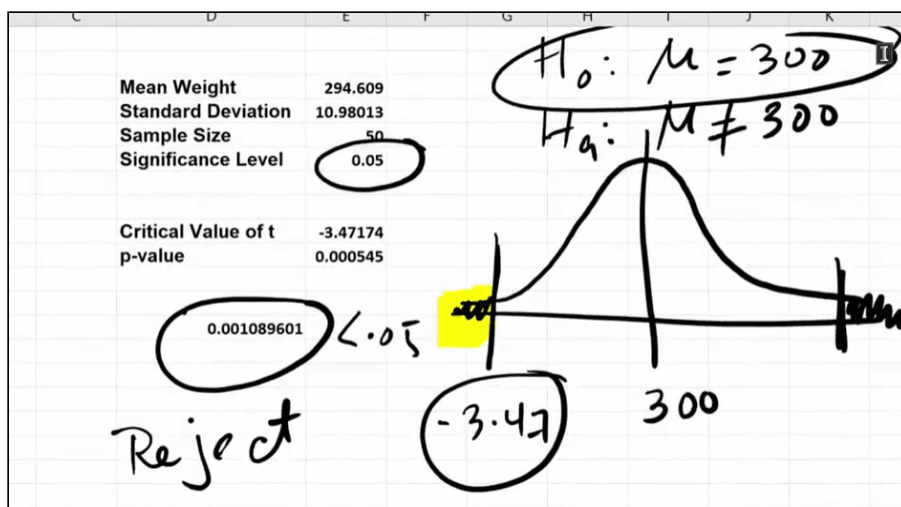
$$H_A: \mu \neq 300$$

Significance Level = 0.05 is circled.

## Transcript

What this is telling me is that probability is .005, but we we're doing a two-tail test. Why? Because I had null hypothesis was that  $\mu$  was equal to 300 and alternate hypothesis is that  $\mu$  is not equal to 300. So, we only found this, but there is a likelihood that we would have made a mistake on the other side also.

## Excel Worksheet (11 of 11) - Slide 64



This slide contains the same Excel worksheet and line graph presented on [Slide 61 - Excel Worksheet \(8 of 11\)](#).

Two times p-value is 0.001089601 and therefore less than 0.05.

[Download the Chocolate Weights excel file \(Refer to 2 Tail Test - Second worksheet\)](#)

## Transcript

So, this .005 has to be doubled before I can compare it to this .05. So, doubling this will return .001 and since this one is less than .05, we will reject the null hypothesis which means the machine is not working correctly and we are not at 300.

## Lesson 1-4 One-Tail Test for Mean

### Lesson 1-4.1 One-Tail Test for Mean

[Media Player for Video](#) 

#### One-tail Mean Test - Slide 65

### ONE-TAIL MEAN TEST

The average debt for the class of 2015 graduates is \$35,000. University of Good Deal believes their students have a lower debt amount. They survey 150 of their graduating seniors to find the average debt and find that, on average, their students graduate with \$33,800 in debt and a standard deviation of \$10,000.

At 5% level of significance, is there enough evidence to support University of Good Deal's claim?



The average debt for the class of 2015 graduates is \$35,000. University of Good Deal believes their students have a lower debt amount. They survey 150 of their graduating seniors to find the average debt and find that, on average, their students graduate with \$33,800 in debt and a standard deviation of \$10,000.

At 5% level of significance, is there enough evidence to support University of Good Deal's claim?

#### Transcript

Now, let's start on how to do the hypothesis test when we do one-tail test. Going back to the student debt, and the Good Deal universities example, it is believed that the average debt for the class of 2015 graduates is \$35,000. University of Good Deal believes that their students have a lower debt amount. They survey 150 of their graduates – seniors, and find the average debt to be 33,800 with a standard deviation of 10,000. At 5% level of significance, is there enough evidence to support Good Deal University's claim?



## Steps in Testing the Mean - Slide 66

### STEPS IN TESTING THE MEAN



1. State  $H_0$  and  $H_a$ .  
 $H_0: \mu \geq 35,000$   
 $H_a: \mu < 35,000$
2. Specify the significance level  $\alpha$ .  
 $\alpha = 0.05$
3. Calculate the p-value for the sample data.  
 $\bar{x} = 33,800$   
 $s = 10,000$   
 $n = 150$
4. Reject or not reject  $H_0$ .

1. State  $H_0$  and  $H_A$ :  $H_0: \mu \geq 35,000$  and  $H_A: \mu < 35,000$
2. Specify the significance level  $\alpha$ :  $\alpha = 0.05$
3. Calculate the p-value for the sample data:  $\bar{x} = 33,800$ ,  $s = 10,000$ , and  $n = 150$
4. Reject or not reject  $H_0$

### Transcript

We will follow the steps for performing the hypothesis testing of stating the hypothesis. Since Good Deal thinks that their students have lower than national average debt, that is the alternate hypothesis here. They need to reject the current belief that the graduates are no different than any other graduates. Specified at the significance of .05, and now, we will calculate the p-value based on sample information. P-value is the probability of finding a sample like we've found. To answer this question, we have to calculate its t-value. So, in this example, we put 33,800, the sample mean, and subtract it from the hypothesized mean of 35,000, and then, the difference is divided by the sample's standard deviation of 10,000, divided by the square root of the sample size of 150. The t-value for this example is negative 1.47, which means the sample we have falls 1.47 standard deviations away from the center of the distribution on the left side.

## Calculate p-value - Slide 67

### CALCULATE p-VALUE

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$
$$t = \frac{33,800 - 35,000}{10,000 / \sqrt{150}} = -1.47$$

=T.DIST(-1.47,149,1)

| D        | E |
|----------|---|
| 0.071835 |   |

p-value = 0.071835 ≈ 0.072



$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

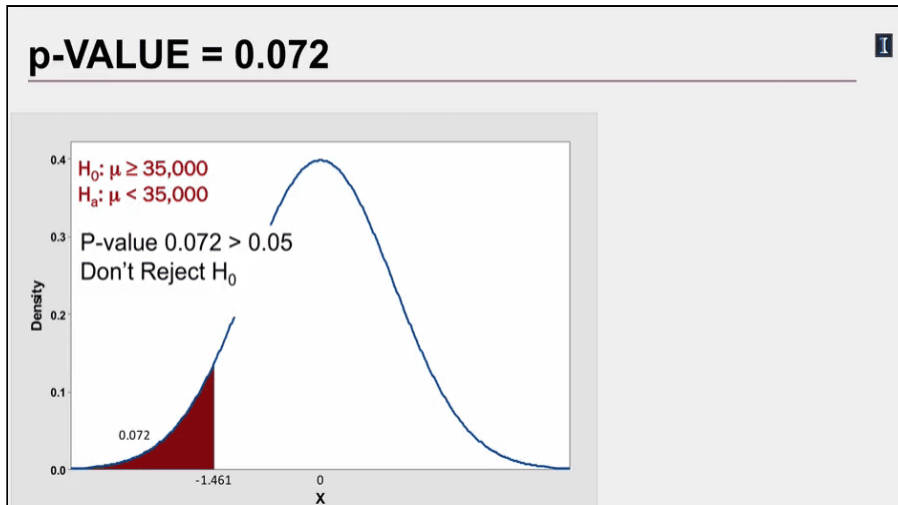
$$t = \frac{33,800 - 35,000}{10,000 / \sqrt{150}} = -1.47$$

$$P\text{-Value} = T.DIST(1.47, 149, 1) \approx 0.072$$

### Transcript

The p-value is found by using the Excel function shown here. This function, again, has three arguments. The first one is the t-value, which you calculated to be negative 1.47. Second argument is the sample size minus 1, which is 149, and the last argument is 1. And we get the p-value of .0718, or roughly about .072. I like pictures, so let me show you what this means in picture.

## p-value = 0.072 - Slide 68



This slide contains a bell curve on graph centered at 0. Shaded area under curve to the left of -1.461. P-value 0.073 greater than 0.05. Don't reject  $H_0$

$$H_0 : \mu \geq 35,000$$

$$H_A : \mu < 35,000$$

### Transcript

Since this is a one-tail test, the entire p-value is on the left side of the mean, but the p-value is not less than .05, which means we don't have a sample that shows enough difference from the national average. So, p-value is more than the  $\alpha$  value. So, we will not reject the null hypothesis. This means Good Deal University has failed to provide enough evidence to show that their graduates are any better off than the rest of the college graduates in the nation.

## Let's Practice - Slide 69

### LET'S PRACTICE

We believe the freshman entering our university have an average ACT score of more than 24.

The university selects 200 students and find the average ACT score to be 24.8 and a standard deviation of 6.1.

At 5% level of significance is there enough evidence to support this claim?



We believe the freshman entering our university have an average ACT score of more than 24.

The university selects 200 students and find the average ACT score to be 24.8 and a standard deviation of 6.1. At 5% level of significance is there enough evidence to support this claim?

### Transcript

This was the other example we looked at in lesson one. Now, let's try to solve it together. State the hypothesis and the level of significance.

## Let's Practice - Solved (1 of 3) - Slide 70

### LET'S PRACTICE – SOLVED

1. State  $H_0$  and  $H_a$ .  
 $H_0: \mu \leq 24$   
 $H_a: \mu > 24$
2. Specify the significance level  $\alpha$ .  
 $\alpha = 0.05$
3. Calculate the p-value for the sample data.  
 $\bar{x} = 24.8$   
 $s = 6.1$   
 $n = 200$
4. Reject or not reject  $H_0$ .



1. State  $H_0$  and  $H_A$ :  $H_0 : \mu \leq 24$  and  $H_A : \mu > 24$

2. Specify the significance level  $\alpha$ :  $\alpha = 0.05$
3. Calculate the p-value for the sample data:  $\bar{x} = 24.8$ ,  $s = 6.1$ , and  $n = 200$
4. Reject or not reject  $H_0$

## Transcript

What is your hypothesis? It should look like this. Remember, the current belief is that the average ACT is greater than 24, not 24 or more. That means the correct representation is  $\mu$  is greater than 24, and since equals sign needs to be in the null hypothesis, we will put this as the alternate. And the null becomes  $\mu$  is less than or equal to 24. The specified level of significance is .05, and now given the following sample information, let's start solving for the p-value. Given the sample statistics, what is the t-value?

## Let's Practice - Solved (2 of 3) - Slide 71

### LET'S PRACTICE – SOLVED

$\bar{x} = 24.8$

$s = 6.1$

$n = 200$

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

$$t = \frac{24.8 - 24}{6.1 / \sqrt{200}} = 1.85$$



$$\bar{x} = 24.8$$

$$s = 6.1$$

$$n = 200$$

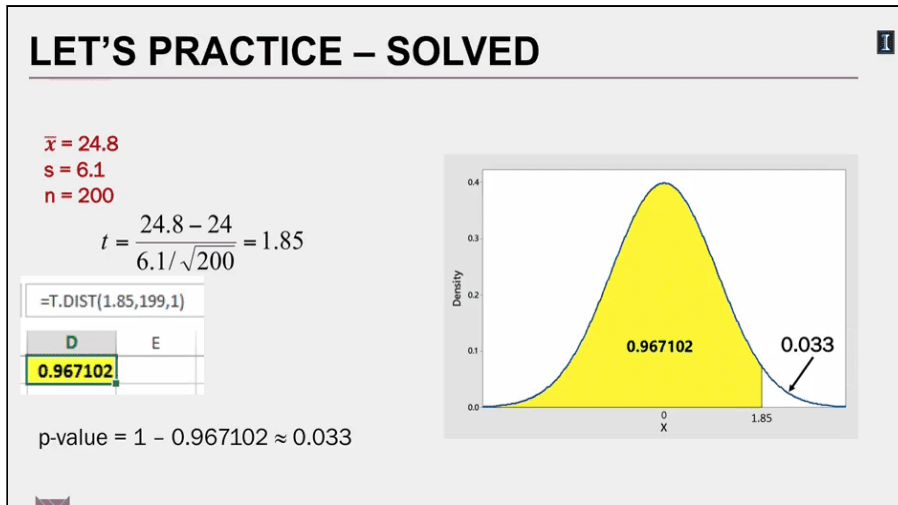
$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

$$t = \frac{24.8 - 24}{6.1 / \sqrt{200}}$$

## Transcript

Putting a value in the equation, we get the t-value of 1.85. So, what is now the p-value?

## Let's Practice - Solved (3 of 3) - Slide 72



$$\bar{x} = 24.8$$

$$s = 6.1$$

$$n = 200$$

$$t = \frac{24.8 - 24}{6.1 / \sqrt{200}} = 1.85$$

This slide contains a bell curve graph with mean of 0 with a shaded area under the curve to the left of 1.85. The tail on the far right under the curve represents the P-value of 0.033.

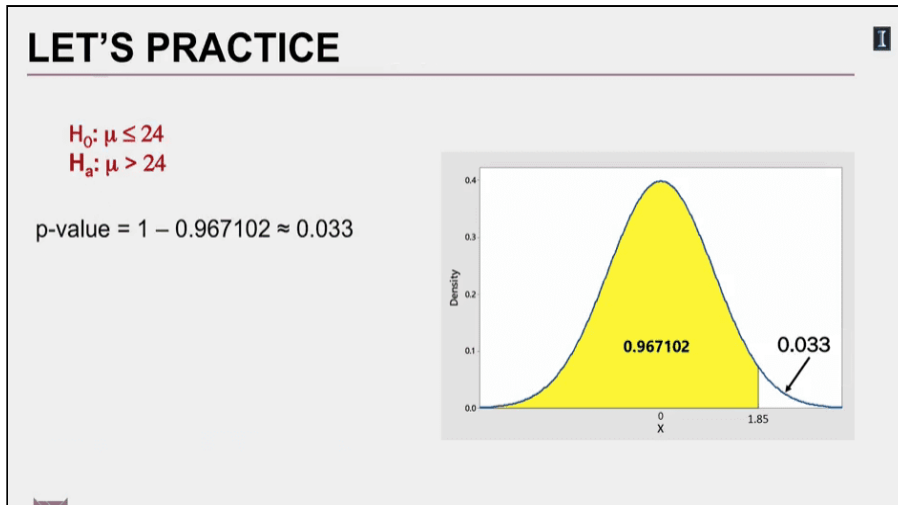
P-value formula in Excel = T.DIST(1.85, 199s 1, 1)

$$\text{p-value} = 1 - 0.967102 \approx 0.033$$

### Transcript

If we use the same function, t.dist, we get the value of .967. That is the area to the left of 1.85.

## Let's Practice (1 of 2) - Slide 73



This slide contains the same bell curve presented on [Slide 72 - Let's Practice - Solved \(3 of 3\)](#).

$$H_0 : \mu \leq 24$$

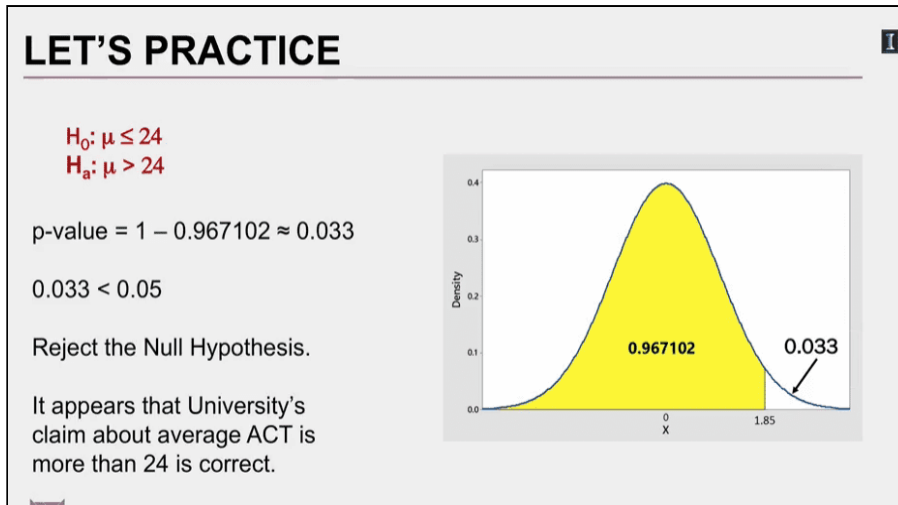
$$H_A : \mu > 24$$

$$\text{p-value} = 1 - 0.967102 \approx 0.033$$

### Transcript

This is the area to the left of the t-value. P-value is the area in the tail, and we have to subtract it from 1, which gives us .033. So, what is your conclusion?

## Let's Practice (2 of 2) - Slide 74



This slide contains the same bell curve presented on [Slide 72 - Let's Practice - Solved \(3 of 3\)](#).

$$H_0 : \mu \leq 24$$

$$H_A : \mu > 24$$

$$\text{p-value} = 1 - 0.967102 \approx 0.033$$

- $0.033 < 0.05$
- Reject the Null Hypothesis.
- It appears that University's claim about average ACT is more than 24 is correct.

### Transcript

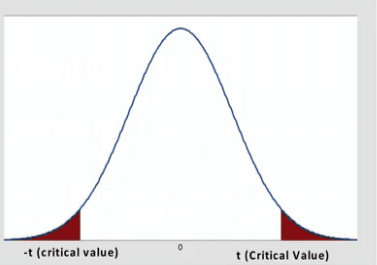
Since p is less than  $\alpha$ , we will reject the null hypothesis in favor of the alternate, which means it seems the university's correct in thinking the average ACT score of the freshmen is higher than 24.



## Possible Types of Hypothesis Testing (1 of 2) - Slide 75

**POSSIBLE TYPES OF HYPOTHESIS TESTING**

**TWO-TAIL**  
 $H_0: \mu = \mu_0$   
 $H_A: \mu \neq \mu_0$       Reject  $H_0$ , if  $(2 \times p) < \alpha$



A woman with dark hair, wearing a dark patterned top, is standing next to the slide content, gesturing with her hands as if explaining the material.

### Two-Tail

This slide contains a bell curve on graph centered at 0. Shaded area under both the left and right tails of the curve.

$$H_0 : \mu = \mu_0$$

$$H_A : \mu \neq \mu_0$$

Reject  $H_0$ , if  $(2 \times p) < \alpha$

### Transcript

When you are testing the mean, then you have three possible sets of hypotheses. If it is a test of equality versus not equal, you have two rejection regions. You will reject if your sample statistic deviates by too much from the mean in either direction. How much is too much of course depends on your level of significance. In case of two-tail, find the p-value, multiply it by 2. If that is less than  $\alpha$ , reject the null hypothesis.

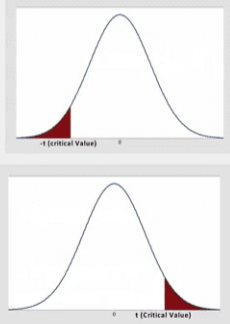
## Possible Types of Hypothesis Testing (2 of 2) - Slide 76

**POSSIBLE TYPES OF HYPOTHESIS TESTING**

**ONE-TAIL**

$H_0: \mu \geq \mu_0$   
 $H_a: \mu < \mu_0$

$H_0: \mu \leq \mu_0$   
 $H_a: \mu > \mu_0$



### One-Tail

This slide contains two bell curves on graph centered at 0. The first bell curve is shaded under the left tail of the curve and the second is shaded under the right tail of the curve.

$$H_0 : \mu \geq \mu_0$$

$$H_A : \mu < \mu_0$$

$$H_0 : \mu \leq \mu_0$$

$$H_A : \mu > \mu_0$$

### Transcript

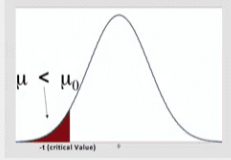
If you are testing greater than or equal versus less than, or less than or equal versus greater than, then you are conducting a one-tail test.

## One-tail Test - Slide 77

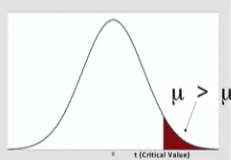
### ONE-TAIL TEST

REJECT  $H_0$ , IF  $p < \alpha$

$H_0: \mu \geq \mu_0$   
 $H_a: \mu < \mu_0$



$H_0: \mu \leq \mu_0$   
 $H_a: \mu > \mu_0$





The slide titled "ONE-TAIL TEST" includes the instruction "REJECT  $H_0$ , IF  $p < \alpha$ ". It displays two normal distribution curves. The top curve represents a left-tail test with  $H_0: \mu \geq \mu_0$  and  $H_a: \mu < \mu_0$ ; the rejection region is the area to the left of the critical value, where  $\mu < \mu_0$ . The bottom curve represents a right-tail test with  $H_0: \mu \leq \mu_0$  and  $H_a: \mu > \mu_0$ ; the rejection region is the area to the right of the critical value, where  $\mu > \mu_0$ . A video of a woman speaking is shown on the right side of the slide.

This slide contains the same bell curves presented on [Slide 76 - Possible Types of Hypothesis Testing \(2 of 2\)](#).

Reject  $H_0$ , if  $p < \alpha$

$$H_0 : \mu \geq \mu_0$$

$$H_A : \mu < \mu_0$$

$$H_0 : \mu \leq \mu_0$$

$$H_A : \mu > \mu_0$$

### Transcript

The rejection region is only on the tail that corresponds to the side of the alternate hypotheses. In case of one-tail test, reject the null hypothesis if  $p$  is less than  $\alpha$ .

## Lesson 1-4.2 Left-Tail Test for Mean in Excel

[Media Player for Video](#) 

## Excel Worksheet (1 of 9) - Slide 78

|                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |   |   |   |   |   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|
| A                                                                                                                                                                                                                                                                                                                                           | B | C | D | E | F | G | H | I | J | K |
| The customer relations department is planning to survey their customers. They want a survey instrument that takes less than 20 minutes to complete. They have given their survey to 65 employees and recorded their Time. At 0.05 level of significance can they claim that average time to complete the survey takes less than 20 minutes? |   |   |   |   |   |   |   |   |   | 1 |

The customer relations department is planning to survey their customers. They want a survey instrument that takes less than 20 minutes to complete. They have given their survey to 65 employees and recorded their Time. At 0.05 level of significance can they claim that average time to **complete the survey takes less than 20 minutes?**

### Transcript

The customer relationship department is planning to survey their customers. They want a survey instrument that takes less than 20 minutes to complete. They have given their survey to 65 employees and recorded their time. At 0.05 level of significance can they claim that the average time to complete the survey takes less than 20 minutes? So, we are doing a hypothesis testing about the mean. Mean here would be the average time it takes to finish a survey. Now, what they want is that they want to have less than 20 minutes to complete the survey. They want  $\mu$  to be less than 20 minutes. And, we want to know if this can be supported with our data. So, since they want it to be less than 20 minutes, this would be our alternative hypothesis. So, they want  $\mu$  to be less than 20 and therefor our null hypothesis will be no, it takes more than 20 minutes to complete.

# Excel Worksheet (2 of 9) - Slide 79

A

B

C

D

E

F

G

H

I

J

The customer relations department is planning to survey their customers. They want a survey instrument that takes less than 20 minutes to complete. They have given their survey to 65 employees and recorded their Time. At 0.05 level of significance can they claim that average time to complete the survey takes less than 20 minutes?

$$H_0: \mu \geq 20$$
$$H_A: \mu < 20$$

$H_0 : \mu \geq 20$

$H_A : \mu < 20$

## Transcript

And, if that's the case then we have to say that the survey is too long and probably customers will not take the time to complete it. So, they have done their testing using their own employees, using 65 employees, and they have found the date and have recorded it here.

# Excel Worksheet (3 of 9) - Slide 80

A

B

C

D

E

F

G

H

I

J

Time

18

20

16

18

18

20

18

12

17

16

19

13

25

20

17

22

10

16

Mean

Standard deviation

Significance level

hypothesized mean

t-value

p-value

=AVERAGE(A2

AVERAGE(number1, [number2], ...)

Ctrl

+

Shift

+

↓

This slide contains a screen shot of an Excel worksheet. Column A is labeled: Time and has 65 numeric entries. Column C has the following labels: Mean, Standard deviation, Significance level, hypothesized mean, t-value, and p-value.

In Column D, next to the label Mean, the formula =AVERAGE is being used.

[Download the Survey excel file \(Refer to Data - First worksheet\)](#)

Transcript

So, I have recorded 65 times here and now I want to do my testing. So first of all, my mean is going to be the average values for these 65 employees.

Excel Worksheet (4 of 9) - Slide 81

| A    | B | C                  | D                                | E | F | G | H | I |
|------|---|--------------------|----------------------------------|---|---|---|---|---|
| Time |   |                    |                                  |   |   |   |   |   |
| 1    |   |                    |                                  |   |   |   |   |   |
| 20   |   | Mean               | 18.03077                         |   |   |   |   |   |
| 16   |   | Standard deviation | =STDEV.S{                        |   |   |   |   |   |
| 18   |   | Significance level | STDEV.S(number1, [number2], ...) |   |   |   |   |   |
| 18   |   | hypothesized mean  |                                  |   |   |   |   |   |
| 20   |   |                    |                                  |   |   |   |   |   |
| 18   |   |                    |                                  |   |   |   |   |   |
| 12   |   | t-value            |                                  |   |   |   |   |   |
| 17   |   | p-value            |                                  |   |   |   |   |   |
| 16   |   |                    |                                  |   |   |   |   |   |
| 19   |   |                    |                                  |   |   |   |   |   |
| 13   |   |                    |                                  |   |   |   |   |   |
| 25   |   |                    |                                  |   |   |   |   |   |
| 20   |   |                    |                                  |   |   |   |   |   |
| 17   |   |                    |                                  |   |   |   |   |   |
| 22   |   |                    |                                  |   |   |   |   |   |
| 10   |   |                    |                                  |   |   |   |   |   |
| 16   |   |                    |                                  |   |   |   |   |   |

This slide contains the same Excel worksheet presented on [Slide 80 - Excel Worksheet \(3 of 9\)](#).

In Column D, next to the label Standard deviation, the formula =STDEV.S is being used.

Transcript

So, I will do control, shift, down, and we find that the average time for these employees was 18 minutes, .03. So, standard deviation. It would be standard deviation dot s and the same data range. So, the standard deviation is about is about 3.71. Significance level we are interested in, .05 and our hypothesis mean is that it would take less than 20 minutes.

## Excel Worksheet (5 of 9) - Slide 82

| Time |  |                    |  |          |  |  |  |  |  |
|------|--|--------------------|--|----------|--|--|--|--|--|
| 18   |  |                    |  |          |  |  |  |  |  |
| 20   |  | Mean               |  | 18.03077 |  |  |  |  |  |
| 16   |  | Standard deviation |  | 3.716388 |  |  |  |  |  |
| 18   |  | Significance level |  | 0.05     |  |  |  |  |  |
| 18   |  | hypothesized mean  |  | 20       |  |  |  |  |  |
| 20   |  |                    |  |          |  |  |  |  |  |
| 18   |  |                    |  |          |  |  |  |  |  |
| 12   |  | t-value            |  |          |  |  |  |  |  |
| 17   |  | p-value            |  |          |  |  |  |  |  |
| 16   |  |                    |  |          |  |  |  |  |  |
| 19   |  |                    |  |          |  |  |  |  |  |
| 13   |  |                    |  |          |  |  |  |  |  |
| 25   |  |                    |  |          |  |  |  |  |  |
| 20   |  |                    |  |          |  |  |  |  |  |
| 17   |  |                    |  |          |  |  |  |  |  |
| 22   |  |                    |  |          |  |  |  |  |  |
| 10   |  |                    |  |          |  |  |  |  |  |
| 16   |  |                    |  |          |  |  |  |  |  |

This slide contains the same Excel worksheet presented on [Slide 80 - Excel Worksheet \(3 of 9\)](#). The following data is shown: Mean: 18.03077, Standard deviation: 3.716388, Significance level: 0.05, hypothesized mean: 20.

### Transcript

So, now I want to calculate the t-value. As before, the t-value is the sample mean, minus hypothesis, mean divided by the standard deviation, divided by the square root of n. And, in this case that would be the 18.03 minus the hypothesis mean of 20, divided by 3.71, divided by the square root of 65. I did not do the count here but there are 65 people in our data.

## Excel Worksheet (6 of 9) - Slide 83

| Time |  |                    |  |                         |  |  |  |  |  |
|------|--|--------------------|--|-------------------------|--|--|--|--|--|
| 18   |  |                    |  |                         |  |  |  |  |  |
| 20   |  | Mean               |  | 18.03077                |  |  |  |  |  |
| 16   |  | Standard deviation |  | 3.716388                |  |  |  |  |  |
| 18   |  | Significance level |  | 0.05                    |  |  |  |  |  |
| 18   |  | hypothesized mean  |  | 20                      |  |  |  |  |  |
| 20   |  |                    |  |                         |  |  |  |  |  |
| 18   |  |                    |  |                         |  |  |  |  |  |
| 12   |  | t-value            |  | = (D3-D6)/(D4/sqrt(65)) |  |  |  |  |  |
| 17   |  | p-value            |  |                         |  |  |  |  |  |
| 16   |  |                    |  |                         |  |  |  |  |  |
| 19   |  |                    |  |                         |  |  |  |  |  |
| 13   |  |                    |  |                         |  |  |  |  |  |
| 25   |  |                    |  |                         |  |  |  |  |  |
| 20   |  |                    |  |                         |  |  |  |  |  |
| 17   |  |                    |  |                         |  |  |  |  |  |
| 22   |  |                    |  |                         |  |  |  |  |  |
| 10   |  |                    |  |                         |  |  |  |  |  |
| 16   |  |                    |  |                         |  |  |  |  |  |
| 25   |  |                    |  |                         |  |  |  |  |  |
| 23   |  |                    |  |                         |  |  |  |  |  |

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

$$= \frac{18.03 - 20}{3.71/\sqrt{65}}$$

This slide contains the same Excel worksheet presented on [Slide 80 - Excel Worksheet \(3 of 9\)](#).

In Column D, next to the label t-value, the formula  $= (D3 - D6) / (D4 / \sqrt{D5})$  is being used.

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} = \frac{18.03 - 20}{3.71 / \sqrt{65}}$$

## Transcript

So, now I am going to just let excel tackle this for me and not myself. So, this will be 80 – 20, divided by standard deviation, divided by the square root of the sample size which was, in this case, 65.

## Excel Worksheet (7 of 9) - Slide 84

| Time |                    |                  |  |
|------|--------------------|------------------|--|
| 18   |                    |                  |  |
| 20   | Mean               | 18.03077         |  |
| 16   | Standard deviation | 3.716388         |  |
| 18   | Significance level | 0.05             |  |
| 18   | hypothesized mean  | 20               |  |
| 20   |                    |                  |  |
| 18   |                    |                  |  |
| 12   | t-value            | -4.27201         |  |
| 17   | p-value            | =T.DIST(D9,64,1) |  |
| 16   |                    |                  |  |
| 19   |                    |                  |  |
| 13   |                    |                  |  |
| 25   |                    |                  |  |
| 20   |                    |                  |  |
| 17   |                    |                  |  |
| 22   |                    |                  |  |
| 10   |                    |                  |  |
| 16   |                    |                  |  |
| 25   |                    |                  |  |
| 23   |                    |                  |  |

Handwritten notes on the right side of the table:

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}} = \frac{18.03 - 20}{3.71 / \sqrt{65}}$$

Excel tooltip for T.DIST:

T.DIST(x, deg, [cumulative])  
 (TRUE) - cumulative distribution function  
 (FALSE) - probability density function  
 T.DIST returns the cumulative distribution function or the probability density function of a student's t-distribution, depending on the value of the cumulative parameter.

This slide contains the same Excel worksheet presented on [Slide 80 - Excel Worksheet \(3 of 9\)](#).

In Column D, next to the label p-value, the formula  $=T.DIST$  is being used.

## Transcript

And, my t-value turns out to be -4.27. So, already I know I am more than 3 standard deviations away. So, this is going to be a rejection process because p-value is going to be very small but we can calculate this. So, let's make this equal to t.dist and are x is -4.27. The next value is degrees of freedom. Degrees of freedom is always your sample size minus 1. And, in this case there were 65 employees that were tested. So, 64. And, we want an accumulative. And, as we see, the p-value comes out to be extremely small.



## Excel Worksheet (8 of 9) - Slide 85

| Time |                    |             |   |
|------|--------------------|-------------|---|
| 18   |                    |             |   |
| 20   | Mean               | 18.03076923 |   |
| 16   | Standard deviation | 3.716387824 |   |
| 18   | Significance level | 0.05        |   |
| 18   | hypothesized mean  | 20          |   |
| 20   |                    |             |   |
| 18   |                    |             |   |
| 12   | t-value            | -4.27201002 |   |
| 17   | p-value            | 3.27938E-05 | + |
| 16   |                    |             |   |
| 19   |                    |             |   |
| 13   |                    |             |   |
| 25   |                    |             |   |
| 20   |                    |             |   |
| 17   |                    |             |   |
| 22   |                    |             |   |
| 10   |                    |             |   |
| 16   |                    |             |   |
| 25   |                    |             |   |

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}$$

$$= \frac{18.03 - 20}{3.71/\sqrt{65}}$$

3.20

This slide contains the same Excel worksheet presented on [Slide 80 - Excel Worksheet \(3 of 9\)](#). The following data is shown: Mean: 18.03077, Standard deviation: 3.716388, Significance level: 0.05, hypothesized mean: 20, t-value: -4.27201002, p-value: 3.27938E-05

[Download the Survey excel file \(Refer to Left Tail Test - Second worksheet\)](#)

## Transcript

And, so this is practically 0 and this is less than half of .05. So, we will end up rejecting the null hypothesis.

## Excel Worksheet (9 of 9) - Slide 86

The customer relations department is planning to survey their customers. They want a survey instrument that takes less than 20 minutes to complete. They have given their survey to 65 employees and recorded their Time. At 0.05 level of significance can they claim that average time to complete the survey takes less than 20 minutes?

$H_0: \mu \geq 20$   
 $H_a: \mu < 20$

P-value < .05 Rejected

P-values is 3.27938E-05. P-value less than .05. Rejected. Therefore,  $H_0: \mu \leq 20$  is true.

## Transcript

So, p-value in this case was smaller than  $\alpha$  and therefore, we would reject the null hypothesis which means what? Which means that we believe that it would take less than 20 minutes. So, if that was what they were looking for, this survey will satisfy that requirement and they can go ahead and send it to their customers.

## Lesson 1-4.3 Right-Tail Test for Mean in Excel

[Media Player for Video](#) 

### Excel Worksheet (1 of 9) - Slide 87

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <p>The mayor of a large metropolitan city has been receiving complaints about high gas prices in the city. This is complexing since the city doesn't have a gas tax that is added to the price, so the mayor has asked one of his staff to look into this matter.</p> <p>Data on the price of regular unleaded gas has been collected on a day when the average for the unleaded gas in the state was \$2.79. At 5% level of significance and based on this data, is there evidence to suggest that gas prices are higher in this city?</p> |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|

The mayor of a large metropolitan city has been receiving complaints about high gas prices in the city. This is complexing since the city doesn't have a gas tax that is added to the price, so the mayor has asked one of his staff to look into this matter.

Data on the price of regular unleaded gas has been collected on a day when the average for the unleaded gas in the state was \$2.79. At 5% level of significance and based on this data, is there evidence to suggest that gas prices are higher in this city?

## Transcript

The mayor of a large metropolitan city has been receiving complaints about high gas prices in the city. This is complexing since the city doesn't have a gas tax that is added to the price. So, the mayor has asked one of his staff to look into this matter. We have data on prices of regular, unleaded gas on a day when the average for unleaded gas in the state was \$2.79. They're going to test this at 5% level of significance to see if there is evidence to suggest that the gas prices are higher in the city or not. So, first of all, we have to form our hypothesis. So, what we see here is that we think the prices in this city is higher, or at least the citizens are saying. So, they're saying that it is higher than \$2.79. And, the null hypothesis, therefore, would be that  $\mu$  is less than or equal \$2.79. So, this is what we would call a right-tailed test. So, remember the sign here always refers to the side of the curve that will have the rejection area in it. So, in this case, we are looking at .05. So, what we are saying is that the significance level is .05. Therefore, we are going to take a sample, and we have done that. And, if that sample falls in the tail area here, it means that it is too unlikely to be randomly chosen to be that way. There must be a cause for that, and that is why we will end up rejecting the null hypothesis if our p-value is very small and falls into the tail.

## Excel Worksheet (2 of 9) - Slide 88

The mayor of a large metropolitan city has been receiving complaints about high gas prices in the city. This is complexing since the city doesn't have a gas tax that is added to the price, so the mayor has asked one of his staff to look into this matter.

Data on the price of regular unleaded gas has been collected on a day when the average for the unleaded gas in the state was \$2.79. At 5% level of significance and based on this data, is there evidence to suggest that gas prices are higher in this city?

$H_0 : \mu \leq 2.79$   
 $H_A : \mu > 2.79$



This slide contains the same text presented on [Slide 87 - Excel Worksheet \(1 of 9\)](#).

$$H_0 : \mu \leq 2.79$$

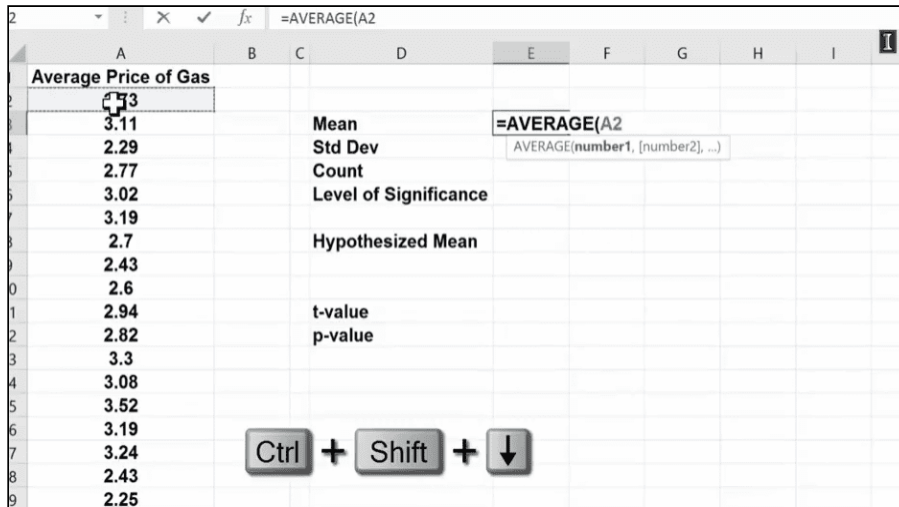
$$H_A : \mu > 2.79$$

Circles greater than symbol in Alternate hypothesis. Draws standard graph and shades in area under the tail of the curve at the far right. Labels this .05.

## Transcript

On the other hand, if you have a sample that falls on this side of the tail (All area under the curve to the left of the right tail), then it is a possible random variation that you see from city to city, and it's not really an indication that this city's any different than any other city in the state.

## Excel Worksheet (3 of 9) - Slide 89



The screenshot shows an Excel worksheet with the following data:

| Row | A: Average Price of Gas | B | C | D                     | E                                | F | G | H | I |
|-----|-------------------------|---|---|-----------------------|----------------------------------|---|---|---|---|
| 2   | 3.11                    |   |   | Mean                  | =AVERAGE(A2                      |   |   |   |   |
| 3   | 2.29                    |   |   | Std Dev               | AVERAGE(number1, [number2], ...) |   |   |   |   |
| 4   | 2.77                    |   |   | Count                 |                                  |   |   |   |   |
| 5   | 3.02                    |   |   | Level of Significance |                                  |   |   |   |   |
| 6   | 3.19                    |   |   |                       |                                  |   |   |   |   |
| 7   | 2.7                     |   |   | Hypothesized Mean     |                                  |   |   |   |   |
| 8   | 2.43                    |   |   |                       |                                  |   |   |   |   |
| 9   | 2.6                     |   |   |                       |                                  |   |   |   |   |
| 10  | 2.94                    |   |   | t-value               |                                  |   |   |   |   |
| 11  | 2.82                    |   |   | p-value               |                                  |   |   |   |   |
| 12  | 3.3                     |   |   |                       |                                  |   |   |   |   |
| 13  | 3.08                    |   |   |                       |                                  |   |   |   |   |
| 14  | 3.52                    |   |   |                       |                                  |   |   |   |   |
| 15  | 3.19                    |   |   |                       |                                  |   |   |   |   |
| 16  | 3.24                    |   |   |                       |                                  |   |   |   |   |
| 17  | 2.43                    |   |   |                       |                                  |   |   |   |   |
| 18  | 2.25                    |   |   |                       |                                  |   |   |   |   |

At the bottom of the worksheet, there is a keyboard shortcut提示: Ctrl + Shift + ↓.

This slide contains a screen shot of an Excel worksheet. Column A is labeled: Average Price of Gas and has 133 numeric entries. Column D has the following labels: Mean, Std Dev, Count, Level of Significance, Hypothesized Mean, t-value, and p-value.

In Column E, next to the label Mean, the formula =AVERAGE is being used.

[Download the Gas Prices excel file \(Refer to Data - First worksheet\)](#) 

## Transcript

So, now, let's look at the data. So, the data is being collected here, so we're going to look at what is the mean, what is the standard deviation, and the count, and so on.

## Excel Worksheet (4 of 9) - Slide 90

|    | A    | B | C | D                     | E      | F | G | H | I |
|----|------|---|---|-----------------------|--------|---|---|---|---|
| 2  | 2.73 |   |   |                       |        |   |   |   |   |
| 3  | 3.11 |   |   |                       | 2.9415 |   |   |   |   |
| 4  | 2.29 |   |   | Mean                  |        |   |   |   |   |
| 5  | 2.77 |   |   | Std Dev               |        |   |   |   |   |
| 6  | 3.02 |   |   | Count                 |        |   |   |   |   |
| 7  | 3.19 |   |   | Level of Significance |        |   |   |   |   |
| 8  | 2.7  |   |   | Hypothesized Mean     |        |   |   |   |   |
| 9  | 2.43 |   |   |                       |        |   |   |   |   |
| 10 | 2.6  |   |   |                       |        |   |   |   |   |
| 11 | 2.94 |   |   | t-value               |        |   |   |   |   |
| 12 | 2.82 |   |   | p-value               |        |   |   |   |   |
| 13 | 3.3  |   |   |                       |        |   |   |   |   |
| 14 | 3.08 |   |   |                       |        |   |   |   |   |
| 15 | 3.52 |   |   |                       |        |   |   |   |   |
| 16 | 3.19 |   |   |                       |        |   |   |   |   |
| 17 | 3.24 |   |   |                       |        |   |   |   |   |
| 18 | 2.43 |   |   |                       |        |   |   |   |   |
| 19 | 2.25 |   |   |                       |        |   |   |   |   |
| 20 | 3.17 |   |   |                       |        |   |   |   |   |

This slide contains the same Excel worksheet presented on [Slide 89 - Excel Worksheet \(3 of 9\)](#).  
Mean now reads: 2.9415.

## Transcript

So, we can count the p-value. So, the mean is simply the average for the values that we see here. So, I'm going to use control shift down, take the entire thing. Return, and right now, it turns out to be \$2.94.

## Excel Worksheet (5 of 9) - Slide 91

The mayor of a large metropolitan city has been receiving complaints about high gas prices in the city. This is complexing since the city doesn't have a gas tax that is added to the price, so the mayor has asked one of his staff to look into this matter.

Data on the price of regular unleaded gas has been collected on a day when the average for the unleaded gas in the state was \$2.79. At 5% level of significance and based on this data, is there evidence to suggest that gas prices are higher in this city?

$$H_0: \mu \leq 2.79$$
$$H_a: \mu > 2.79$$

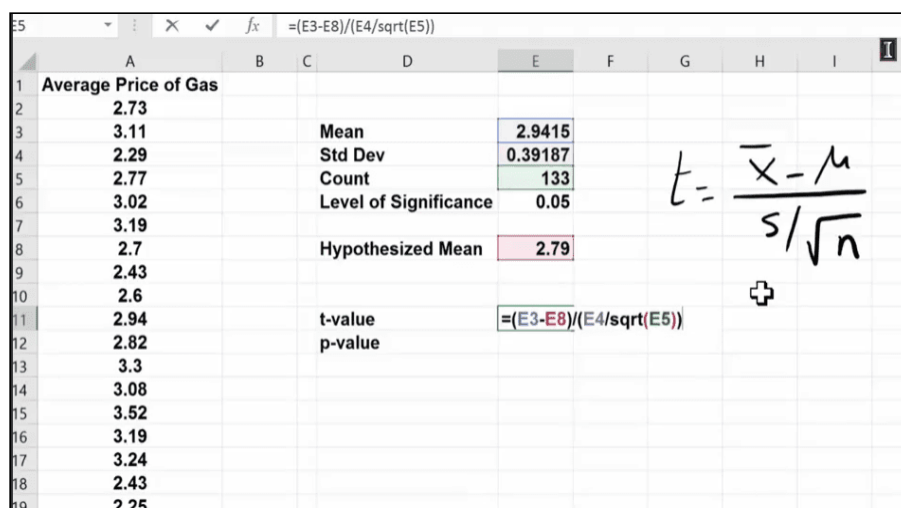

This slide contains the same text presented on [Slide 87 - Excel Worksheet \(1 of 9\)](#).

Level of significance indicated as 0.05.

## Transcript

Now, \$2.94 is slightly higher than \$2.79, but as we have seen before, just because it's higher doesn't mean that this is statistically different. So, we need to do the hypothesis testing to establish that that 5% level of significance. So, let's go back to our data. Standard deviation is stdev.s. I will pick this, and then, I will pick the entire range of the data and close the parentheses. Count will tell me the number of datasets that we have here. So, we have 133 gas stations that we have collected data from. Level of significance, we said it was .05. Hypothesized mean is that it's \$2.79 because on this day, the price of gas in the state, average price of regular gas in the state was \$2.79. So, now, I'm ready to calculate the t-value. So, again, the t-value is  $\bar{x} - \mu$  divided by standard deviation divided by score which have n. So, now, I am going to use this formula right here and enter the actual representation of it here.

## Excel Worksheet (6 of 9) - Slide 92



|    | A                    | B | C                     | D | E                       | F | G | H | I |
|----|----------------------|---|-----------------------|---|-------------------------|---|---|---|---|
| 1  | Average Price of Gas |   |                       |   |                         |   |   |   |   |
| 2  | 2.73                 |   |                       |   |                         |   |   |   |   |
| 3  | 3.11                 |   | Mean                  |   | 2.9415                  |   |   |   |   |
| 4  | 2.29                 |   | Std Dev               |   | 0.39187                 |   |   |   |   |
| 5  | 2.77                 |   | Count                 |   | 133                     |   |   |   |   |
| 6  | 3.02                 |   | Level of Significance |   | 0.05                    |   |   |   |   |
| 7  | 3.19                 |   |                       |   |                         |   |   |   |   |
| 8  | 2.7                  |   | Hypothesized Mean     |   | 2.79                    |   |   |   |   |
| 9  | 2.43                 |   |                       |   |                         |   |   |   |   |
| 10 | 2.6                  |   |                       |   |                         |   |   |   |   |
| 11 | 2.94                 |   | t-value               |   | = (E3-E8)/(E4/sqrt(E5)) |   |   |   |   |
| 12 | 2.82                 |   | p-value               |   |                         |   |   |   |   |
| 13 | 3.3                  |   |                       |   |                         |   |   |   |   |
| 14 | 3.08                 |   |                       |   |                         |   |   |   |   |
| 15 | 3.52                 |   |                       |   |                         |   |   |   |   |
| 16 | 3.19                 |   |                       |   |                         |   |   |   |   |
| 17 | 3.24                 |   |                       |   |                         |   |   |   |   |
| 18 | 2.43                 |   |                       |   |                         |   |   |   |   |
| 19 | 2.25                 |   |                       |   |                         |   |   |   |   |

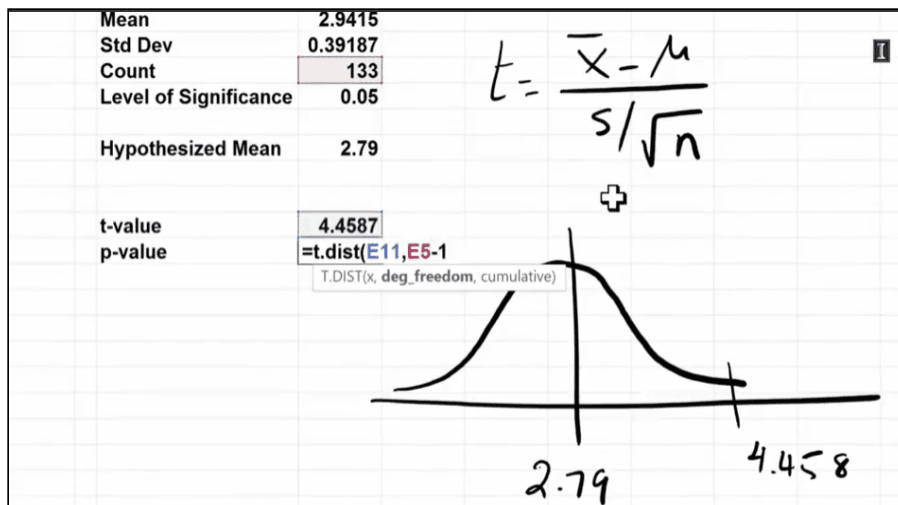
This slide contains the same Excel worksheet presented on [Slide 89 - Excel Worksheet \(3 of 9\)](#). The following data is shown: Mean: 2.9415, Std Dev: 0.39187, Count: 133, Level of Significance: 0.05, Hypothesized Mean: 2.79.

In Column E, next to the label t-value, the formula  $=(E3-E8)/(E4/\text{sqrt}(E5))$  is being used.

## Transcript

So, it would be equal to parentheses  $\bar{x}$  is right here, that's my sample mean  $-\mu$ , that's my hypothesized mean, it's right here. Divided by, again, another parenthesis, by the standard deviation which I have here. Divided by the square root of  $n$  in this case is 133. Count is really my number of datasets. So, this would be the equation that I have written on a corner about  $t$ . So, this represents all the values that goes into that equation, and  $t$  value turns out to be 4.458. So, again, looking at it from a perspective of our problem is that we think that \$2.79 is here, and if this is a normal distribution, we have found a sample on this day that falls 4.458 away from the mean. So, you know already that is way into the tail. So, even without really giving you the  $p$ -value, if you're that far off, you are way into the tail. The, it's practically zero chances that you would find this thing randomly. It is so unheard of that it probably will point to a cause. So, it's not random. So, I would end up rejecting, but let me just show you how we would calculate the  $p$ -value.

## Excel Worksheet (7 of 9) - Slide 93

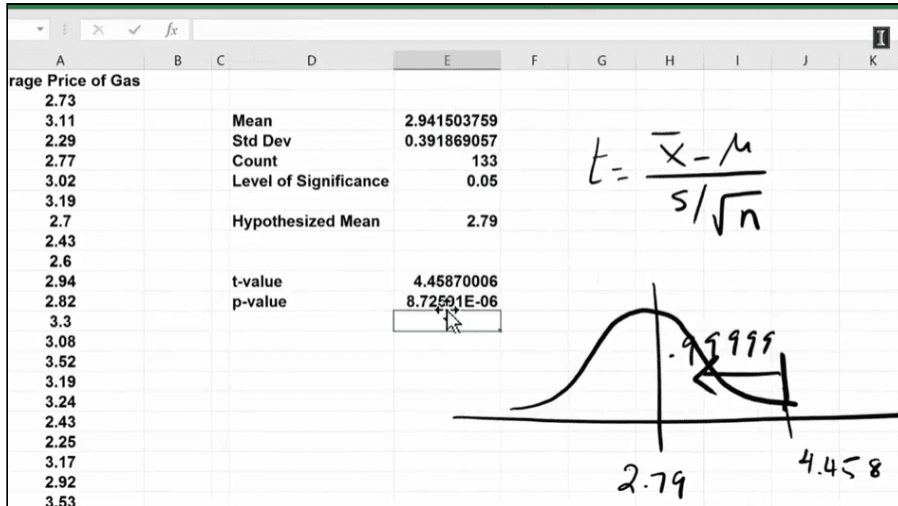


This slide contains a standard graph with peak at 2.79 on horizontal axis. 4.458 line at the far right under the tail.  $p$ -value: Equals T.DIST(E11, E5 minus 1, 1), which is 0.99999.

## Transcript

And, this is the one that I want you to pay attention to. When you're doing a right-tail test, if I put the  $t$ .dist, just let me show you what happens. If I put just  $t$ .dist, and I put this value along with my degrees of freedom which is  $133 - 1$ . And, the last part is one. You would get a very large number. Why? Because this is returning this side. So, this is .99999, the value you see here. So, when you have a right-tail test, you have to make sure that you enter this value as 1 minus (Clicks on  $p$ -value 0.99999 in column E. 8.72591E-06).

## Excel Worksheet (8 of 9) - Slide 94



This slide contains the same Excel worksheet presented on [Slide 89 - Excel Worksheet \(3 of 9\)](#). The following data is shown: Mean: 2.941503759, Std Dev: 0.391869057, Count: 133, Level of Significance: 0.05, Hypothesized Mean: 2.79, t-value: 4.45870006, p-value: 8.72591E-06.

[Download the Gas Prices excel file \(Refer to Right Tail Test - Second worksheet\)](#)

### Transcript

And then, you get the value that is really small because it's 1 minus this that you saw. So, this is practically zero. So, going back, our p-values is extremely small. If your p-value's small, you'll end up rejecting the null hypothesis in favor of the alternate. So, citizens are right by saying that the gas prices in the city is higher than the state, and there should be reasons why they're doing this.



## Excel Worksheet (9 of 9) - Slide 95

The mayor of a large metropolitan city has been receiving complaints about high gas prices in the city. This is perplexing since the city doesn't have a gas tax that is added to the price, so the mayor has asked one of his staff to look into this matter.

Data on the price of regular unleaded gas has been collected on a day when the average for the unleaded gas in the state was \$2.79. At 5% level of significance and based on this data, there is evidence to suggest that gas prices are higher in this city?

Handwritten notes:

$$H_0: \mu \leq 2.79$$
$$H_a: \mu > 2.79$$

A hand-drawn normal distribution curve is shown with a vertical line at 2.79. The area to the right of this line is shaded and labeled with .05, representing the significance level.

This slide contains the same information presented on [Slide 91 - Excel Worksheet \(5 of 9\)](#).

### Transcript

If there is now city tax, then the city manager needs to find out what is it that they're doing, and this could be a case of price gouging, for example, which is actually punishable. So, this is how we would test this. Now, again, I just wanted you to pay attention that if this is a positive value, you're not getting a p value that is .99999. You have to look at 1 minus.

# Lesson 1-5 Testing the Proportion

## Lesson 1-5.1 Testing the Proportion

[Media Player for Video](#) 

### In the News - Slide 96

#### IN THE NEWS

A study presented by the American Academy of Neurology (AAN) states that:

"More than 40 percent of retired National Football League players ... had signs of traumatic brain injury based on sensitive MRI scans called diffusion tensor imaging."



(American Academy of Neurology, 2016)

A study presented by the American Academy of Neurology (AAN) states that: "More than 40 percent of retired National Football League players... had signs of traumatic brain injury based on sensitive MRI scans called diffusion tensor imaging."

(American Academy of Neurology, 2016)

### Transcript

A study presented at the American Academy of Neurology states that more than 40% of retired National Football League players had signs of traumatic brain injury based on sensitive MRI scan. So, when you read this study like this, we see the researchers are making a claim about a population proportion based on a sample. If we were to challenge this finding, then we will be conducting a hypothesis testing by the population proportion.

## Steps in Testing the Proportion - Slide 97

### STEPS IN TESTING THE PROPORTION 1

1. State  $H_0$  and  $H_a$ .
2. Specify the significance level  $\alpha$ .
3. Calculate the p-value for the sample data.
4. Reject or not reject  $H_0$ .



1. State  $H_0$  and  $H_A$
2. Specify the significance level  $\alpha$
3. Calculate the p-value for the sample data
4. Reject or not reject  $H_0$

### Transcript

The steps taken will not be any different than what we have seen when we are doing a hypothesis testing about the mean. There are a few differences in notation as well as how statistics is calculated and that is what we will learn in this lesson. So, let's start with the first step which is stating the null and the alternate hypothesis. Using the claim made about the NFL players, let  $p$  represent the proportion of players who had signs of traumatic brain injury. Looking at the current belief, which is more than 40% have this problem, then the correct notation would be  $p$  is greater than 0.40, but I can't make this the null hypothesis. Why not? Do you remember? We have to have the equal sign in the null hypothesis. When our belief – when our current belief doesn't allow for that, then it becomes the alternate hypothesis and the null will be its complement which is  $p$  is less than or equal to 0.40.

## Example - State the Null Hypothesis - Slide 98

### EXAMPLE – STATE THE NULL HYPOTHESIS 1

Let  $p$  = proportion of players who have signs of traumatic brain injury

**"More than 40 percent** of retired National Football League players ... had signs of traumatic brain injury ..."

$p > 0.40$

$H_0: p \leq 0.40$   
 $H_a: p > 0.40$



(American Academy of Neurology, 2016)

Let  $p$  = proportion of players who have signs of traumatic brain injury

**"More than 40 percent** of retired National Football League players... had signs of traumatic brain injury..."

$p > 0.40$

$H_0 : p \leq 0.40$

$H_A : p > 0.40$

(American Academy of Neurology, 2016)

### Transcript

These hypotheses lead to a one-sided test. There are a lot of legal implications of such studies and studies like this will not go unchallenged. But challenging it means taking yet another sample and possibly different tests in order to reject the findings.

## Example - Slide 99

**EXAMPLE**

Every month the Bureau of Statistics releases many reports. One such report is our nation's unemployment rate. For the month of March 2016 the average unemployment rate was reported as 5%.

Some economists disagree with how the government reports this number, and they think the unemployment rate is actually worse.



(Bureau of Statistics, 2016)

Every month, the Bureau of Statistic releases many reports. One such report is our nation's unemployment rate. For the month of March 2016 the average unemployment rate was reported as 5%.

Some economists disagree with how the government reports this number, and they think the unemployment rate is actually worse.

(Bureau of Statistics, 2016)

### Transcript

Every month the Bureau of Statistics releases many reports. One such report is our nation's unemployment rate. For the month of March 2016, the average unemployment rate is reported as 5%. So, economists disagree with how the government reports this number and they think it's actually worse, meaning that it's higher than what is being reported. Now, let  $p$  represent the proportion of unemployed. And government is saying for March it was 5%. The challenge is that the proportion of unemployment is higher, so it's easier to write the alternate hypothesis first which is  $p$  is greater than 5% and then the null.

## Example - State the Null Hypothesis - Slide 100

### EXAMPLE – STATE THE NULL HYPOTHESIS

Let  $p$  = proportion unemployed

$p = .05$

$H_0: p \leq 0.05$

$H_a: p > 0.05$



(Bureau of Statistics, 2016)

Let  $p$  = proportion unemployed

$$p = 0.05$$

$$H_0 : p \leq 0.05$$

$$H_A : p > 0.05$$

(Bureau of Statistics, 2016)

### Transcript

Once again, these hypotheses lead to a one-sided test. You may find that one-tail tests are a lot more common than two-tail tests. But let me show you an example for a two-tail test.

## Example - Slide 101

### EXAMPLE



Dextrose is a prescription medication that is used to increase a person's blood sugar. One form of the medication is a 30% Dextrose injection.

(Drugs.com, n.d.)



Dextrose is a prescription medication that is used to increase a person's blood sugar. One form of the medication is a 30% Dextrose injection.

(Drugs.com, n.d.)

### Transcript

Dextrose is a prescription medication which is used to increase a person's blood sugar. One form of the medication is a 30% dextrose injection. Imagine if you were in charge of quality control for this. What would you like to see if you took a sample and measured the concentration of its active ingredients? Let  $p$  represent the percentage of the active ingredient in each vial.

## Example - State the Null Hypothesis - Slide 102

### EXAMPLE – STATE THE NULL HYPOTHESIS

Let  $p$  = the percentage of the active ingredient in each vial

$p = .30$

$H_0: p = 0.30$



<https://www.washingtonpost.com/news/moning-mix/wp/2016/04/12/40-percent-of-former-nfl-players-suffer-from-brain-damage/>



Let  $p$  = the percentage of the active ingredient in each vial

$$p = 0.30$$

$$H_0 : p = 0.30$$

## Transcript

In this case, the label says 30%, you want it to be 30%, right? No more, no less, anything different can result in disastrous consequences. So, the null hypothesis should show what is in the vial matches the label, so  $p$  is equal to 30%. And the samples are taken to test this hypothesis which means alternate is  $p$  is not 30%. This will be a case of a two-tail test.

## Example 1 - Slide 103

**EXAMPLE 1**

An HR director is concerned about her aging workforce and what will happen to the company if these employees decide to retire at about the same time. The manager believes that at least 18% of the employees are at or above the retirement age. To check the validity of this belief, a sample of 500 employees was examined, which showed the proportion to be 15%. Testing at the 5% level of significance, is the manager right?



An HR director is concerned about her aging workforce and what will happen to the company if these employees decide to retire at about the same time. The manager believes that at least 18% of the employees are at or above the retirement age. To check the validity of this belief, a sample of 500 employees was examined, which showed the proportion to be 15%. Testing at the 5% level of significance, is the manager right?

## Transcript

Now, consider the following example. An HR director is concerned about her aging workforce and what will happen to the company if these employees decide to retire at about the same time. The manager believes that at least 18% of the employees are at or above their retirement age. To check validity of disbelief, a sample of 500 employees was examined which showed the proportion to be 15%. Testing at the 5% level of significance, is the manager, right? Now, let's help this manager analyze the data.



## Steps in Testing the Proportion - Slide 104

### STEPS IN TESTING THE PROPORTION

1. State  $H_0$  and  $H_a$ .  
 $H_0: p \geq 0.18$   
 $H_a: p < 0.18$
2. Specify the significance level  $\alpha$ .  
 $\alpha = 0.05$
3. Calculate the p-value for the sample data.
4. Reject or not reject  $H_0$ .



1. State  $H_0$  and  $H_A$ :  
 $H_0 : p \geq 0.18$   
and  $H_A : p < 0.18$
2. Specify the significance level  $\alpha$ :  $\alpha = 0.05$
3. Calculate the p-value for the sample data
4. Reject or not reject  $H_0$

### Transcript

Begin with stating the hypothesis. The manager believes that at least 18% are at or near retirement age. This will translate to null hypothesis of p is greater than or equal to 18%. Again, the belief is that at least 18% are near or at retirement age. An alternate is p is less than 18%. Testing at 5% level of significance means that alpha is 0.05.

## Example 1 - Continued - Slide 105

**EXAMPLE 1 – CONTINUED**

Based on  $n = 500$ ,  $\hat{p} = 0.15$

Assuming  $H_0$  is true, the sampling distribution of  $\hat{p}$  is approximately normal with:

$p = 0.18$ , and

Std. error =  $\sqrt{\frac{p_0 \times (1 - p_0)}{n}}$

S.E. =  $\sqrt{\frac{0.18 \times (1 - 0.18)}{500}} = 0.0172$



Based on  $n = 500$ ,  
 $\hat{p} = 0.15$

Assuming  $H_0$  is true, the sampling distribution of  $\hat{p}$  is approximately normal with:

$p = 0.18$ , and

$$\text{Std. error} = \sqrt{\frac{p_0 \times (1 - p_0)}{n}}$$

$$\text{S.E.} = \sqrt{\frac{0.18 \times (1 - 0.18)}{500}} = 0.0172$$

### Transcript

Now, we are ready to calculate the p value for our sample which tells us what is the chance of getting a sample that differs from the hypothesized proportion by as much as this one if the null hypothesis is true. Based on sample size of 500, and the sample proportion  $\hat{p}$  has of 15%, and assuming null is true, the sampling distribution of sample proportion is approximately normal centered at  $p$  of 0.18, 18%. This is the hypothesized proportion which is denoted as  $p$  subscript 0. And standard error, these sample proportions, again, this is to be sampled over and over again. The distribution of sample proportions will have a spread which is its standard error. Standard error is calculated based on this equation which we learned in the previous course when we learned about Central Limit Theorem.

Substituting the values for the notations, 18% is the  $p$  subscript 0 which is the hypothesized proportion, we get the value of 0.0172. Now, we need to find z-value which represents how many

standard errors is this samples proportion from the hypothesized portion of 18%. For testing, the population proportion, the test statistics z is calculated as follows. Substituting the values in the notation, sample proportion p had this 0.15, the hypothesized proportion p subscript 0 is 18% and the denominator is the standard error we just calculated in the last Slide to be 0.172, and this gives us the z-value of negative 1.744.

## Example 1 - Z Value - Slide 106

### EXAMPLE 1 – Z VALUE 1

For testing the population proportion, the test statistics (z-value) is:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 \times (1 - p_0)}{n}}}$$

$$z = \frac{0.15 - 0.18}{0.0172}$$

$$= -1.744$$


For testing the population proportion, the test statistics (z-value) is:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 \times (1 - p_0)}{n}}}$$

$$z = \frac{0.15 - 0.18}{0.0172} = -1.744$$

### Transcript

So, our samples proportion falls 1.744 standard errors away on the left side from the 0.18 hypothesized proportion. What are the chances of finding a sample like we did? Answer to this will be the p-value.

## Example 1 - Continued (1 of 2) - Slide 107

### EXAMPLE 1 – CONTINUED

**p-value**  
Use NORM.S.DIST

`=NORM.S.DIST(`

`NORM.S.DIST(z, cumulative)`

`=NORM.S.DIST(-1.744,1)`

|                  | D     | E | F |
|------------------|-------|---|---|
| <b>p-value =</b> | 0.041 |   |   |



### p-value

Use NORM.S.DIST

The Slide shows the Excel function for calculating the p-value "NORM.S.DIST(z, cumulative)." The function used to calculate the p-value is NORM.S.DIST(-1.744, 1). The outcome of this function is 0.041.

### Transcript

To find the p-value, we will use norm.s.dist function in Excel which is two arguments, z and cumulative. We just calculated z to be, negative 1.744 and cumulative is true for us and we enter it as 1. And we get the value p of 0.041. Now, that we have completed steps 1-3, we are ready to make the decision.

## Example 1 - Continued (2 of 2) - Slide 108

**EXAMPLE 1 – CONTINUED**



1. State  $H_0$  and  $H_a$ .  
 $H_0: p \geq 0.18$   
 $H_a: p < 0.18$
2. Specify the significance level  $\alpha$ .  
 $\alpha = 0.05$
3. Calculate the p-value for the sample data. **p-value = 0.041**
4. Reject or not reject  $H_0$ .  
**p-value = 0.041 < 0.05 → Reject  $H_0$**

1. State  $H_0$  and  $H_A$ :  $H_0 : p \geq 0.18$  and  $H_A : p < 0.18$
2. Specify the significance level  $\alpha$ :  $\alpha = 0.05$
3. Calculate the p-value for the sample data: **p-value = 0.041**
4. Reject or not reject  $H_0$ : **p-value = 0.041 < 0.05 → Reject  $H_0$**

### Transcript

And since the p-value is less than  $\alpha$ , we reject the null hypothesis based on our sample study we can't support the managers belief that at least 18% of the employees are at or near retirement age. Now, let's practice together.

## Let's Practice - Solved (1 of 3) - Slide 109

**LET'S PRACTICE – SOLVED**

A drug company claims that only 2% of patients suffer any major side effects by taking one of its drugs. You are asked to check the validity of this claim at 1% level of significance. You start by surveying 1000 patients and find that 26 have reported major side effects. What is your conclusion?

**p**: proportion of patients who had side effects

1. State  $H_0$  and  $H_a$ .  
 $H_0: p \leq 0.02$   
 $H_a: p > 0.02$
2. Specify the significance level  $\alpha$ .  
 $\alpha = 0.01$
3. Calculate the p-value for the sample data.
4. Reject or not reject  $H_0$ .

A drug company claims that only 2% of patients suffer any major side effects by taking one of its drugs. You are asked to check the validity of this claim at 1% level of significance. You start by surveying 1000 patients and find that 26 have reported major side effects. What is your conclusion?

$p$ : proportion of patients who had side effects.

1. State  $H_0$  and  $H_A$ :  $H_0 : p \leq 0.02$  and  $H_A : p > 0.02$
2. Specify the significance level  $\alpha$ :  $\alpha = 0.01$
3. Calculate the p-value for the sample data
4. Reject or not reject  $H_0$

## Transcript

A drug company claims that only 2% of patients suffer any major side effects by taking one of their drugs. You are asked to check the validity of this claim at 1% of the significance. You start by serving 1,000 patients and find that 26 have reported major side effects. What is your conclusion? Start by stating the hypothesis and the value of  $\alpha$ . We start by defining  $p$  as the proportion of patients who had some side effects. Based on what you see, the claim is that  $p$  is 2%. But we are not just interested in saying that it's not. We are interested in knowing if the drug company is low balling this figure. In other words, they are investigating this to see if the proportion is higher than 2% which means the alternate of the claim is that  $p$  is greater than 2%. Thus, the null is,  $p$  is less than or equal to 2% and  $\alpha$  is 0.01. So, now let's move on to finding the p-value.

## Let's Practice - Solved (2 of 3) - Slide 110

### LET'S PRACTICE – SOLVED

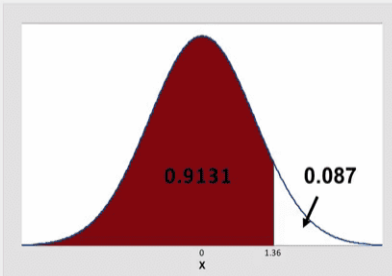
**CALCULATE THE P-VALUE FOR THE SAMPLE DATA.**

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 \times (1 - p_0)}{n}}} = \frac{0.026 - 0.02}{\sqrt{\frac{0.02 \times (1 - 0.02)}{1000}}} = 1.36$$

=NORM.S.DIST[1.36,1]

| D     | E |
|-------|---|
| 0.913 |   |

**p-value = 1 - 0.913 = 0.087**



**Calculate the p-value for the sample data.**

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0 \times (1 - p_0)}{n}}} = \frac{0.026 - 0.02}{\sqrt{\frac{0.02 \times (1 - 0.02)}{1000}}} = 1.36$$

This slide contains a bell curve with shaded area under curve labeled 0.9131. Tail under curve on right labeled 0.087.

$$p \text{ value} = 1 - 0.913 = 0.087$$

## Transcript

For our study, we had 26 out of 1,000 who reported some major side effects and that means our sample proportion is 2.6%. Now, we can calculate z-value and that is 1.36. Since our z-value is positive, Excel returns the probability as 0.913. Again, this is the area being returned. We are interested in the area in the tail. That is the p-value, probability of finding a sample like ours. That many or more standards away from the center of the distribution. And in this case, that would be 1 minus 0.913 or 0.087. So, now that you have this, we can make a decision.

## Let's Practice - Solved (3 of 3) - Slide 111

### LET'S PRACTICE – SOLVED

1. State  $H_0$  and  $H_a$ .  
 $H_0: p \leq 0.02$   
 $H_a: p > 0.02$
2. Specify the significance level  $\alpha$ .  
 $\alpha = 0.01$
3. Calculate the p-value for the sample data.  
 $p = 0.087$
4. Reject or not reject  $H_0$ .  
 $p > \alpha$



1. State  $H_0$  and  $H_A$ :  $H_0 : p \leq 0.02$  and  $H_A : p > 0.02$
2. Specify the significance level  $\alpha$ :  $\alpha = 0.01$
3. Calculate the p-value for the sample data:  $p = 0.087$
4. Reject or not reject  $H_0$ :  $p > \alpha$

## Transcript

What is your conclusion? Reject or not reject the null hypothesis? Since  $p$  is larger than 0.01, they will not reject the null hypothesis. So, we retain the belief that less than 2% of people taking this medication, will suffer major side effects. Before we end this topic, I want to make you aware of a few things. Statistical significance does not mean that you have made an important or meaningful discovery. This is setting the probability of Type I error. Rejecting a null hypothesis which is actually true. Take this value, keeping both errors in mind. Remember, reducing  $\alpha$  will increase the probability of Type II error, not rejecting a null hypothesis when it's false. Not rejecting a null hypothesis, does not mean it is true. Remember, Lance Armstrong? While there was suspicion around him using drugs, it took more than a decade to actually reject the null hypothesis. In his case, the assumption was that he was innocent and not using anything. The reason scientists were not able to reject this innocent hypothesis was, because they didn't have tests that were good enough to detect the illegal performance and enhancers which he was using.

## Concluding Remarks - Slide 112

### CONCLUDING REMARKS

Statistical significance does not mean that you have made an important or meaningful discovery.

Not rejecting the null hypothesis does not mean it is true.

The size of the sample affects the p-value of a test. With enough data, a trivial difference from  $H_0$  leads to a statistically significant outcome.



Statistical significance does not mean that you have made an important or meaningful discovery.

Not rejecting the null hypothesis does not mean it is true.

The size of the sample affects the p-value of a test. With enough data, a trivial difference from  $H_0$  leads to a statistically significant outcome.



## Transcript

Size of the sample effects the p-value of a test with enough data, a trivial difference from the null hypothesis leads to statistically significant outcome or the other way around which means not all statistical significance have substantive importance. If you really get to understand what is going on, when we do hypothesis testing then you can design your questions in sampling better which will yield more meaningful results and understanding the principals of hypothesis testing, also means that you would also be able to legitimately question when studies you're looking at have failed to do this test correctly and does create erroneous insight. The more you learn on which factors influences the strengths of statistical methods used, the more likely that you have investigated the right questions and have arrived at the right solution.

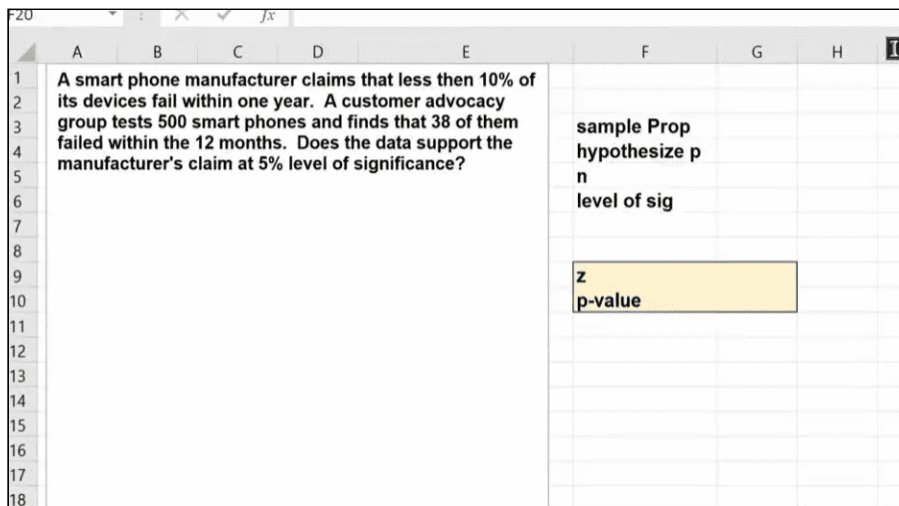
### Academic Citations:

- American Academy of Neurology. (2016, April 11). *Study: More than 40 percent of retired NFL players had brain injury* [Press release]. Retrieved from [https://www.aan.com/pressRoom/Home/PressRelease/1453?tid=a\\_inl](https://www.aan.com/pressRoom/Home/PressRelease/1453?tid=a_inl).
- Bureau of Labor Statistics. (2016, April 23). *Labor force statistics from the current population survey* [Data set]. Retrieved on April 23, 2016 from <http://data.bls.gov/timeseries/LNS14000000>.
- Drugs.com. (n.d.). *Hypertonic dextrose injection*. Retrieved from <http://www.drugs.com/pro/hypertonic-dextrose-injection.html>

## Lesson 1-5.2 Left-Tail Test for Proportion in Excel

[Media Player for Video](#) 

### Excel Worksheet (1 of 6) - Slide 113



|    | A                                                                                                                                                                                                                                                                             | B | C | D | E | F             | G | H | I |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---------------|---|---|---|
| 1  | A smart phone manufacturer claims that less then 10% of its devices fail within one year. A customer advocacy group tests 500 smart phones and finds that 38 of them failed within the 12 months. Does the data support the manufacturer's claim at 5% level of significance? |   |   |   |   | sample Prop   |   |   |   |
| 2  |                                                                                                                                                                                                                                                                               |   |   |   |   | hypothesize p |   |   |   |
| 3  |                                                                                                                                                                                                                                                                               |   |   |   |   | n             |   |   |   |
| 4  |                                                                                                                                                                                                                                                                               |   |   |   |   | level of sig  |   |   |   |
| 5  |                                                                                                                                                                                                                                                                               |   |   |   |   |               |   |   |   |
| 6  |                                                                                                                                                                                                                                                                               |   |   |   |   |               |   |   |   |
| 7  |                                                                                                                                                                                                                                                                               |   |   |   |   |               |   |   |   |
| 8  |                                                                                                                                                                                                                                                                               |   |   |   |   |               |   |   |   |
| 9  |                                                                                                                                                                                                                                                                               |   |   |   |   | z             |   |   |   |
| 10 |                                                                                                                                                                                                                                                                               |   |   |   |   | p-value       |   |   |   |
| 11 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |   |   |   |
| 12 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |   |   |   |
| 13 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |   |   |   |
| 14 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |   |   |   |
| 15 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |   |   |   |
| 16 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |   |   |   |
| 17 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |   |   |   |
| 18 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |   |   |   |

A smart phone manufacturer claims that less than 10% of its devices fail within one year. A customer advocacy group tests 500 smart phones and finds that 38 of them failed within the 12 months. Does the data support the manufacturer's claim at 5% level of significance?

This slide contains a screen shot of an Excel worksheet. Column F contains the following labels: sample Prop, hypothesize p, n, level of sig, z, and p-value.

## Transcript

A smart phone manufacturer claims that less than 10% of its devices fail within one year. A customer advocacy group tests 500 smart phones and find that 30 of them fail within the 12 months. Does the data support the manufacturers claim at 5% level of significance?

## Excel Worksheet (2 of 6) - Slide 114

|    | A                                                                                                                                                                                                                                                                             | B | C | D | E | F             | G            | H | I | J |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---------------|--------------|---|---|---|
| 1  | A smart phone manufacturer claims that less than 10% of its devices fail within one year. A customer advocacy group tests 500 smart phones and finds that 30 of them failed within the 12 months. Does the data support the manufacturer's claim at 5% level of significance? |   |   |   |   |               |              |   |   |   |
| 2  |                                                                                                                                                                                                                                                                               |   |   |   |   | sample Prop   | 0.076        |   |   |   |
| 3  |                                                                                                                                                                                                                                                                               |   |   |   |   | hypothesize p | 0.1          |   |   |   |
| 4  |                                                                                                                                                                                                                                                                               |   |   |   |   | n             | 500          |   |   |   |
| 5  |                                                                                                                                                                                                                                                                               |   |   |   |   | level of sig  | 0.05         |   |   |   |
| 6  |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 7  |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 8  |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 9  |                                                                                                                                                                                                                                                                               |   |   |   |   | z             | = (G3-G4)/SQ |   |   |   |
| 10 |                                                                                                                                                                                                                                                                               |   |   |   |   | p-value       |              |   |   |   |
| 11 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 12 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 13 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 14 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 15 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 16 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 17 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 18 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 19 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 20 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 21 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |
| 22 |                                                                                                                                                                                                                                                                               |   |   |   |   |               |              |   |   |   |

$$H_0 : p \geq 0.10$$

$$H_A : p < 0.10$$

$$n \times p \geq 5, n \times (1 - p) \geq 5$$

This slide contains the same Excel worksheet presented on [Slide 113 - Excel Worksheet \(1 of 6\)](#). The following data is shown: sample Prop: 0.076, hypothesize p: 0.1, n:500, level of sig: 0.05.

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

## Transcript

So, we want to do this as a hypothesis test. The claim is that less than will fail. So, as you can see, less than does not have the equality sign in it, it doesn't say 10% or less, it says less than 10%, so we have to put now the claim that the manufacturer makes as the ultimate hypothesis. So, the ultimate hypothesis is that proportion level fail is less than 10%, therefore, our null is that the proportion would be greater than or equal to the 10%. One of the things that I have not mentioned in my lessons is that when you're doing a testing on proportion, proportion is not a continuous variable. For example, a smart phone either fails or it doesn't fail, so it's a zero-one, it's a discrete variable, not a continuous variable. But if you have a large enough sample where  $n$  times  $p$  and  $n$  times  $1$  minus  $p$  is greater than or equal to 5, then you can use normal distribution to approximate for what is really known as a binomial distribution. So, I have not mentioned it because I have been saying to you that we will be using large data sets, which in most cases to have a valid study you would need a large data set and, remember, large is not that big of a sample size, anything above 30 most of the time is considered large enough. So, if these two conditions hold, we will use normal distribution which means we will use  $z$ -value as measure of how many standard errors away is our sample from what we thought it should be.

So, now that we have this, we can go ahead and calculate it and what we are looking for is this value of  $z$  and  $p$  value. Now,  $z$  is calculated by taking the  $\hat{x}$ , which is the sample proportion minus  $p$  subscript zero, which is the hypothesized proportion and dividing that by the standard error, which is just  $p$  subscript zero times  $1$  minus  $p$  subscript zero divided by  $n$  when you take the entire square root. So, I'm going to use the right-hand side to find these values. So, sample proportion which is your  $\hat{x}$  is 38 out of the 500. So, in our sample, 7.6 failed, but we don't know if this is enough to support the manufacturers claim or not, so we're going to check that. The hypothesized proportion is 10% and, in this case, is 500 cell phones that were tested or smart phones that were tested, the level of significance is .05, so now  $z$  is going to be, and I'm going to use this formula to do the calculations.

## Excel Worksheet (3 of 6) - Slide 115

The screenshot shows an Excel worksheet with the following data:

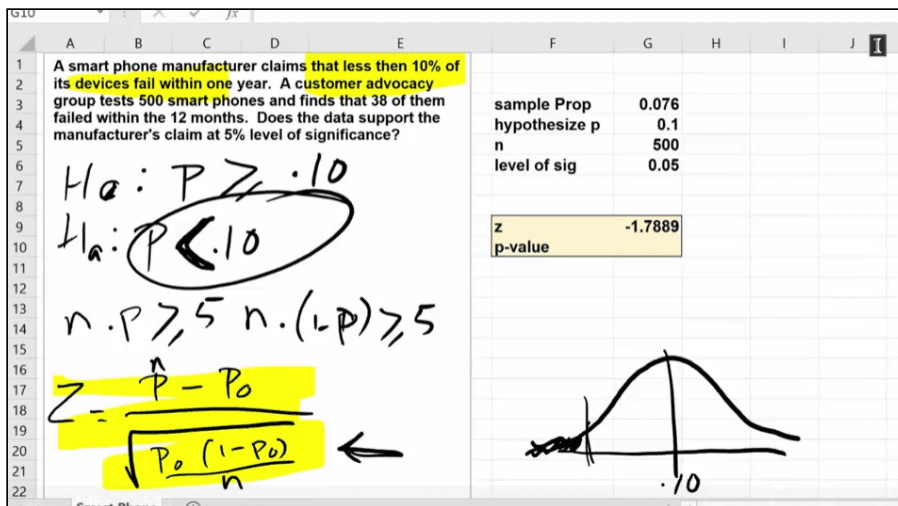
|                                                                             | E | F             | G                        | H | I | J |
|-----------------------------------------------------------------------------|---|---------------|--------------------------|---|---|---|
| less than 10% of<br>er advocacy<br>at 38 of them<br>a support the<br>cance? |   |               |                          |   |   |   |
|                                                                             |   | sample Prop   | 0.076                    |   |   |   |
|                                                                             |   | hypothesize p | 0.1                      |   |   |   |
|                                                                             |   | n             | 500                      |   |   |   |
|                                                                             |   | level of sig  | 0.05                     |   |   |   |
|                                                                             |   |               |                          |   |   |   |
|                                                                             |   | z             | =(G3-G4)/sqrt(G4*(1-G4)/ |   |   |   |
|                                                                             |   | p-value       | G5)/SQRT(number)         |   |   |   |

This slide contains the same Excel worksheet presented on [Slide 113 - Excel Worksheet \(1 of 6\)](#). In Column G, next to the label  $z$ , the formula  $=(G3-G4)/\text{sqrt}(G4*(1-G4)/G5)$  is being used.

## Transcript

So,  $z$  is, and I'm going to use parentheses to not make any errors here. So,  $\hat{p}$  which is right here, minus hypothesized proportion, so that's my numerator, divided by the square root of  $p$  subscript zero, which is hypothesized proportion times 1 minus  $p$  subscript zero and dividing this value by the sample size of 500, closing the parentheses, and you get the value of negative 1.7889. So, now what are we looking at? We are looking at – we are looking at samples that will be normally distributed and you would expect that, if this is the true proportion of failure, we would get this as the mean of the sampling distribution and the sampling distribution would look normal and the standard error, which is right here, this denominator is going to give me the shape. Now, it's a left tail, because we have a less than sign here, so it's a left tail, so anything that deviates more than this 5% chance of finding, then we will say that it will fall into a rejection region, we will reject it.

## Excel Worksheet (4 of 6) - Slide 116



This slide contains the same information presented on [Slide 114 - Excel Worksheet \(2 of 6\)](#). The  $z$  value now reads: -1.7889.

Standard error is  $z$  value. Left-tail because  $H_a$  is less than.

## Transcript

So, if we deviate too much, we will end up rejecting. That's a small peak. The sample that you've taken is so rare, you should not assume that you're just unlucky. The first assumption should be that probably the claim, the null hypothesis is not true. So, now we have to find the  $p$ -value here. So, to find the  $p$  value, we will use the normal distribution, which is norm.s.dist.

## Excel Worksheet (5 of 6) - Slide 117

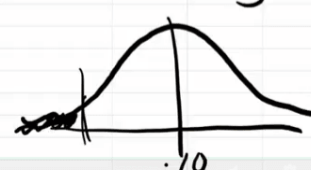
| D                                                                                                                                       | E | F                                                                                                                   | G                 | H | I | J |
|-----------------------------------------------------------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------|-------------------|---|---|---|
| claims that less than 10% of<br>A customer advocacy<br>and finds that 38 of them<br>Does the data support the<br>level of significance? |   | sample Prop                                                                                                         | 0.076             |   |   |   |
|                                                                                                                                         |   | hypothesize p                                                                                                       | 0.1               |   |   |   |
|                                                                                                                                         |   | n                                                                                                                   | 500               |   |   |   |
|                                                                                                                                         |   | level of sig                                                                                                        | 0.05              |   |   |   |
|                                                                                                                                         |   | z                                                                                                                   | -1.7889           |   |   |   |
|                                                                                                                                         |   | p-value                                                                                                             | =NORM.S.DIST(G9,1 |   |   |   |
|                                                                                                                                         |   | NORM.S.DIST returns the cumulative distribution function (TRUE - cumulative distribu<br>FALSE - probability density |                   |   |   |   |

This slide contains the same information presented on [Slide 114 - Excel Worksheet \(2 of 6\)](#). In Column G, next to the label p-value, the formula =norm.s.dist is being used.

### Transcript

That means that I'm just going to give it a z-value and then say 1 and see what I get. I get a p-value of .03, so what does that mean?

## Excel Worksheet (6 of 6) - Slide 118

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>A smart phone manufacturer claims that less than 10% of its devices fail within one year. A customer advocacy group tests 500 smart phones and finds that 38 of them failed within the 12 months. Does the data support the manufacturer's claim at 5% level of significance?</p> <p><math>H_a: P &lt; .10</math></p> <p><math>n \cdot P \geq 5 \quad n \cdot (1-P) \geq 5</math></p> <p><math>z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}</math></p> |  | <p>sample Prop 0.076</p> <p>hypothesize p 0.1</p> <p>n 500</p> <p>level of sig 0.05</p> <p>z -1.7889</p> <p>p-value 0.03682</p> <p><math>P &lt; \alpha</math> Reject</p>  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

This slide contains the same information presented on [Slide 114 - Excel Worksheet \(2 of 6\)](#). The p-value now reads: 0.03682.

[Download the Smart Phones excel file \(Refer to Sheet 1 - First worksheet\)](#) 

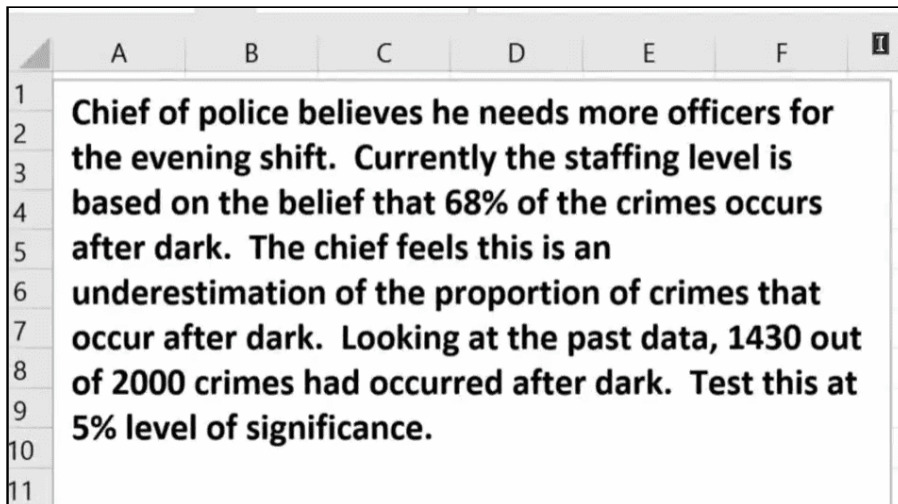
## Transcript

P-value is less than  $\alpha$ , therefore, you will reject in all hypothesis. If you reject in all hypothesis, it means we must accept the claim that less than 10% of the smart phones from this manufacturer will fail within the first year.

## Lesson 1-5.3 Right-Tail Test for Proportion in Excel

[Media Player for Video](#) 

### Excel Worksheet (1 of 5) - Slide 119



The screenshot shows an Excel worksheet with columns A through F and rows 1 through 11. A text box is overlaid on the worksheet, containing the following text:

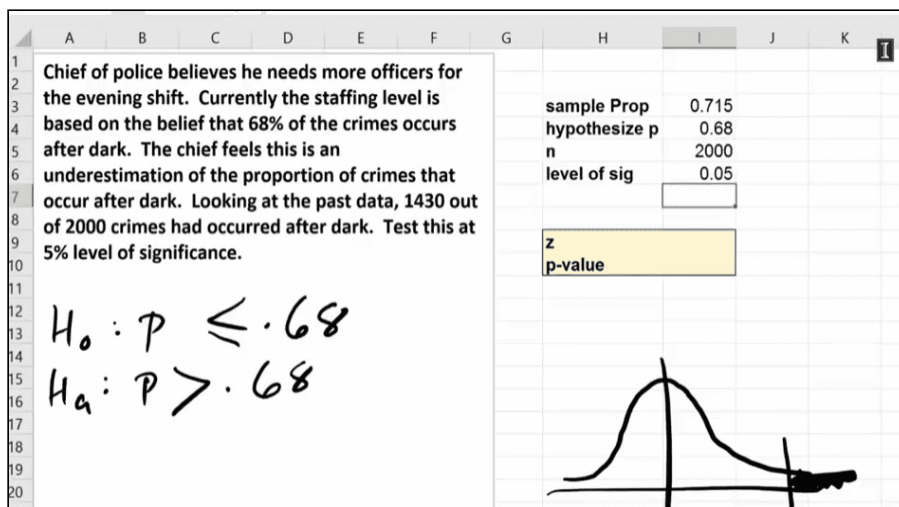
|    | A                                                                                                                                                                                                                                                                                                                                                                                           | B | C | D | E | F |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|
| 1  | <b>Chief of police believes he needs more officers for the evening shift. Currently the staffing level is based on the belief that 68% of the crimes occurs after dark. The chief feels this is an underestimation of the proportion of crimes that occur after dark. Looking at the past data, 1430 out of 2000 crimes had occurred after dark. Test this at 5% level of significance.</b> |   |   |   |   |   |
| 2  |                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |
| 3  |                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |
| 4  |                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |
| 5  |                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |
| 6  |                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |
| 7  |                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |
| 8  |                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |
| 9  |                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |
| 10 |                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |
| 11 |                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |   |

Chief of police believes he needs more officers for the evening shift. Currently the staffing level is based on the belief that 68% of the crimes occurs after dark. The chief feels this is an underestimation of the proportion of crimes that occur after dark. Looking at the past data, 1430 out of 2000 crimes had occurred after dark. Test this at 5% level of significance.

## Transcript

Chief of police believes he needs more officers for the evening shift. Currently the staffing level is based on the belief that 68% of the crimes occur after dark. The chief feels this is an underestimation of the proportion of crimes that occur after dark. Looking at the past data, 1,430 out of 2,000 crimes had occurred after dark. You want to test this at 5% level of significance. So again, this is a test about proportion. And first thing we need to do is write out our hypotheses. So, to write our null hypothesis and alternate hypothesis, we're going to look at the statements that are being made. First of all, we think that there is a 68% crime occurs after dark. But the chief thinks that this is an underestimation. So, the chief believes that this is larger than 68%. Therefore, our null is  $p$  is less than or equal to the 68%. And we're going to test this at 5% level of significance. So once again, we have a one tailed test and this time, our rejection region on the right side. So therefore, it's a right tailed test. If we have a sample that falls into this tail – if the prop of finding a sample like ours is smaller than .05, which is in the tail, then you'll end up rejecting the null hypothesis. Otherwise, we will return the null hypothesis. So now let's see how this will work out. So, what was our sample proportion?

## Excel Worksheet (2 of 5) - Slide 120



This slide contains the same text presented on [Slide 119 - Excel Worksheet \(1 of 5\)](#).

Below the text are the following formulas:

$$H_0 : p \leq 0.68$$

$$H_A : p > 0.68$$

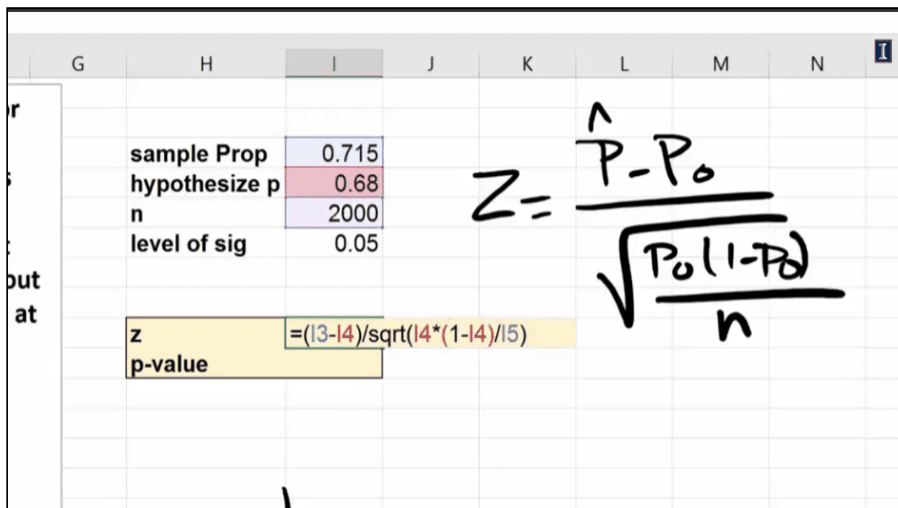
This slide contains a screen shot of an Excel worksheet. Columns H and I contain the following data: sample Prop: 0.715, hypothesize p: 0.68, n: 2000, level of sig: 0.05. Column H also contains the following labels: z and p-value. Below this data is a bell curve with the area under the curve on the far right shaded.



## Transcript

Sample proportion was that 1,430 out of the last 2,000 have occurred after dark, so that's about 71.5%. And again, we see that that is larger than 68%, but we have to make sure that this is a statistically significant difference. So, our hypothesis is that it is 68% unless we are shown otherwise. We looked at 2,000 different crimes and the level of significance that we are looking at is .05. So once again we are going to calculate our z-value. And z, once again, is  $\hat{p}$  minus  $p_0$  divided by  $p_0$  times 1 minus  $p_0$  divided by the sample size.

## Excel Worksheet (3 of 5) - Slide 121



|               | G | H | I                            | J | K | L | M | N |
|---------------|---|---|------------------------------|---|---|---|---|---|
| sample Prop   |   |   | 0.715                        |   |   |   |   |   |
| hypothesize p |   |   | 0.68                         |   |   |   |   |   |
| n             |   |   | 2000                         |   |   |   |   |   |
| level of sig  |   |   | 0.05                         |   |   |   |   |   |
| z             |   |   | = (I3-I4)/sqrt(I4*(1-I4)/I5) |   |   |   |   |   |
| p-value       |   |   |                              |   |   |   |   |   |

This slide contains the same Excel worksheet presented on [Slide 120 - Excel Worksheet \(2 of 5\)](#). In Column I, next to the label z, the formula `= (I3-I4)/sqrt(I4*(1-I4)/I5)` is being used.

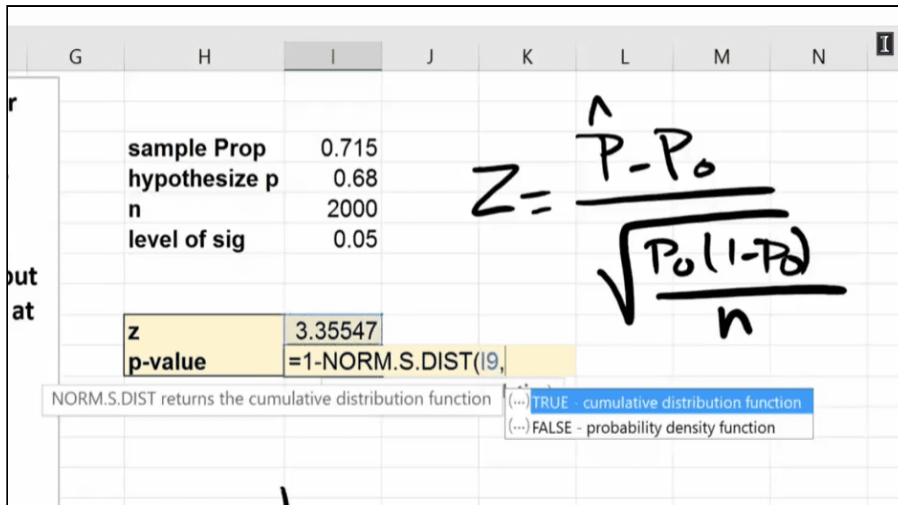
$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

## Transcript

Following that, we will say it's equal to – this is  $\hat{p}$  sample proportion minus hypothesis  $p_0$  – that's the numerator divided by square root of  $p_0$  times 1 minus  $p_0$  divided by  $n$ . And enclose the parentheses and you get a value that is 3.335. Again, we're into the tail.



## Excel Worksheet (4 of 5) - Slide 122

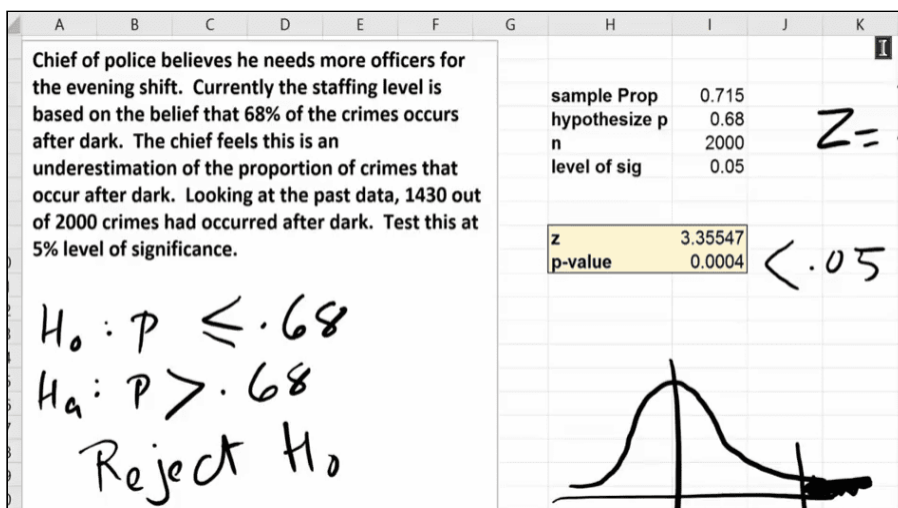


This slide contains the same Excel worksheet and equation presented on [Slide 121 - Excel Worksheet \(3 of 5\)](#). In Column I, next to the label p-value, the formula =1-norm.s.dist is being used.

### Transcript

So, the p-value is therefore 1- norm.s.dist because again, I'm looking for the value to the right of the tail so if I don't put 1 minus, value to the left will be given. So, the z that I'm looking for is this and one and the answer is .004. this is less than the .05 and therefore we will reject the null hypothesis.

## Excel Worksheet (5 of 5) - Slide 123



This slide contains the same Excel worksheet and bell graph presented on [Slide 120 - Excel Worksheet \(2 of 5\)](#). The z value now reads 3.35547 and the p-value reads 0.0004.

$$H_0 : p \leq 0.68$$

$$H_A : p > 0.68$$

Reject  $H_0$

[Download the Chief of Police excel file \(Refer to Sheet 1 - First worksheet\)](#) 

## Transcript

Which means the chief is actually correct – that there is more crime than believed happening after dark so if he needs more staffing to curb the crime, then he should be given the extra staffing.

## Lesson 1-5.4 Two-Tail Test for Proportion in Excel

[Media Player for Video](#) 

### Excel Worksheet (1 of 7) - Slide 124

| A                                                                                                                                                                                                                                                                                    | B | C | D | E | F | G | H             | I |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---------------|---|
| A direct-mail marketing firm has started using an App to distribute coupons. Out of 1000 randomly selected customers that were using the App, 94 bought the product. They are wondering if this rate is different from their 7% success rate? Test this at 5% level of significance. |   |   |   |   |   |   | sample Prop   |   |
| $H_0 : p = .07$                                                                                                                                                                                                                                                                      |   |   |   |   |   |   | hypothesize p |   |
| $H_a : p \neq .07$                                                                                                                                                                                                                                                                   |   |   |   |   |   |   | n             |   |
|                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   | level of sig  |   |
|                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   | z             |   |
|                                                                                                                                                                                                                                                                                      |   |   |   |   |   |   | p-value       |   |

This slide contains a screen shot of an Excel worksheet. Column H contains the following labels: sample Prop, hypothesize p, n, level of sig, z, and p-value.

A direct-mail marketing firm has started using an App to distribute coupons. Out of 1000 randomly selected customers that were using the App, 94 bought the product. They are wondering if this rate is different from their 7% success rate? Test this at 5% level of significance.

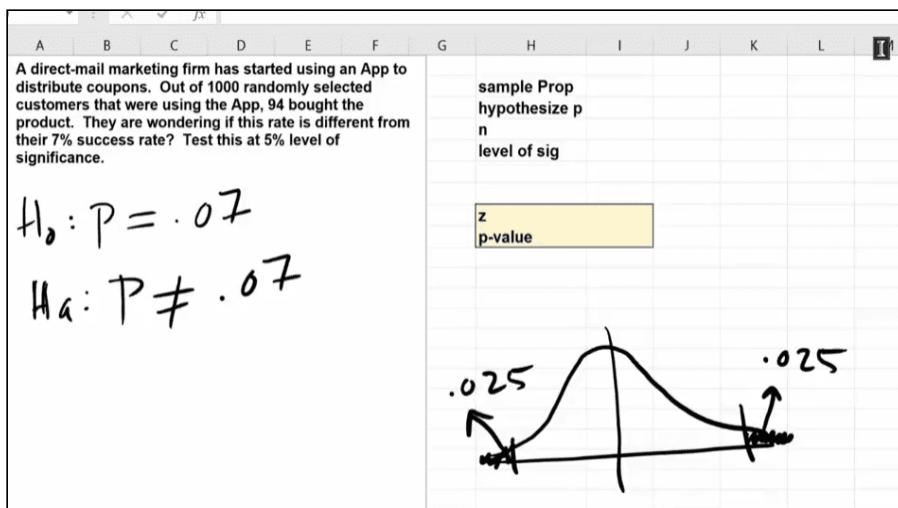
$$H_0 : p = 0.07$$

$$H_A : p \neq 0.07$$

## Transcript

A direct-mail marketing firm has started using an app to distribute coupons. Out of 1,000 randomly selected customers that were using the app, 94 bought the product. They're wondering if this rate is different from their 7% success rate. And you want to test this at 5% level of significance. So, we're going to do hypothesis testing again here. The difference here is that we are not assuming that the app is more successful or less successful. We just want to see if it's any different. So, to rewrite this problem as a set of hypotheses, we would say that the norm hypothesis is that the proportion of people who will buy will not be any different than what we have had before which is seven percent. And the alternate is that this, this new way of marketing might be different. So, it's the test of equality versus inequality, which means that it's a two-tail test. Everything else will remain the same except how the  $\alpha$  is split up.

## Excel Worksheet (2 of 7) - Slide 125



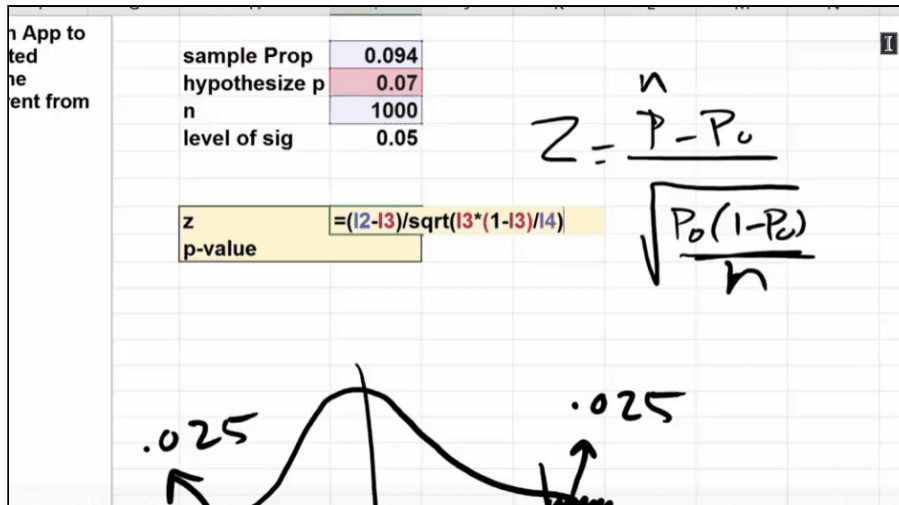
This slide contains the same Excel worksheet and equations presented on [Slide 124 - Excel Worksheet \(1 of 7\)](#).

This slide contains a bell curve that crosses the center line and is shaded at both the far right and far left ends. The shaded portions are labeled: .025.

## Transcript

The  $\alpha$  in this case is going to be split up in both sides of the tail. So, we are different if we have more people buying or less people buying and that's five percent that you're talking about is going to be right here.

## Excel Worksheet (3 of 7) - Slide 126



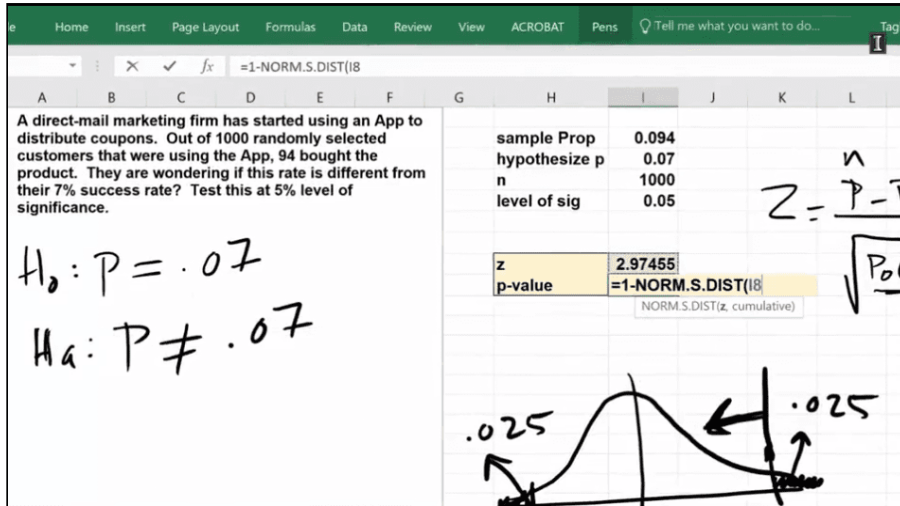
This slide contains the same Excel worksheet presented on [Slide 124 - Excel Worksheet \(1 of 7\)](#). The following data has been added: sample Prop: 0.094, hypothesize p: 0.07, n: 1000, level of sig: 0.05

In column I, next to the label z, the formula `= (I2-I3)/sqrt(I3*(1-I3)/I4)` is being used.

### Transcript

So when we calculate our key value and we compare it to  $\alpha$ , we have one of two options. And I will talk about it when I get to it. So one of the things that I want to do is that first of all find out what was the sample proportion here. And the sample proportion here is that 94 out of 1,000 purchased this product. So then 9.4% of the customers who received this coupon on their app bought the product. And they want to know if .094 is different than that 7%. The sample size that they were looking at was 1,000 customers that they had tested this on. And level of significance here is .05. So now we can calculate z, and z will be calculated by taking  $\hat{p} - p_{\text{subscript zero}}$ , divided by the square root of  $p_{\text{subscript zero}} \times 1 - p_{\text{subscript zero}}$ , divided by n. And I'm going to just calculate that right here. So it's equal to parentheses  $\hat{p}$ , that's our  $\hat{p} - p_{\text{subscript zero}}$  which is right here. Divided by square root of  $p_{\text{subscript zero}}$ . Remember, this is  $p_{\text{subscript zero}}$  or seven times one minus  $p_{\text{subscript zero}}$ . And the whole thing divided by sample size of 1,000. And close the parentheses. And the z-value we get is 2.97. So, the hypothesized proportion was 7%. But we got a sample that is 9.4%, and that is to the right of the mean, and it will be somewhere here.

## Excel Worksheet (4 of 7) - Slide 127

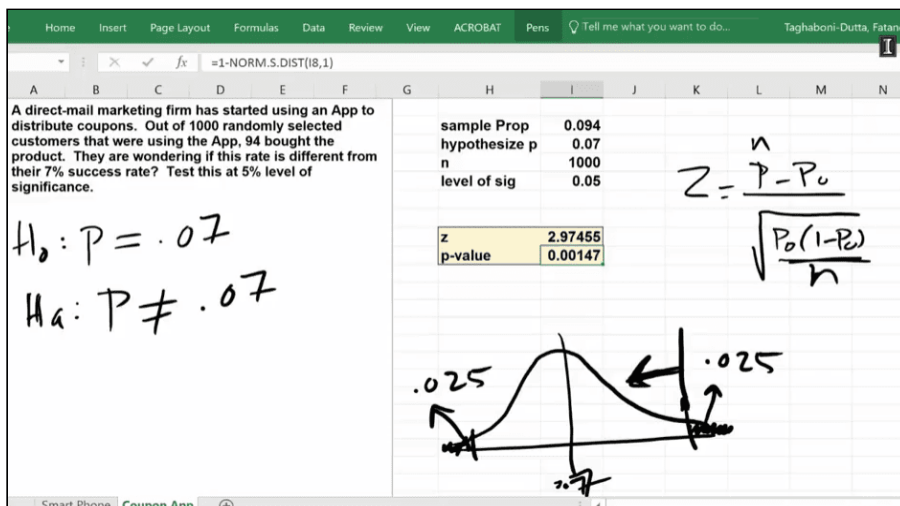


This slide contains the same Excel worksheet presented on [Slide 124 - Excel Worksheet \(1 of 7\)](#). In Column I, next to the label p-value, the formula =1-norm.s.dist is being used.

### Transcript

So in order for me to know what is the p-value, remember the p-value if I put norm.dist is going to return everything to the left of it. So, in order for me to get what is in the tail, I need to do a one-minus. So I'm going to say it's equal to one-minus norm.s.dist, and then take the z-value and one. This is going to return .001.

## Excel Worksheet (5 of 7) - Slide 128

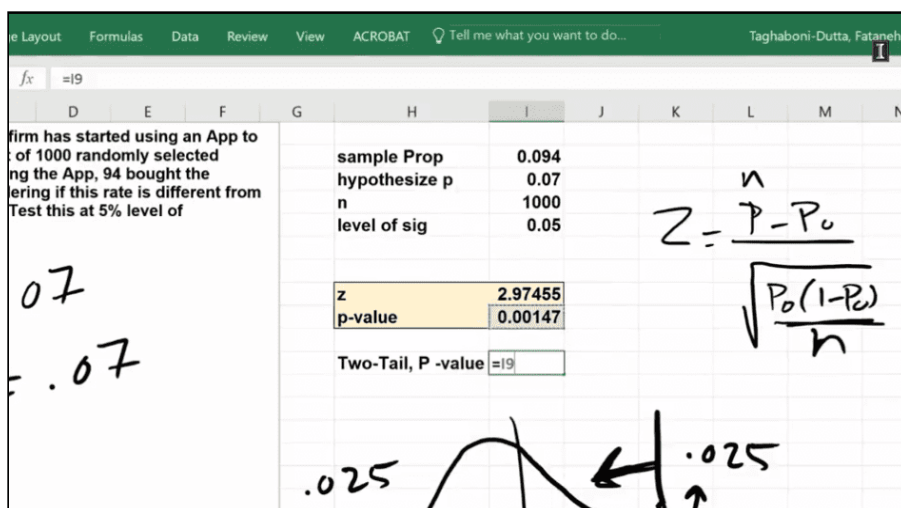


This slide contains the same Excel worksheet presented on [Slide 124 - Excel Worksheet \(1 of 7\)](#). The z value now reads: 2.97455 and the p-value reads: 0.00147.

## Transcript

Now, this is what is in the right tail. But remember because you are doing a two-tail test you have two options here. One is you take this value, this p-value because it's a two-tail test: .001 would be on the right-tail, .001 would be on the left tail. These are the probability of finding samples that are more than 2.97 standard errors away from the mean of the sampling distribution. So, they are, they can be either on the right side or the left side because we will reject the null hypothesis on either end. So, you have two options. One is to take .0011 for seven, and multiply it by two, and then compare it to alpha of .05. Or you can just compare the p-value that you see here with the half of alpha. Either one, doesn't make a difference.

## Excel Worksheet (6 of 7) - Slide 129

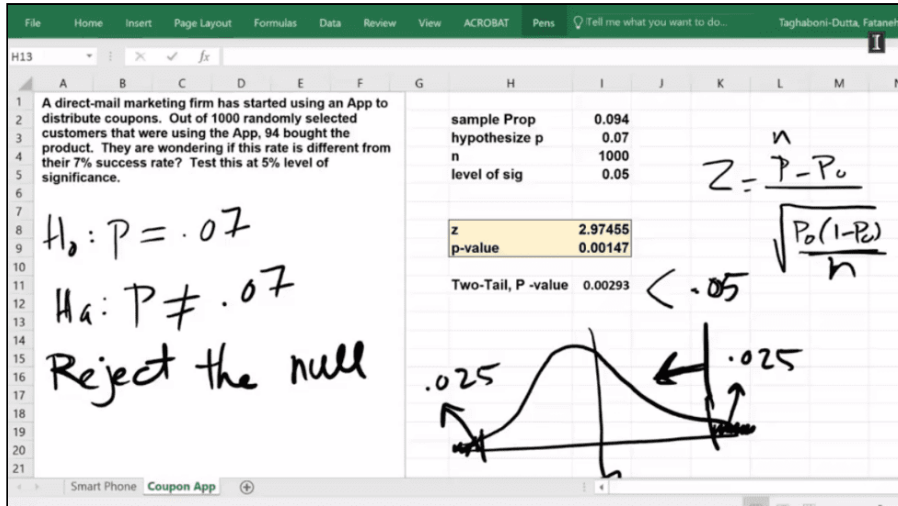


This slide contains the same Excel worksheet presented on [Slide 124 - Excel Worksheet \(1 of 7\)](#). The label: Two-Tail, P-value has been added to column H.

## Transcript

So, to be consistent with my, my teaching in the PowerPoints, I'm going to say since it's a two-tail test, the actual p-value that I'm going to use is going to be this multiplied by two.

## Excel Worksheet (7 of 7) - Slide 130



This slide contains the same Excel worksheet presented on [Slide 124 - Excel Worksheet \(1 of 7\)](#). The Two-Tail, P-value now reads: 0.00293.

[Download the Marketing Firm excel file \(Refer to Sheet 1 - First worksheet\)](#)

### Transcript

Since this is less than my significance level of .05, I will reject the null hypothesis. So, we will reject the null hypothesis which means there is a difference between the ads that have gone out using this app, versus their old method of advertising. Which probably was just the mailed-in rebate or coupon, which was just a mailed-in coupon.